



Advanced Combustion

— Chapter 3. Combustion Reaction Kinetics (Part 2)

Lei Li (李磊) , Fei Qi (齐飞)

Outline of Chapter 3



Part 1

- Fundamentals of Chemical Kinetics
- Reaction Kinetics Theory — Collision Theory
- Reaction Kinetics Theory — Transition State Theory
- Reaction Kinetics Theory — Unimolecular Reaction Theory
- Combustion Reaction Kinetics

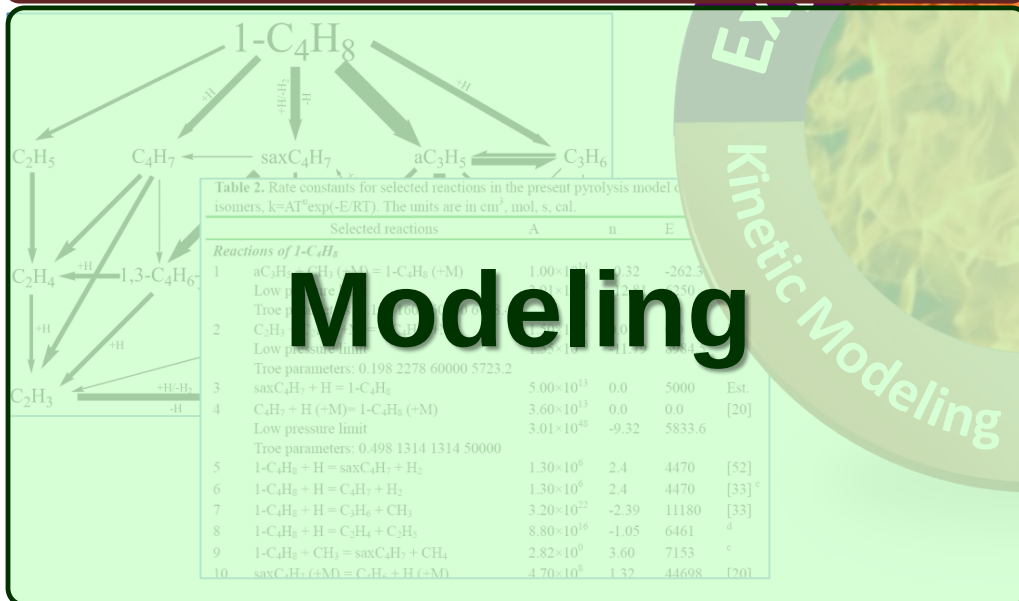
Part 2

- **Research Methods of Combustion Reaction Kinetics**
 - Theoretical methods
 - Experimental Methods (*Reactor and Diagnostics*)
 - Kinetic Modeling Methods
- **Applications of Combustion Chemical Kinetics**

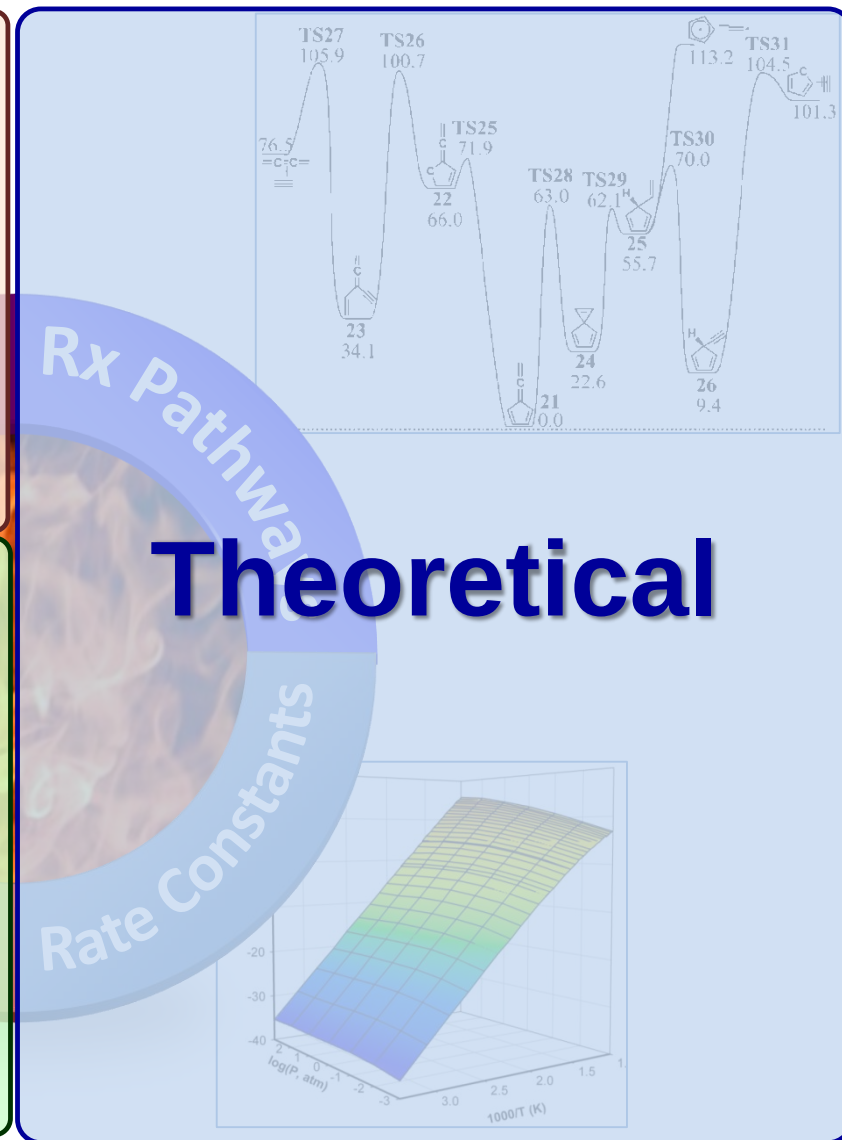
Research Methods

- Flow reactor
- Jet-stirred reactor
- Premixed/nonpremixed flame
- Shock tube
- Constant volume bomb
- ...

Experimental



Modeling



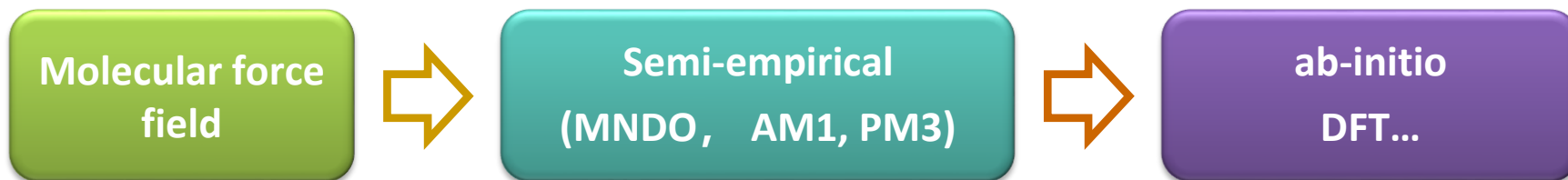
Theoretical Methods



Targets of theoretical study

- | | |
|-----------------------------------|----------------------------|
| ■ Potential energy surface: | Energy of reaction pathway |
| ■ Rate constant calculation: | Kinetic parameters |
| ■ Thermodynamic data calculation: | Thermodynamic parameters |

Development of quantum chemical computation methods:

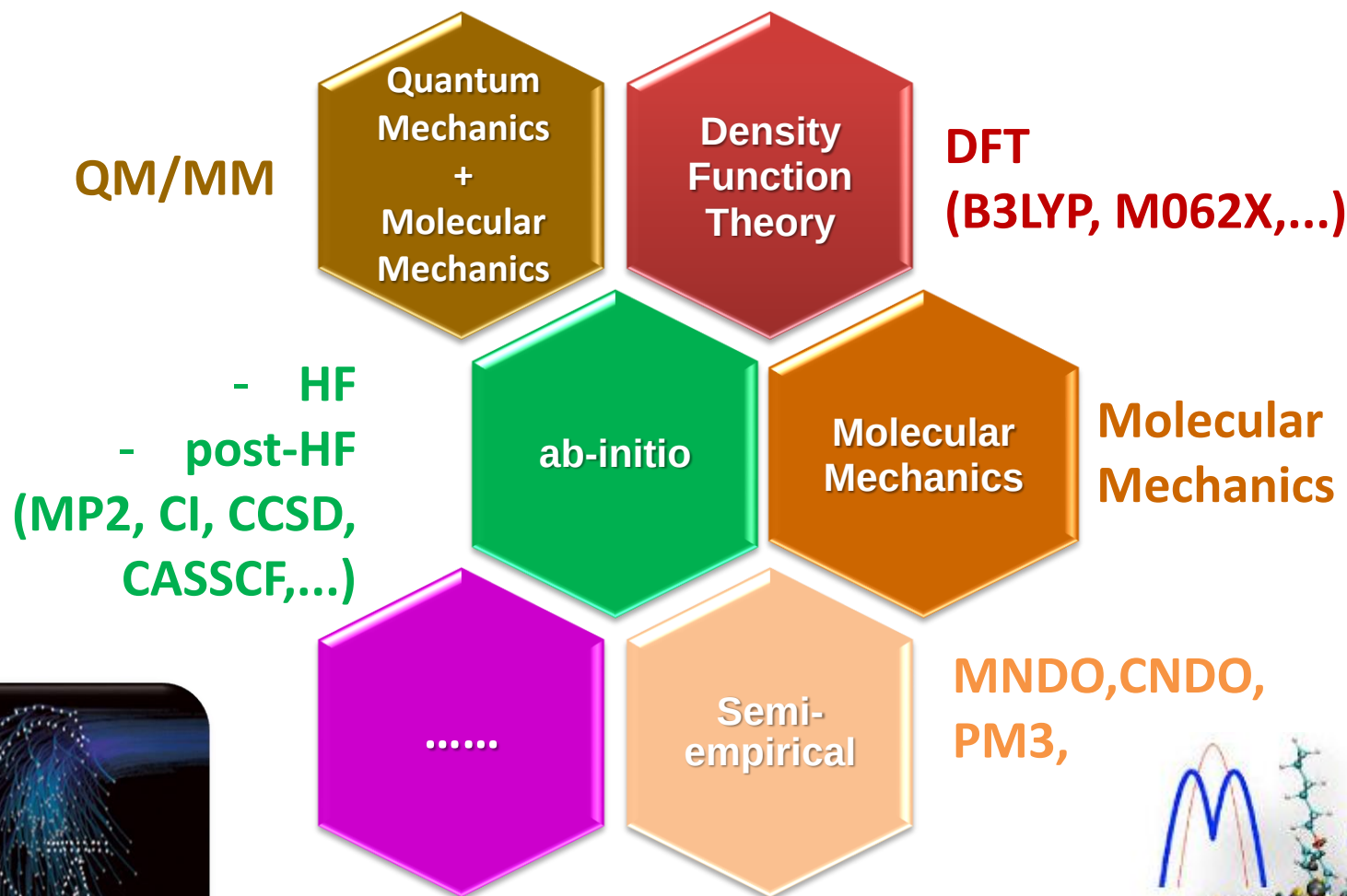


Development of kinetic theory:

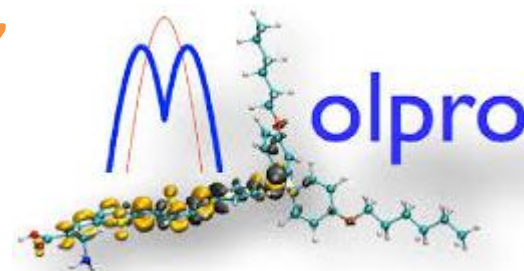


Theoretical Methods—Quantum Chemistry Methods

Quantum Chemistry Methods



Gaussian



Molpro

Theoretical Methods—Quantum Chemistry Methods



Quantum Chemistry Methods

**Molecular Force
Field method...**



**Semi-empirical
(MNDO, AM1,
PM3)**



**ab-initio
Density Function
Theory**

Based on numbers of experimental and computation data; fast, usually used for large system such as bio-macromolecules.

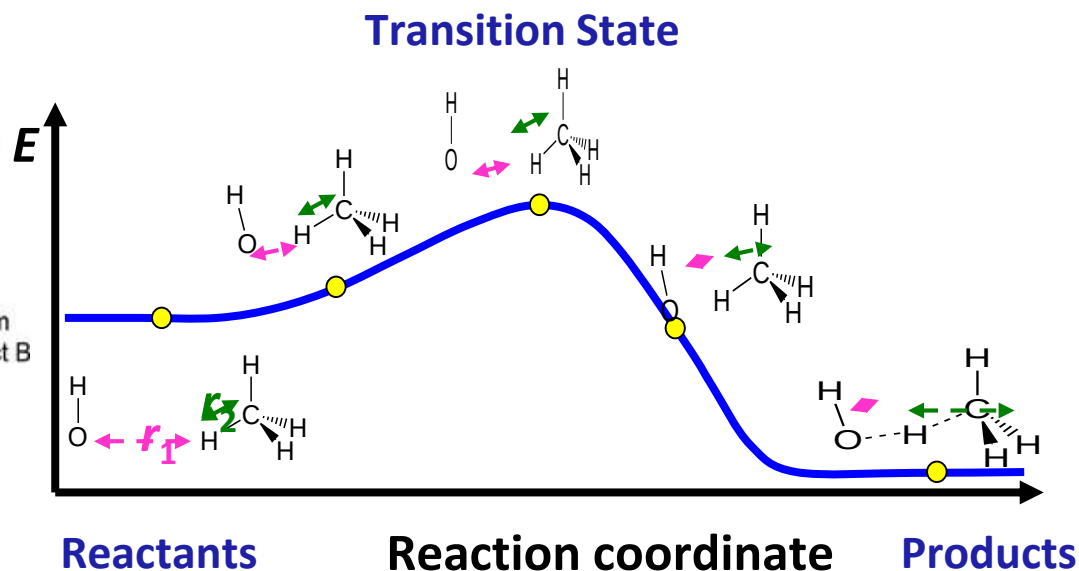
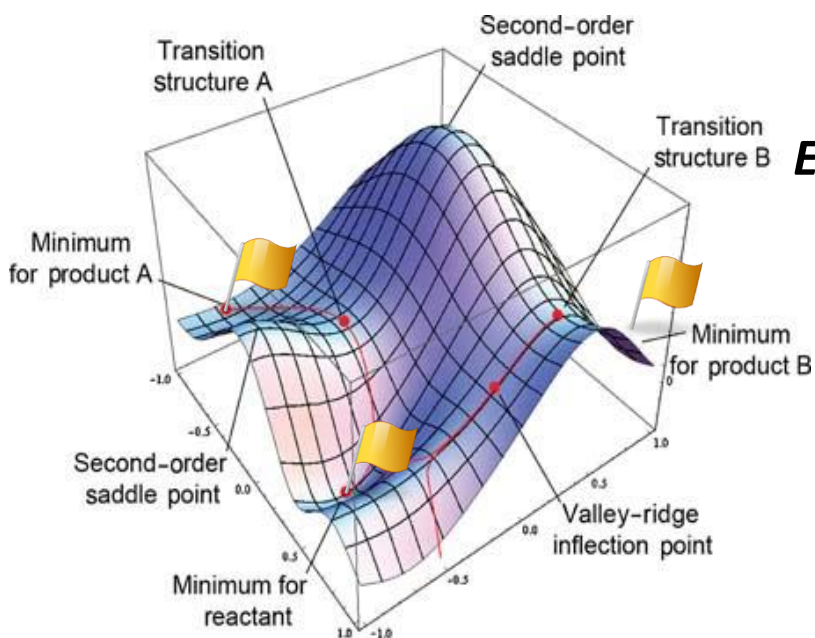
Base on the HF formalism with many approximations; low computation cost; usually used for treating large molecules.

Directly from theoretical principles without any empirical or semi-empirical parameters; accurate, high computation cost.

Theoretical Methods—Quantum Chemistry Methods

PES and Reaction Pathways

Search all possible pathways in combustion reactions



Theoretical Methods—Kinetics Methods



Kinetics Theory

■ Master Equation method (Solves pressure-dependent cases)

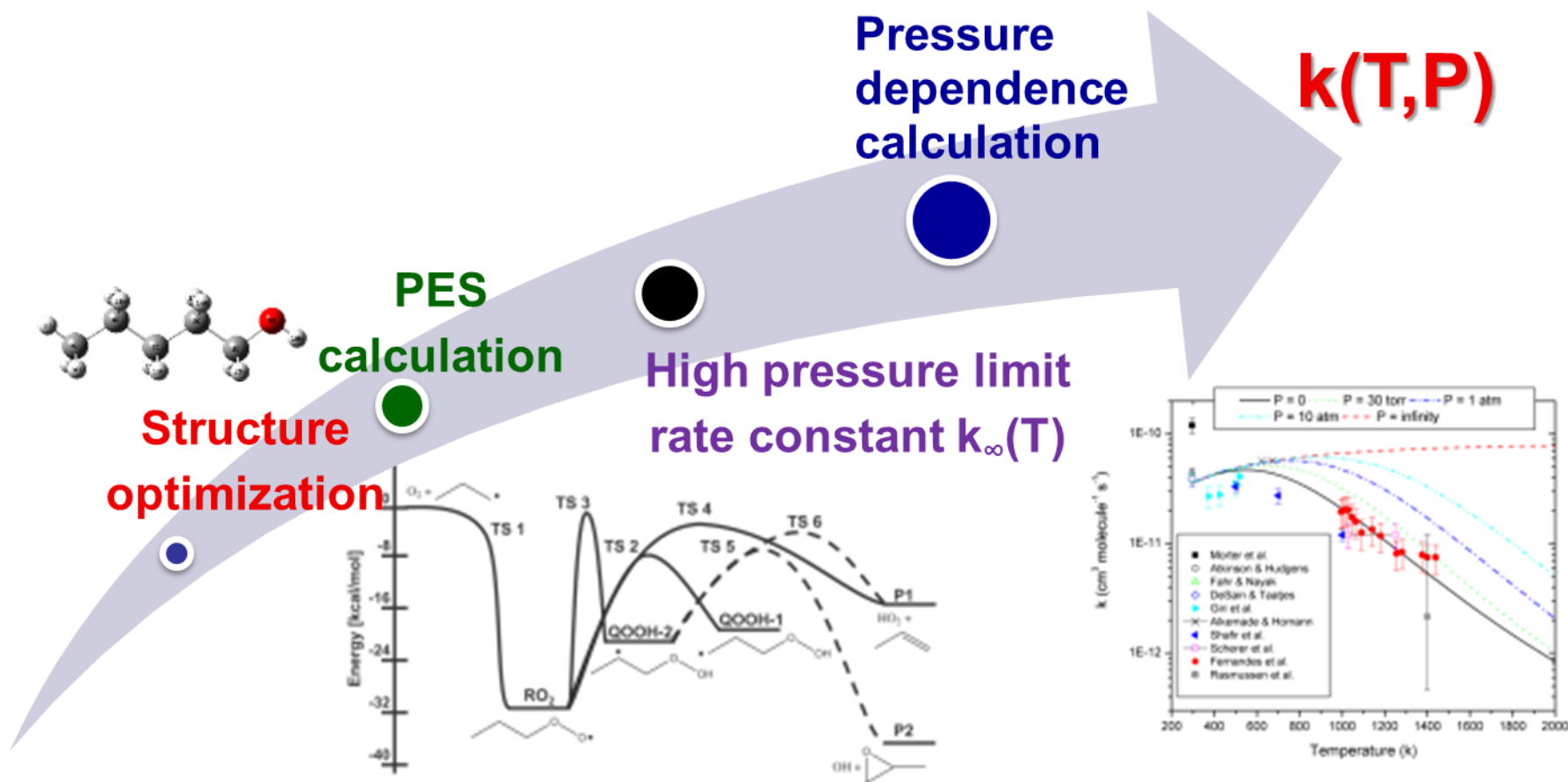
- Reaction process is separated as a collision step and a reaction step.
- Provides a quantitative description of the interaction between collisional energy transfer and chemical reaction for dissociation, isomerization and association processes in combustion.
- By solving ME, pressure-dependent rate constants $k(T,P)$ are got.

■ RRKM-ME method

- Chemical reaction step is treated with the RRKM theory.
- Combining RRKM theory and ME method can predict the rate constants accurately.
- The uncertainty is within a factor of 2.

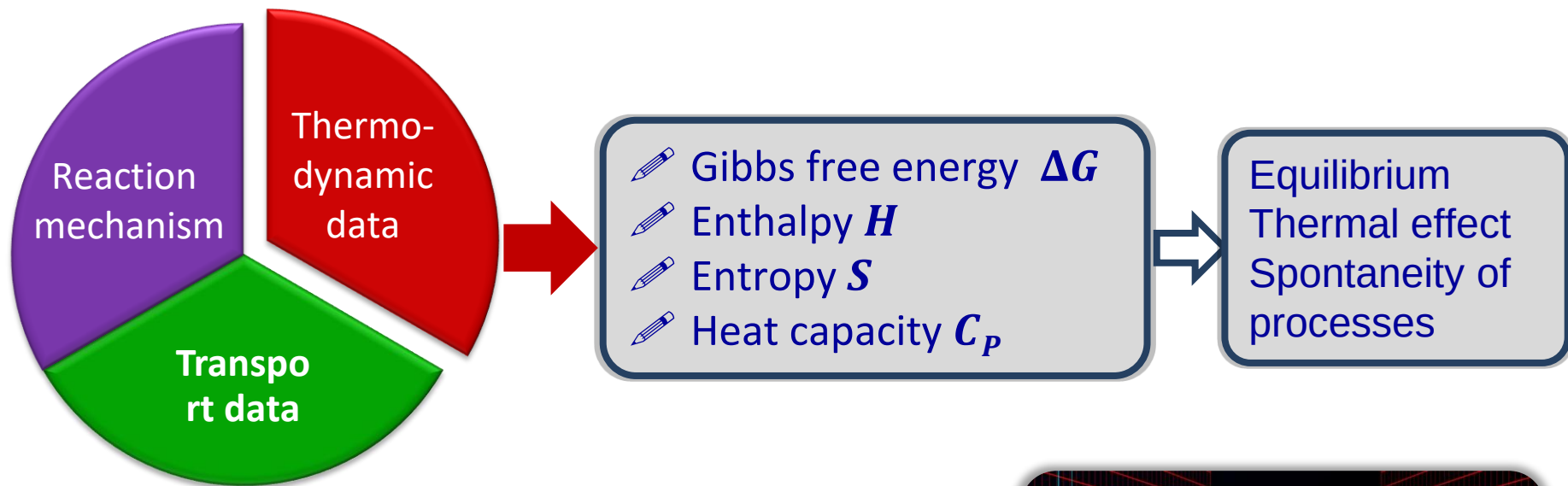
Theoretical Methods

The Flow Chart of Rate Constant Calculation



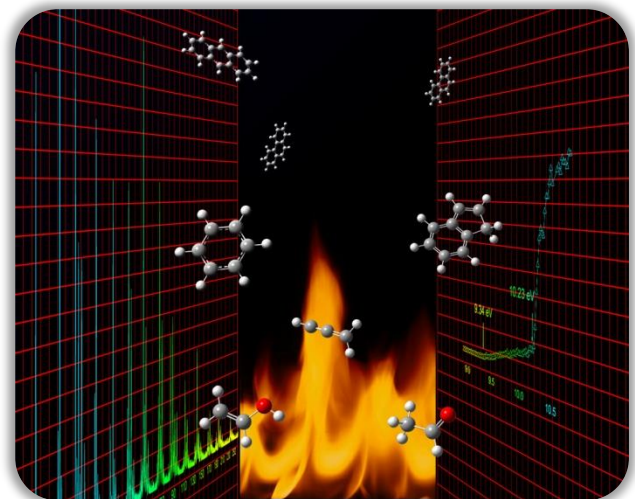
Theoretical Methods

Thermodynamic Properties Calculation



☐ Based on Partition function

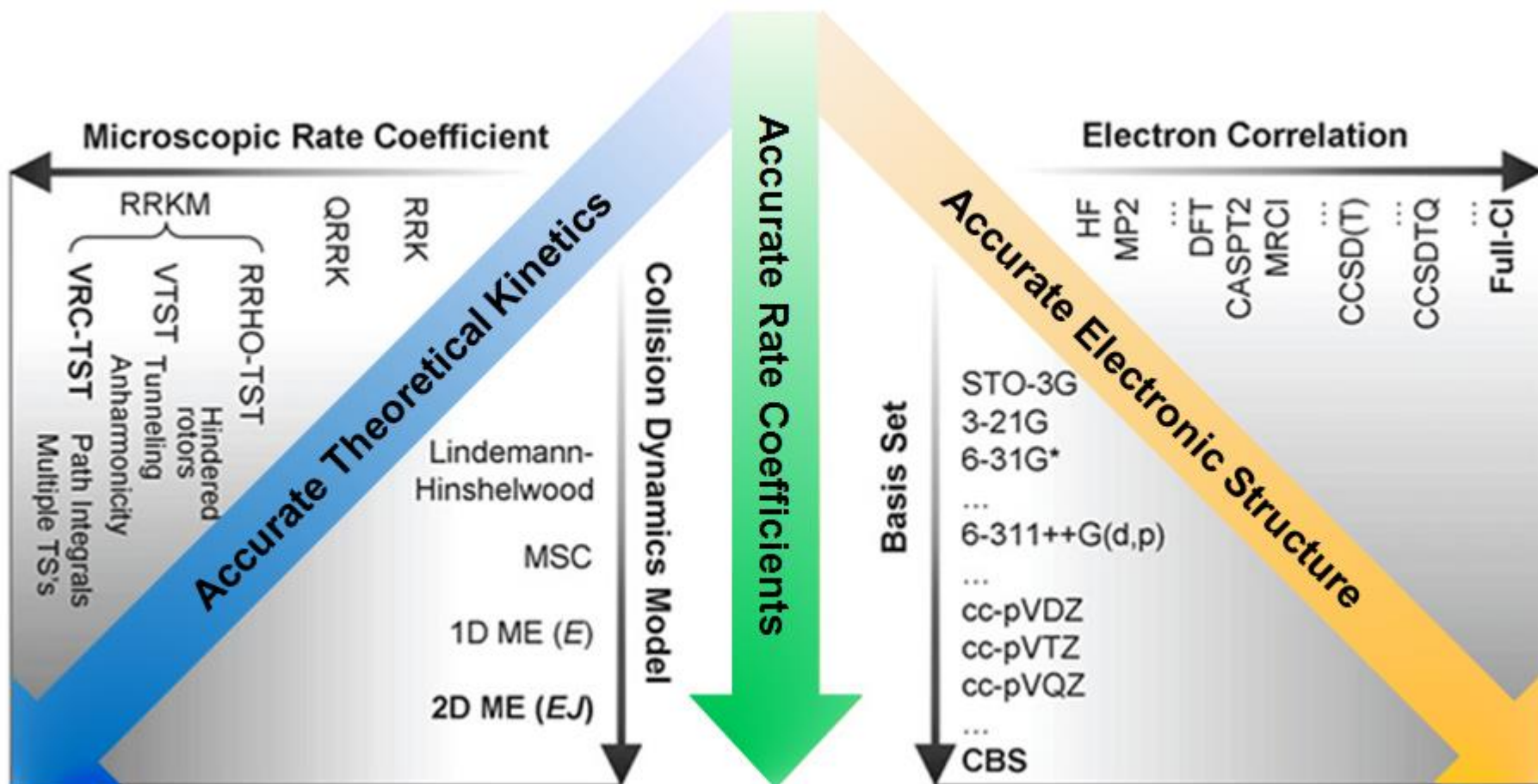
- Compute $\Delta H_{f, 298K}$, S , C_p
- Compute T-dependent thermodynamic parameters



Theoretical Methods



ab-initio QC + Kinetics theory——more accurate rate parameters



Experimental Methods



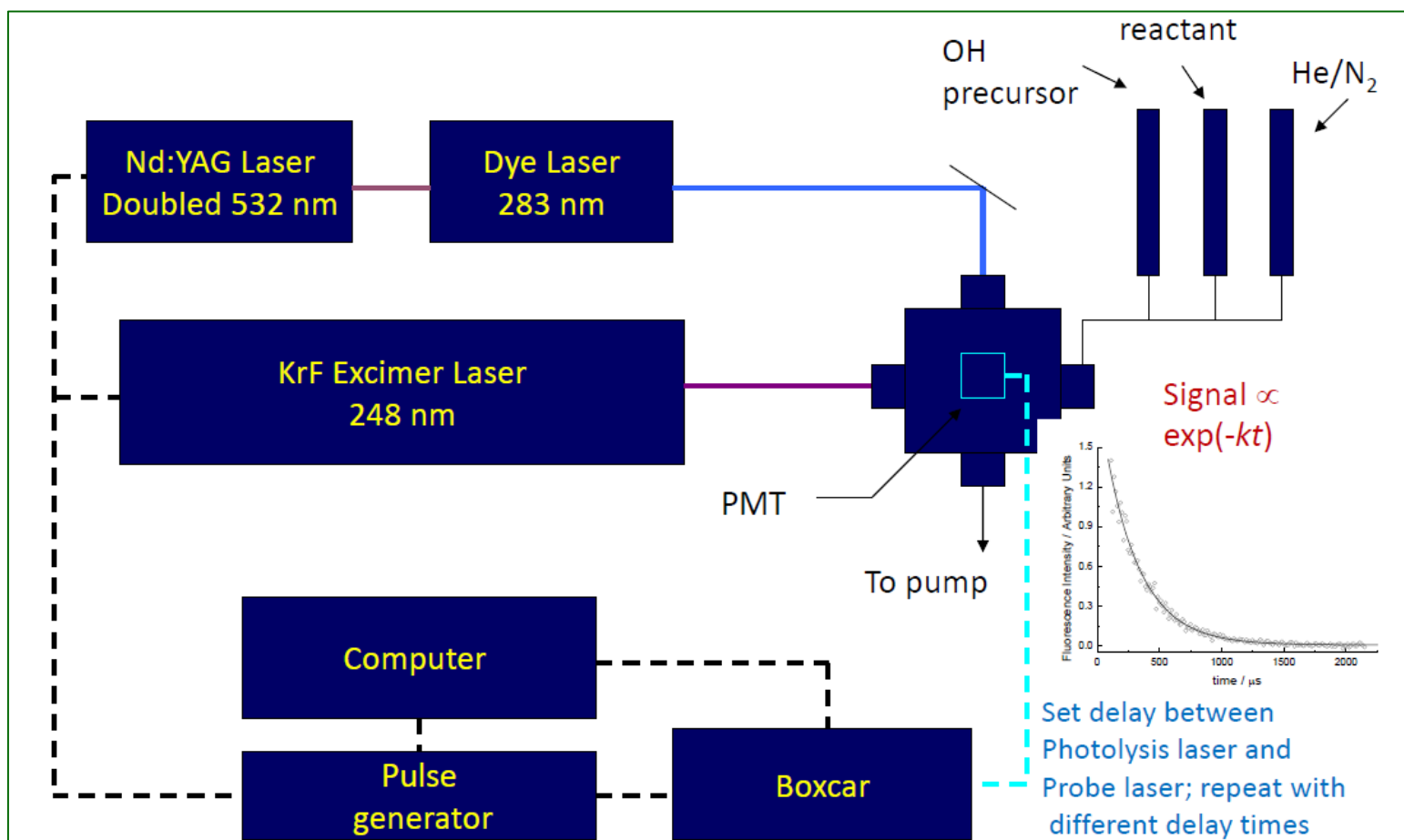
Measurement of Elementary Reactions

- Separate experiment for each elementary reaction step.
- The aim is to **determine the rate coefficient** of a single reaction step at a given temperature, pressure and bath gas.
- Experimental methods
 - Traditional flash photolysis, pulsed laser photolysis (PLP)
 - Microwave discharge, high temperature (shock tube)
- Diagnostic methods for products
 - Laser induced fluorescence (LIF)
 - Resonance fluorescence (RF)
 - Atom resonantly absorption spectroscopy (ARAS)
 - Laser absorption spectroscopy (LAS)
 - Electron-impact ionization mass spectrometry (EIMS)
 - Synchrotron VUV photoionization mass spectrometry (SVUV-PIMS)

Experimental Methods—Reactors

Measurement of Elementary Reactions

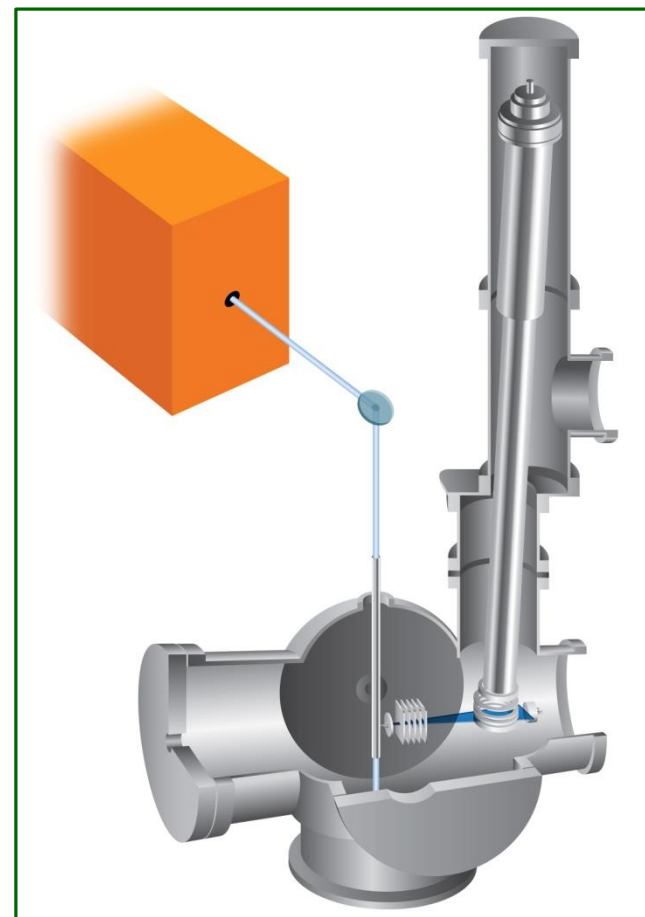
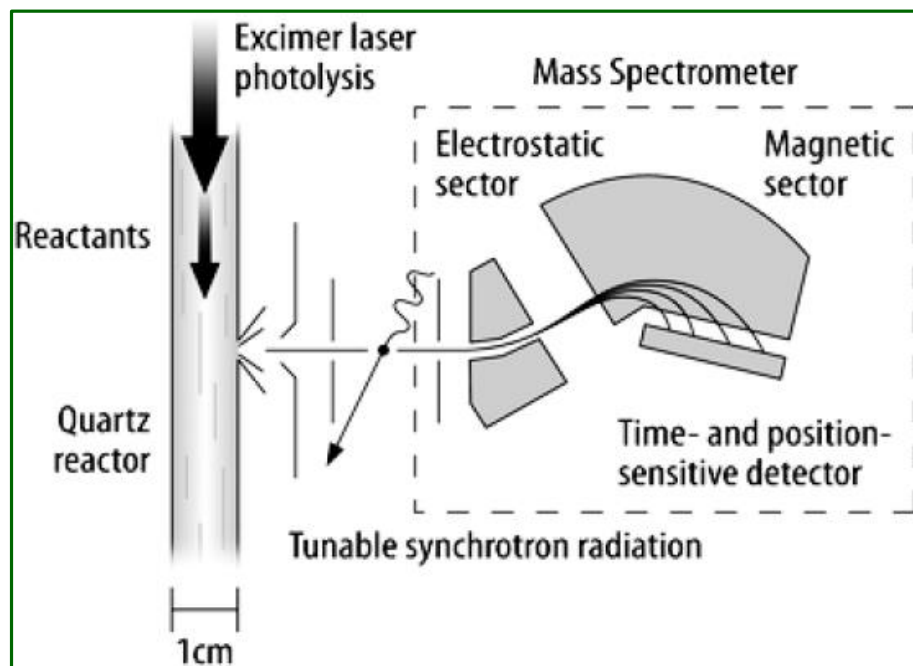
Laser flash photolysis (LFP) / laser induced fluorescence (LIF)



Experimental Methods—Reactors

Measurement of Elementary Reactions

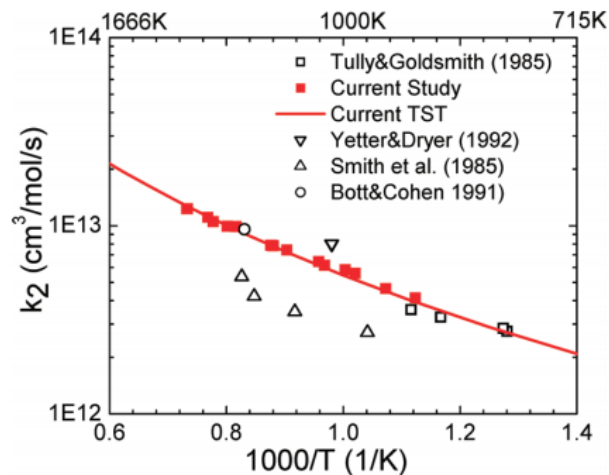
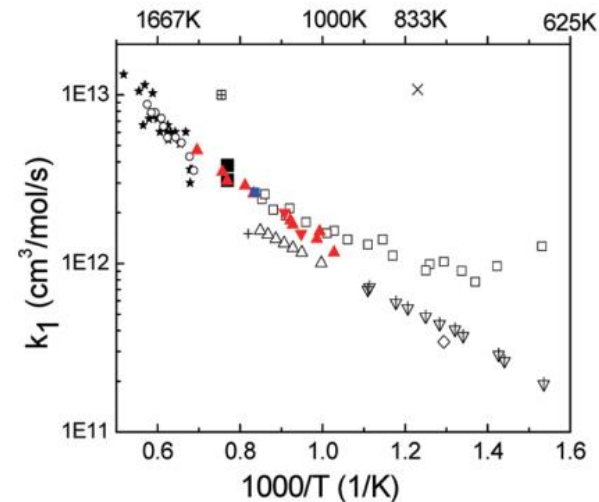
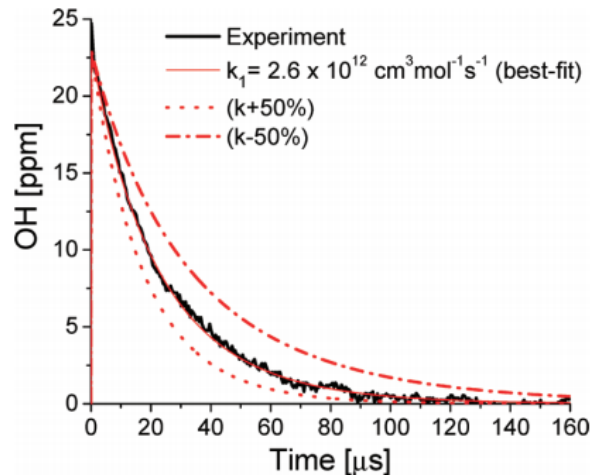
Laser flash photolysis (LFP) + SVUV-PIMS



Experimental Methods—Reactors

Measurement of Elementary Reactions

Shock tube + laser absorption



Experimental Methods—Reactors



Speciation and Global Parameters

- Can be interpreted only with the simulation of a detailed reaction mechanism
- Speciation for pyrolysis, oxidation and flames using laser spectroscopy, gas chromatography, and mass spectrometry
 - Shock tube/flow reactor
 - Jet stirred reactor
 - Laminar premixed flame
 - Coflow/counterflow diffusion flame
 - ...
- Global combustion parameters
 - Ignition delay times
 - Laminar flame speeds
 - Flame extinction measurements

Experimental Methods—Reactors



Speciation and Global Parameters

- ◆ Ideal reactors are designed to measure species concentration or global parameters:
 - Isobaric (等压的)
 - Isothermal or adiabatic (绝热的)
 - Steady state (稳态)
 - Simple flow system
 - Well mixed or plug flow conditions



Experimental Methods—Reactors

Plug Flow Reactor



Velocity
profile

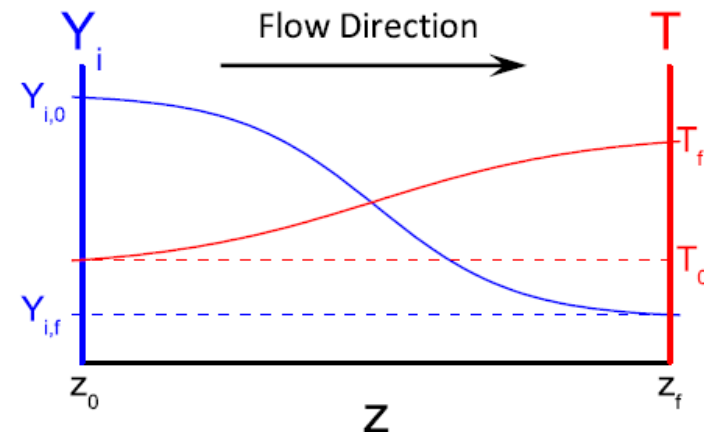
Concentration
profile

Temperature
profile

$$\begin{aligned}
 T &= T(x) \\
 [X_i] &= [X_i](x) \\
 P &= P(x) \\
 V &= u(x)
 \end{aligned}$$

Ideality:

- ◆ Steady flow
- ◆ No mixing in the axial direction, molecular and/or turbulent mass diffusion is negligible in the flow direction, $D_a = \infty$
- ◆ No gradient in the radial direction, uniform properties in the direction perpendicular to the flow (one dimensional)
- ◆ No wall reactions



Governing Equations:

$$\rho U_{pl} \frac{dY_i}{dz} + \dot{w} = 0$$

Species

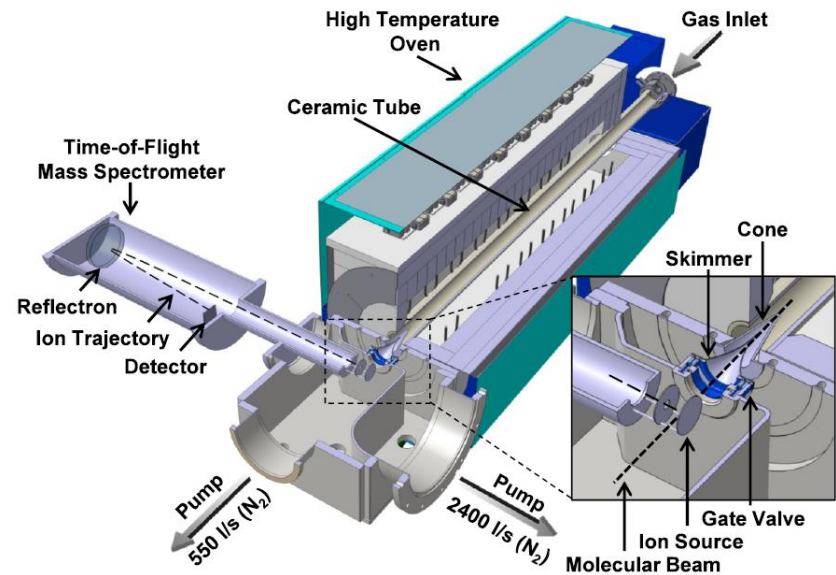
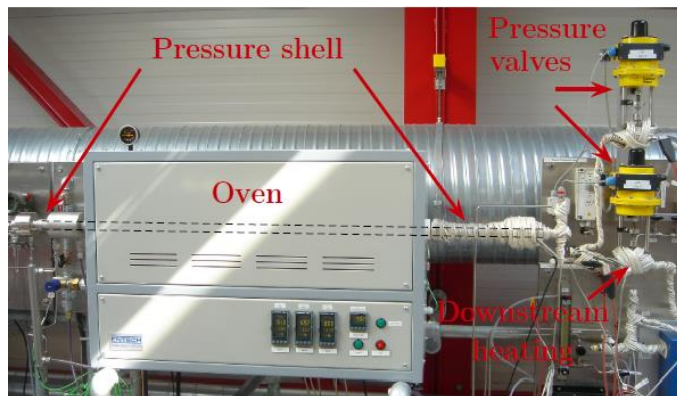
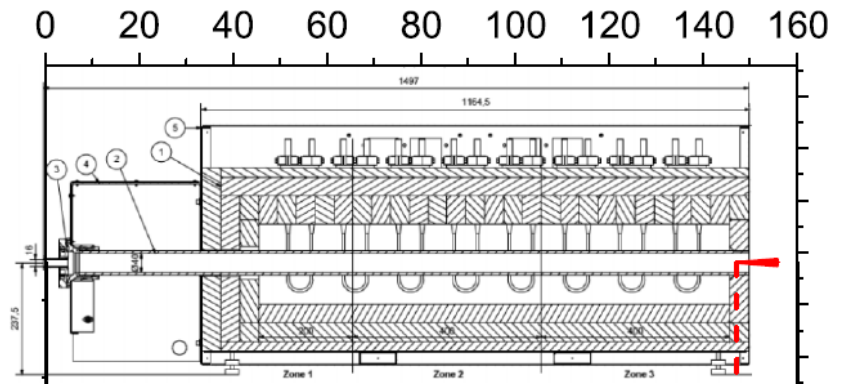
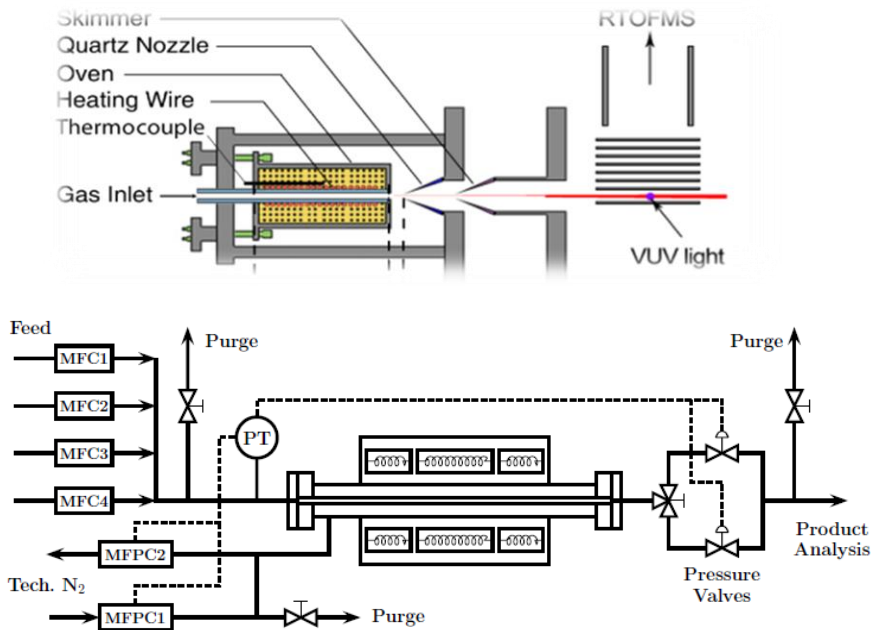
$$\rho U_{pl} c_p \frac{dT}{dz} + \dot{w} Q = 0$$

Energy

Boundary Conditions: $Y_i(z_0) = Y_{i,0}$ $T(z_0) = T_0$

Experimental Methods—Reactors

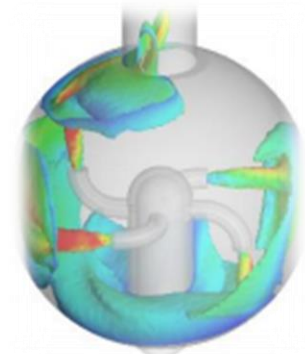
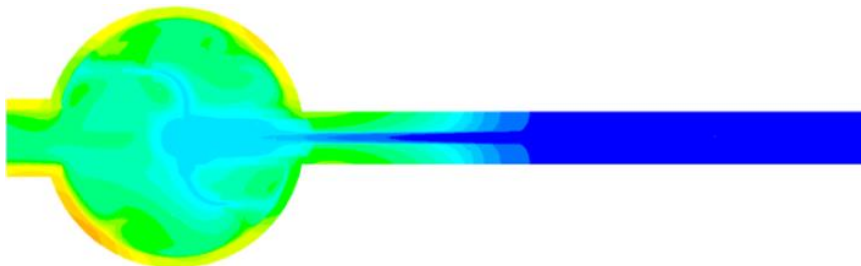
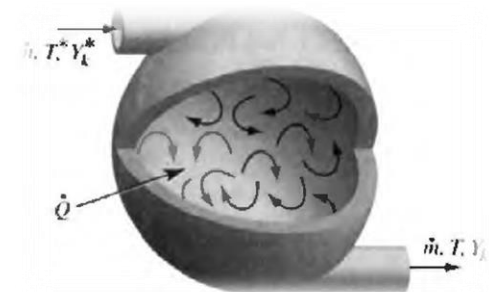
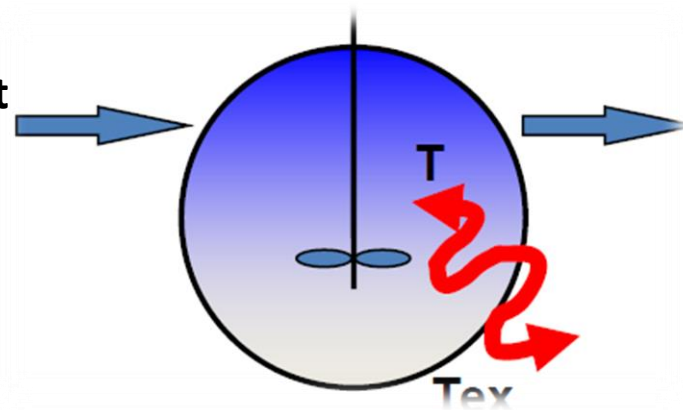
Plug Flow Reactor



Experimental Methods—Reactors

Perfectly Stirred Reactor

- ◆ The composition in the reactor is assumed to be that of the effluent stream and therefore all the reaction occurs at this constant composition
- ◆ Rapid mixing and well defined/characterized residence time
- ◆ Uniform temperature and species distribution
- ◆ Minimum fuel loading and heat release from oxidation
- ◆ Catalytic wall effects are negligible



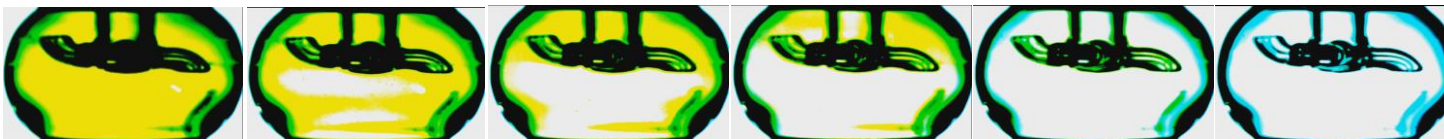
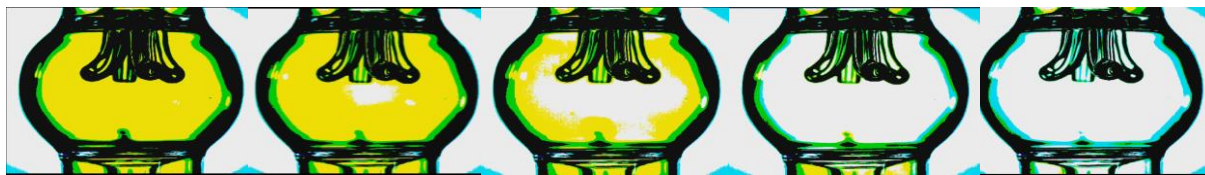
Experimental Methods—Reactors

Perfectly Stirred Reactor

Features:

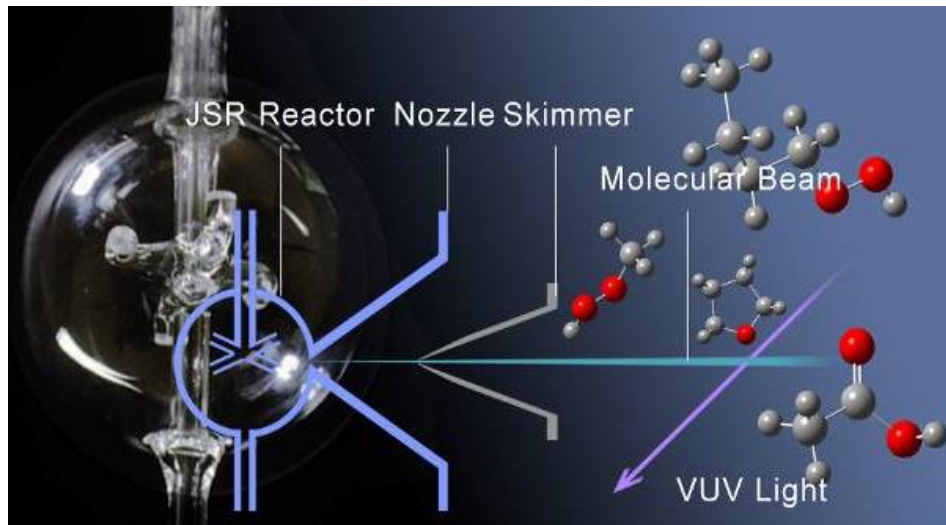
- ◆ Four nozzles to create free jets, lead to turbulent mixing
- ◆ Velocity of the jets at the outlet of the nozzles must not exceed the speed of sound
- ◆ Jets provide a good mixing of the whole gas phase in the reactor, provide very intense internal recycle streams

Many types of design

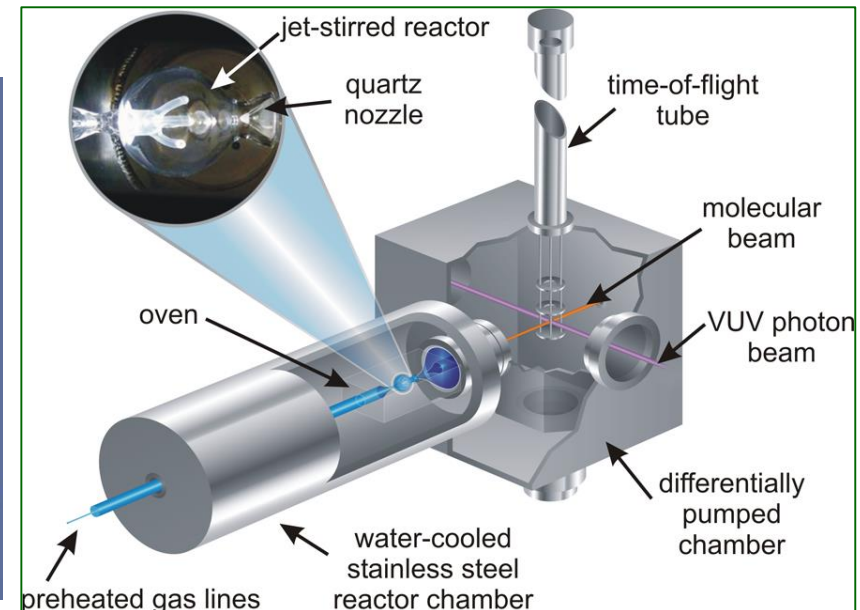


Experimental Methods—Reactors

Perfectly Stirred Reactor NSRL



ALS

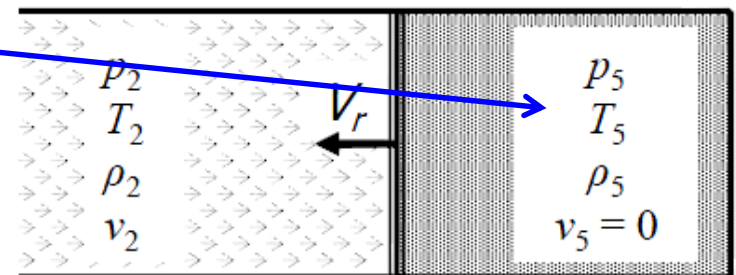
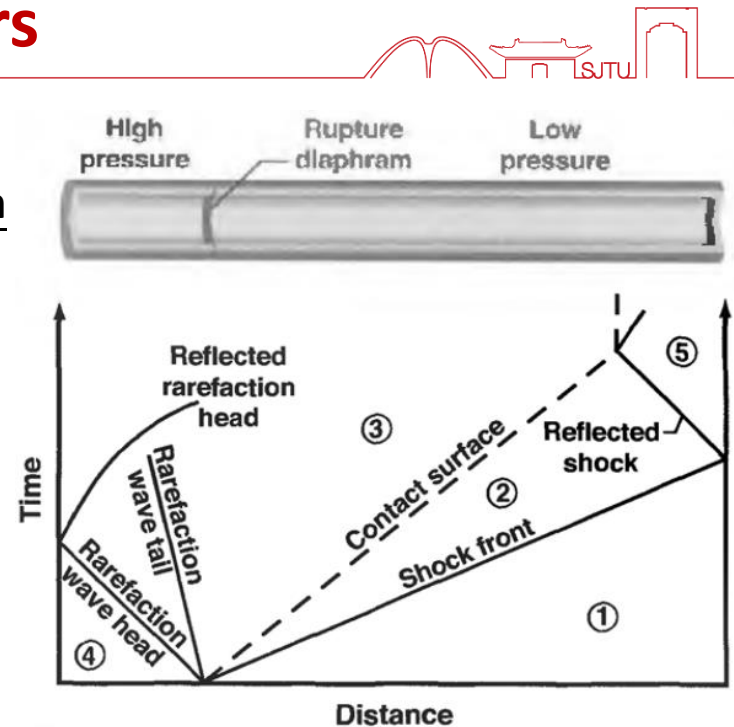
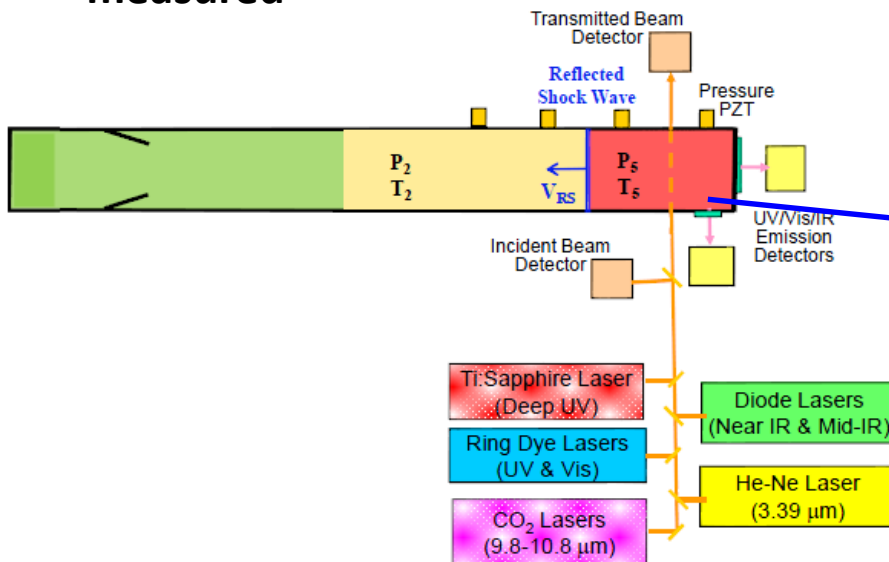


- ◆ The residence time of the gas in the reactor is actually a mean residence time
- ◆ The gas is mixed by the gas jets + slow reaction: spatially homogeneous mixture, process is controlled by kinetics
- ◆ The composition of the outlet gas is analyzed using GC-MS, FTIR, SVUV-PIMS and optical spectroscopy methods

Experimental Methods—Reactors

Shock Tube

- ◆ Close to ideal isobaric or isometric system can be treated as batch reactor
- ◆ Reaction is very fast, usually be treated as constant volume batch reactor
- ◆ Homogeneous system
- ◆ Initial temperature and pressure can be measured



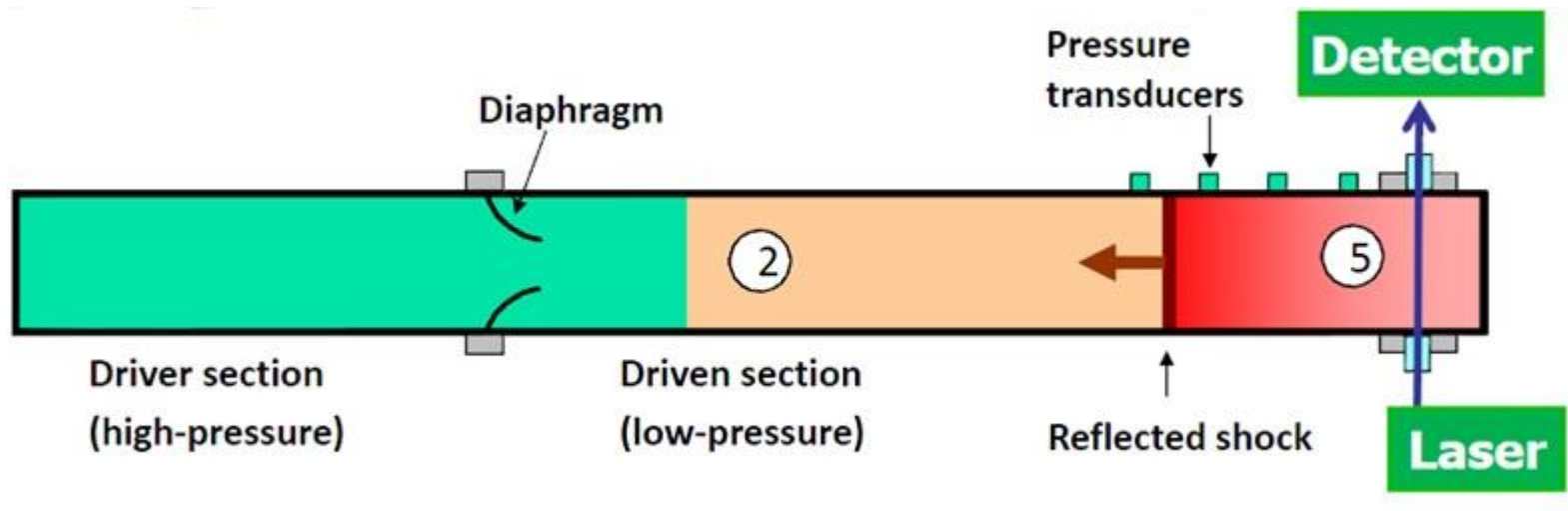
Reflected shock

Experimental Methods—Reactors



Shock Tube

Working principles of shock tube



- ◆ Open the diaphragm, the shock wave is formed driven by high-pressure gas
- ◆ Driven gas is quickly compressed, and heated by forward and reflected shock waves
- ◆ Diagnosed by lasers and pressure transducers
- ◆ Initial temperature and pressure can be measured accurately

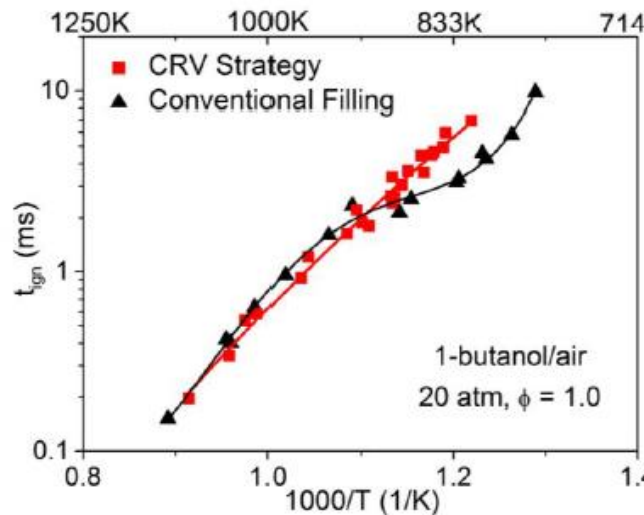
Experimental Methods—Reactors



Shock Tube

Three Types of Shock Tube Experiments

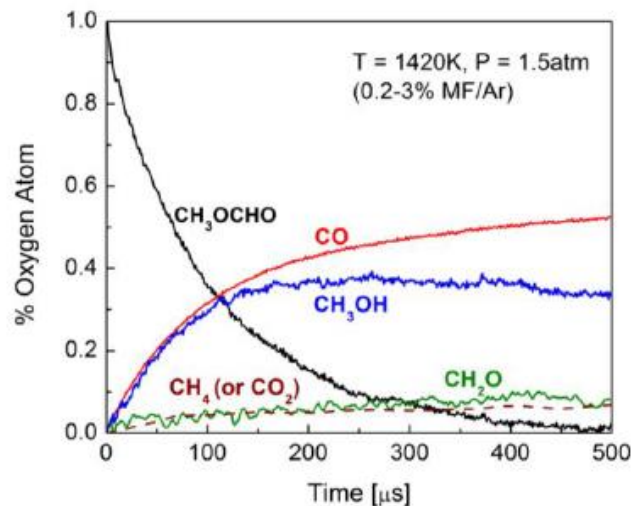
Ignition Delay Times



New CRV strategy

- ◆ Provides well-defined constant P data

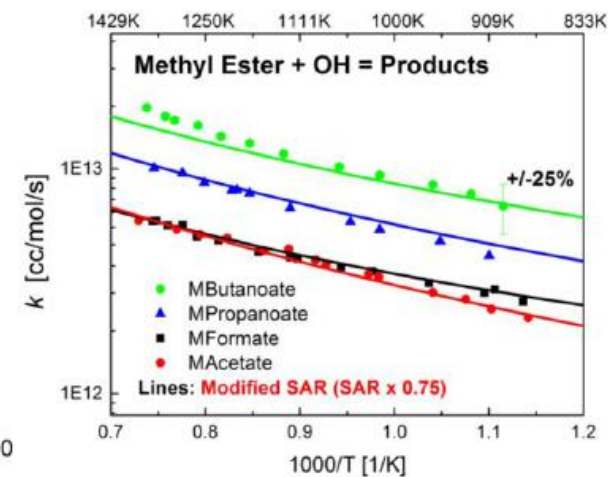
Speciation



Multi-wavelength strategy

- ◆ Provides unique multi-species database for model refinement

Rate Constants

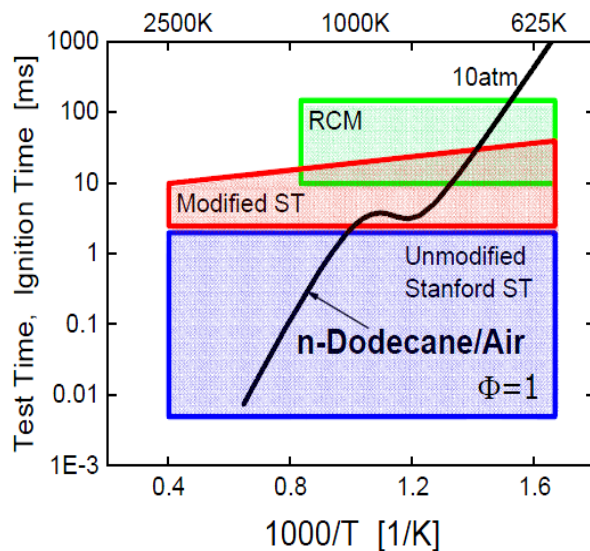
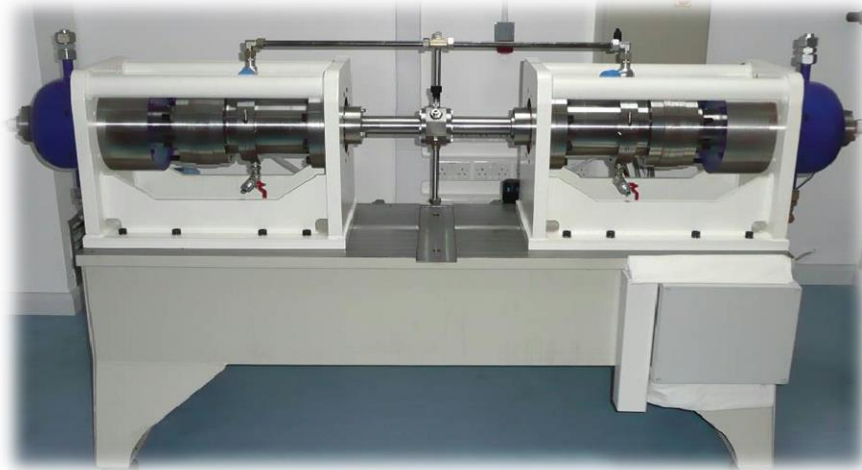


Laser absorption

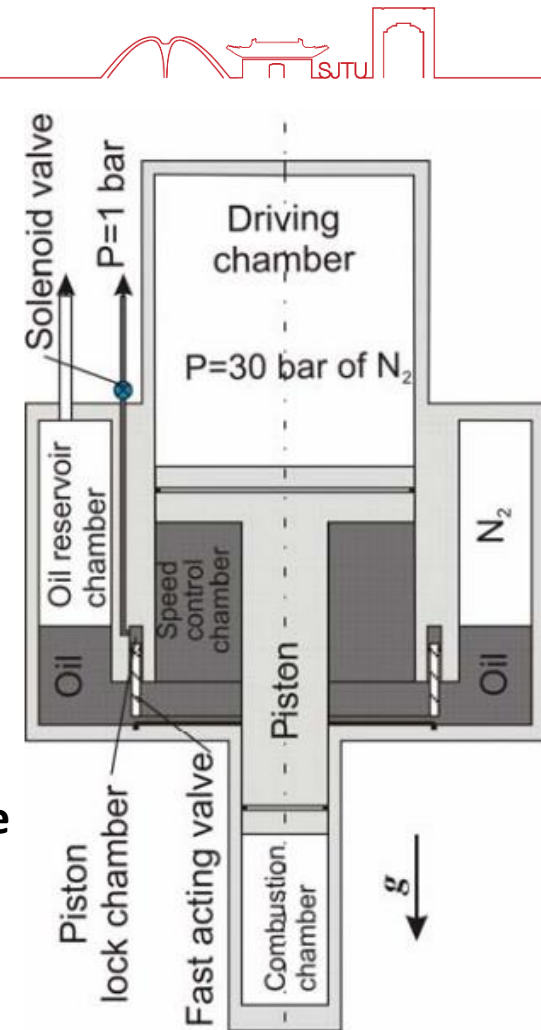
- ◆ Measurements of elementary reactions

Experimental Methods—Reactors

Rapid Compression Machine



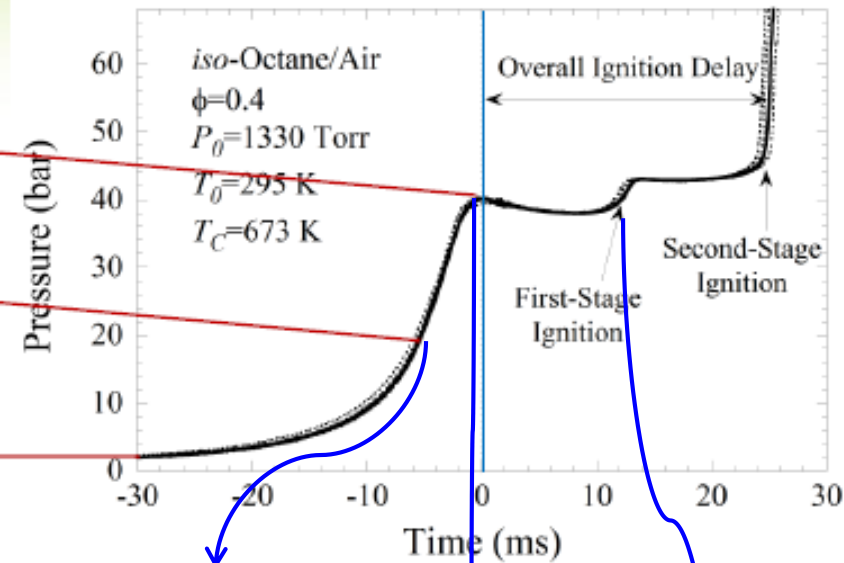
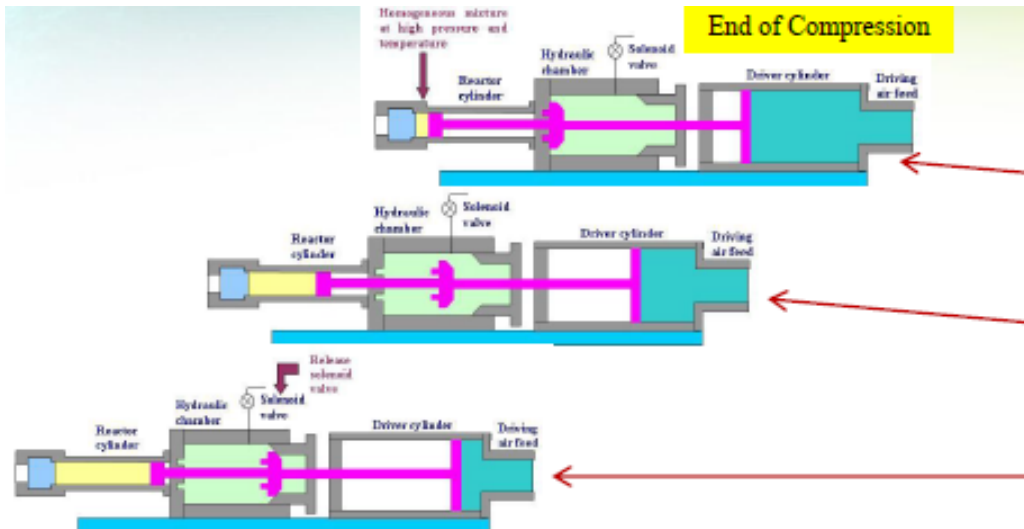
- ◆ CI engine-like condition
- ◆ Measure low temperature ignition delay times or species concentration



Experimental Methods—Reactors

Rapid Compression Machine

Working principles of RCM



- ◆ Pressure is measured directly to deduce ignition delays
- ◆ Temperature is measured via TDLAS

Compression

End of
compression

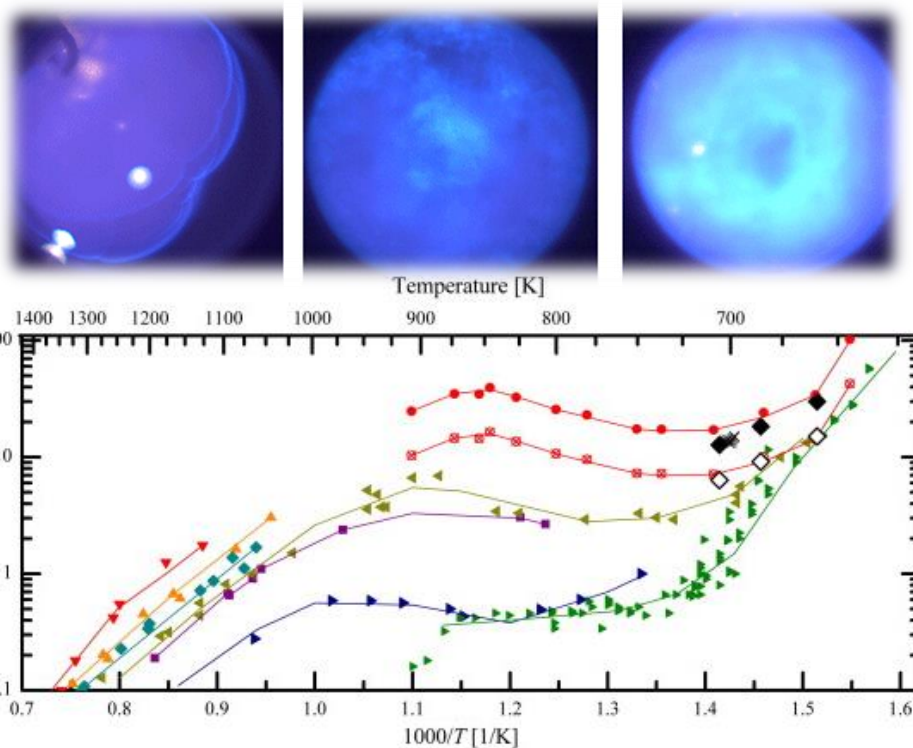
Pressure increase caused by
chemical reactions

Experimental Methods—Reactors

Rapid Compression Machine

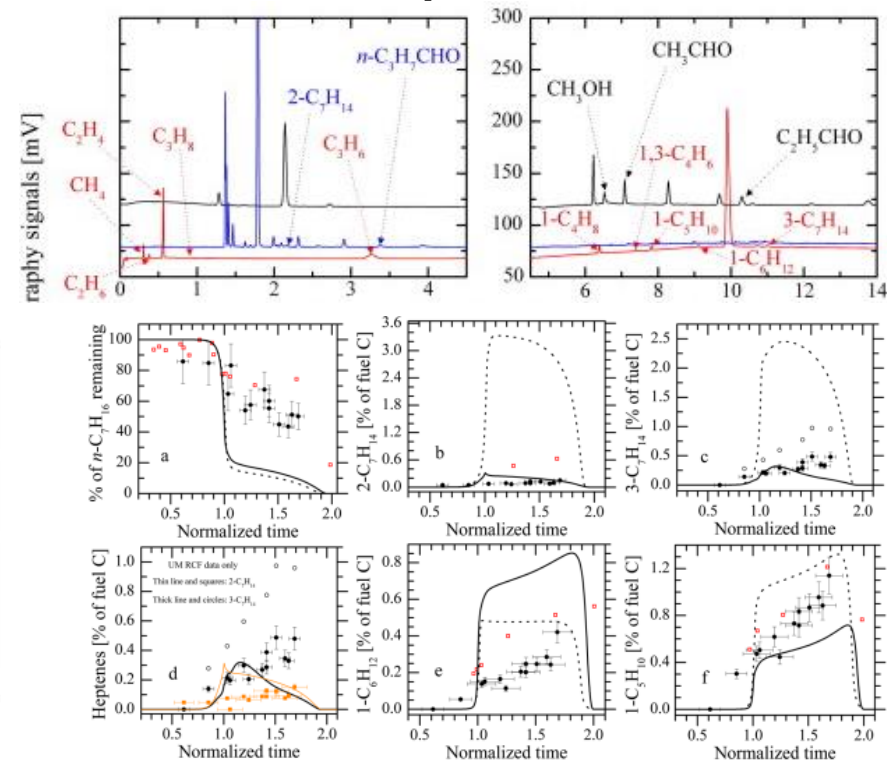
Two Types of RCM Experiments

Ignition Delay Times



- ◆ Pressure transducers to measure pressure

Speciation



- ◆ High-speed gas sampling and gas chromatography

Experimental Methods—Reactors



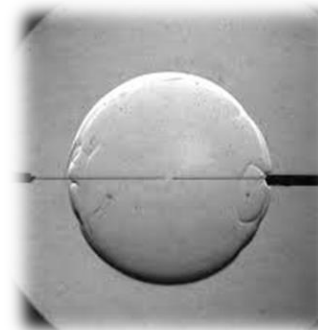
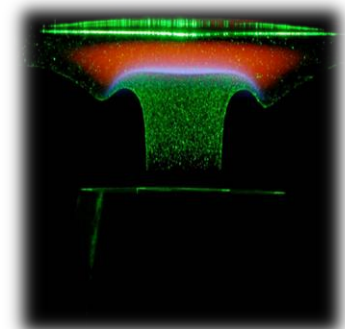
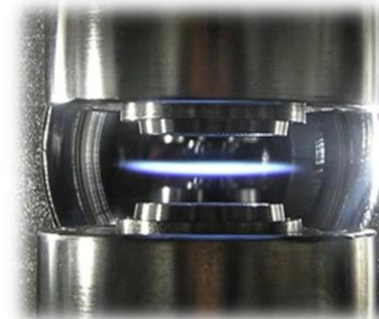
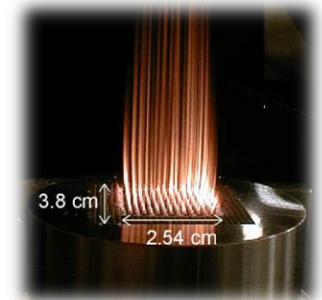
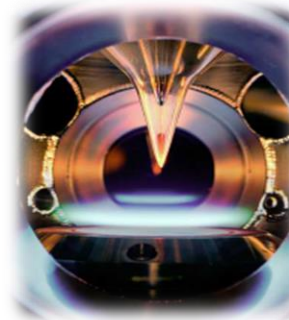
Speciation and Global Parameters

1D flames =>

- ◆ Premixed flat flames (burner stabilized or freely propagating flames)
- ◆ Counterflow (premixed/partially premixed/diffusive) flames
- ◆ Isolated fuel droplet in microgravity

2D laminar flames =>

- ◆ Co-flow flame
- ◆ Conical or 2D Bunsen burner, stagnation flames
- ◆ Require CFD simulations



Experimental Methods—Reactors

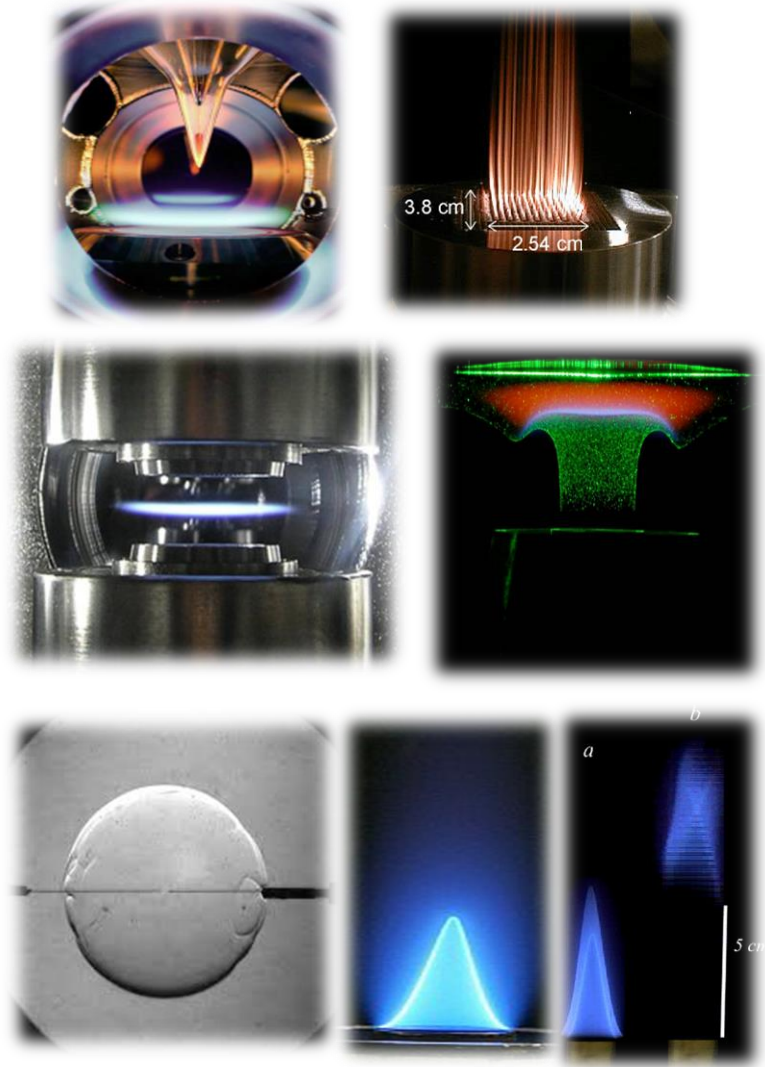


1D flames =>

- ◆ Steady, clearly define boundary conditions
- ◆ The flames that can be modeled using a 1D model are extremely useful for studying the kinetics of combustion at high temperature and in presence of diffusion of heat and mass.

Investigation

- ◆ Composition profiles (speciation)
- ◆ Flame speed measurements
- ◆ Ignition experiments
- ◆ Extinction experiments
- ◆ Dynamic response of flames to forced oscillations

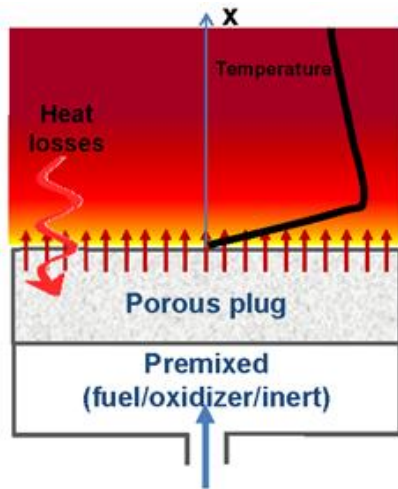


Experimental Methods—Reactors

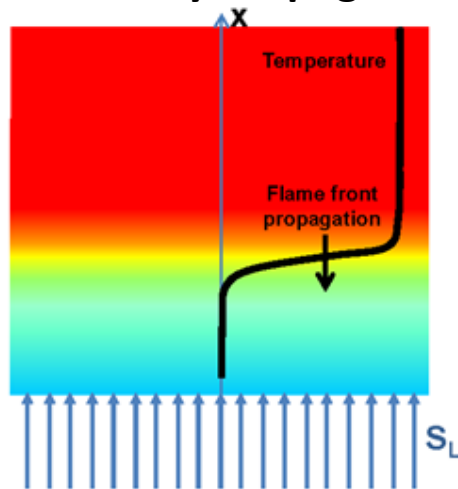


Laminar Premixed Flames

1. Burner Stabilized



2. Freely Propagation



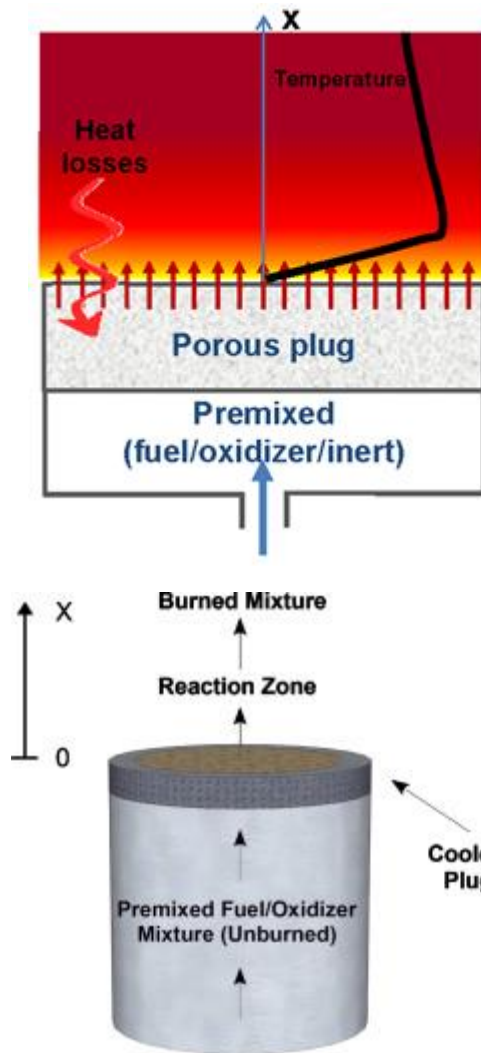
❏ The propagation speed of the premixed flame with respect to the unburned gases is called the burning velocity or laminar flame speed S_L

- ◆ The conservation equations are the same, but the boundary conditions are different
- ◆ For burner-stabilized flames the cold flow velocity (or mass flow rate) and composition are known, and vanishing gradients are imposed at the hot boundary
- ◆ For freely propagating flames, cold flow velocity is an eigenvalue and must be determined as part of the solution

Experimental Methods—Reactors



Laminar Premixed Flames

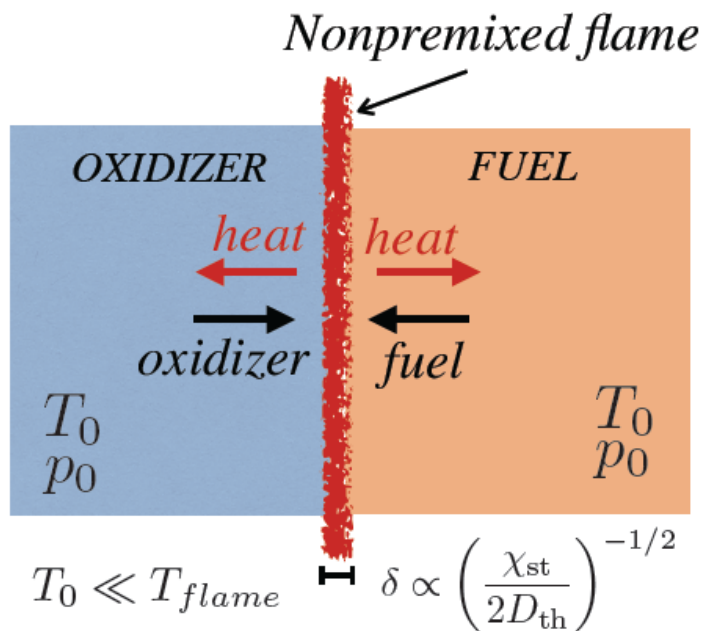


Burner-stabilized Flame

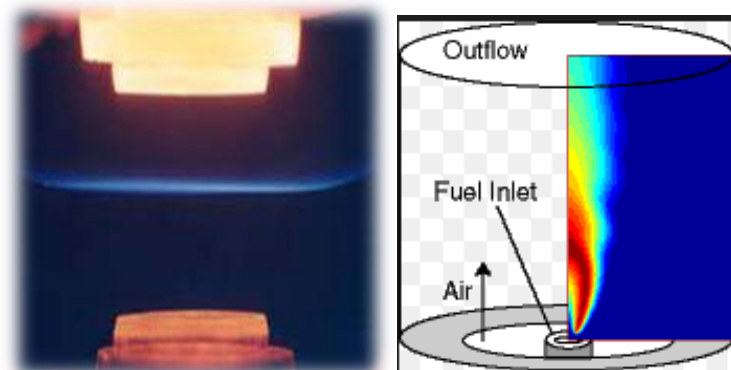
- ◆ The Burner-stabilized Flame Model is often used for analyzing species profiles in flame experiments.
- ◆ The mass flow rate through the burner is known and it is slower than the flame speed of the mixture. Therefore, a flame is stabilized over the burner surface, which is heated by the flame.
- ◆ Because of the significant heat losses (difficult to estimate) in the premixed burner-stabilized flames, the measured temperature profile is usually used as input in the numerical simulations.
- ◆ Comparing the results of simulations using assigned and calculated temperatures, can provide some indication of the heat losses to the burner and to the environment (conductive and radiative).

Experimental Methods—Reactors

Laminar Diffusion Flames



- ◆ Fuel and oxidizer are not mixed prior to reacting.
- ◆ Mixing must bring reactants into the reaction zone fast enough for combustion to proceed.
- ◆ Two boundaries must be considered: oxidizer on the left and fuel on the right.
- ◆ Temperature is maximum in the reaction zone, which is confined to a region where the mixture fraction is stoichiometric.
- ◆ The thickness of the flame is controlled by the gradient of the mixing layer.



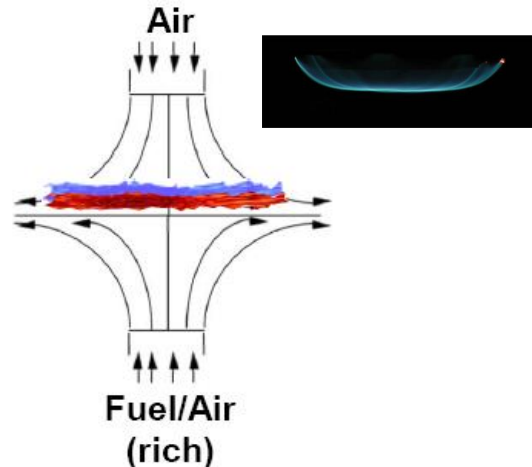
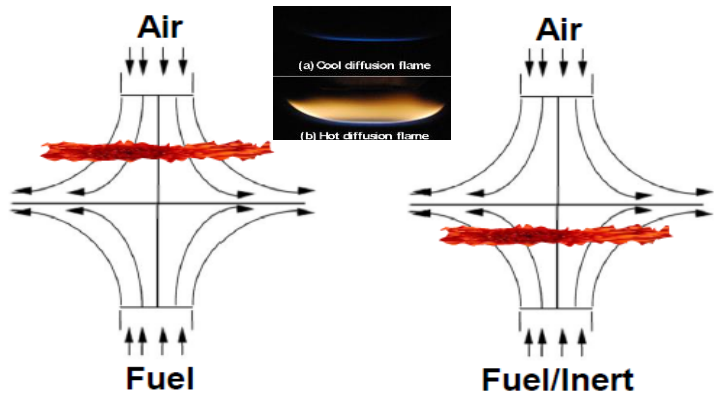
Experimental Methods—Reactors

Laminar Diffusion Flames

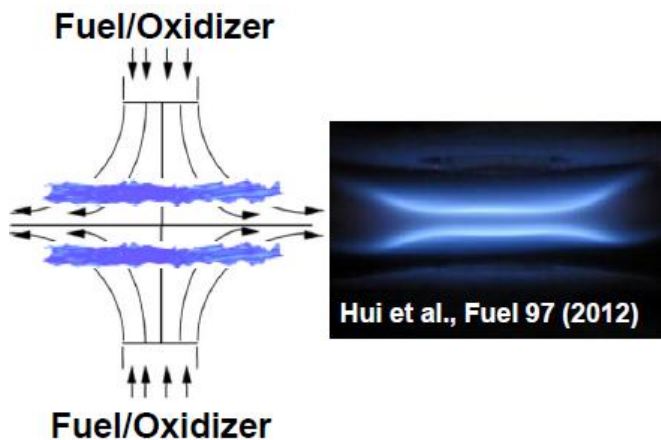
Partially premixed flame

Partially premixed flame

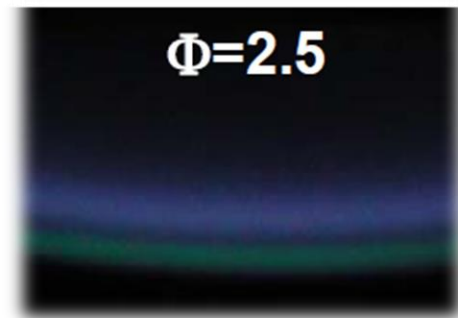
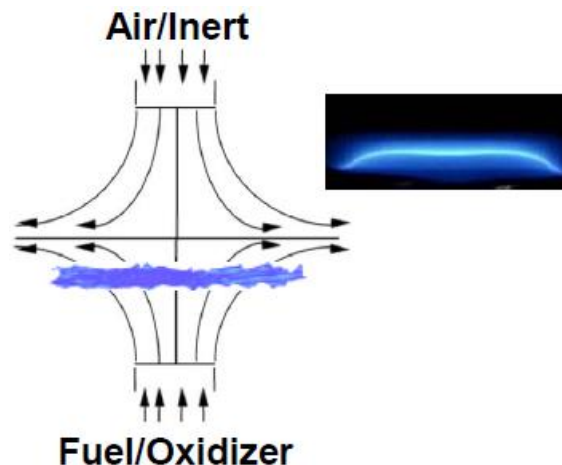
Non premixed flame



Twin premixed flame



Single premixed flame

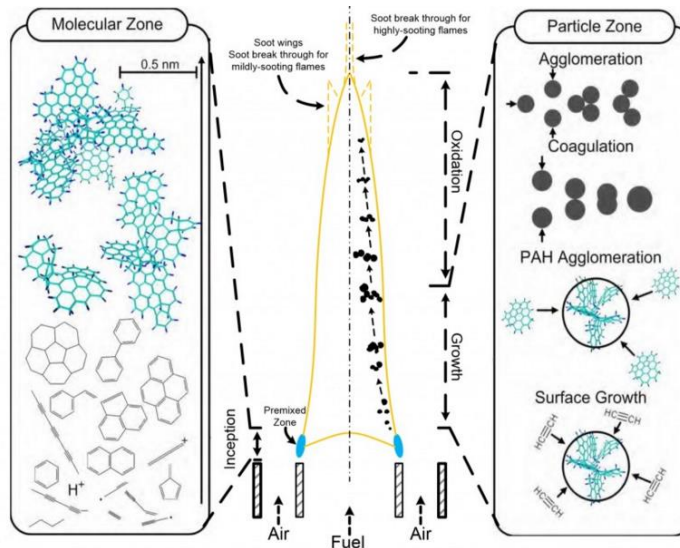
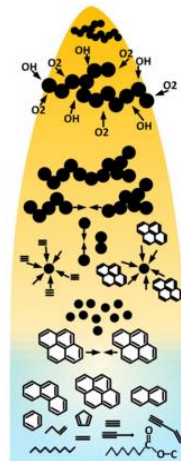
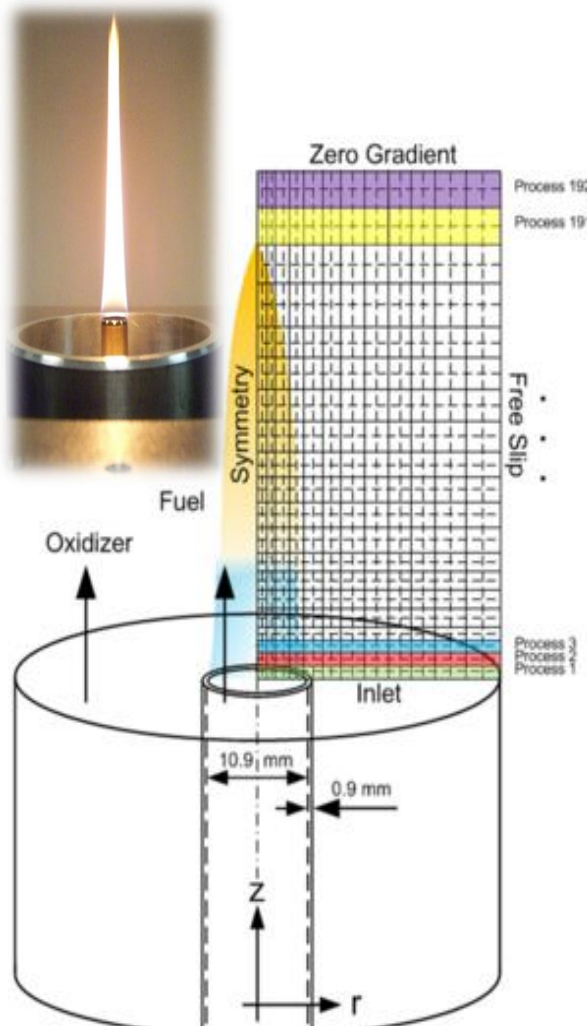


Green → C2
chemiluminescence in
premixed zone

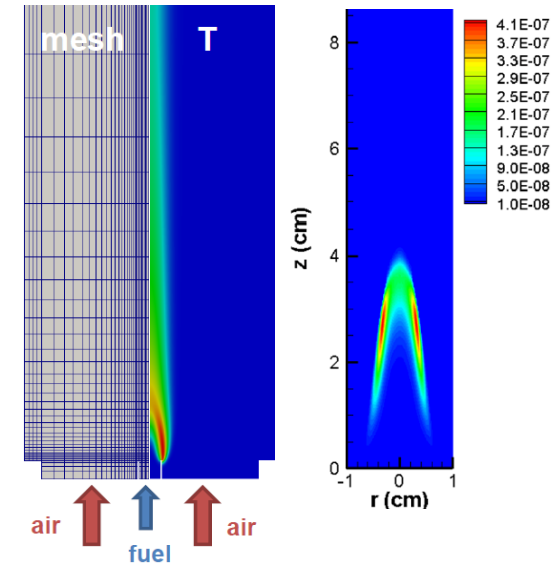
Blue → CO oxidation in the
non-premixed zone

Experimental Methods—Reactors

Laminar Diffusion Flames



Co-flow flame

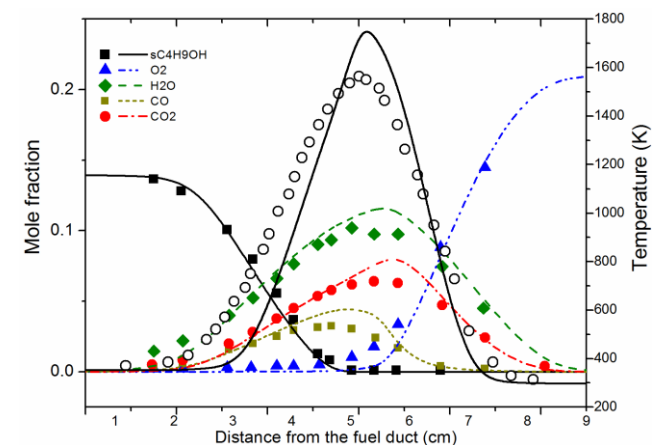
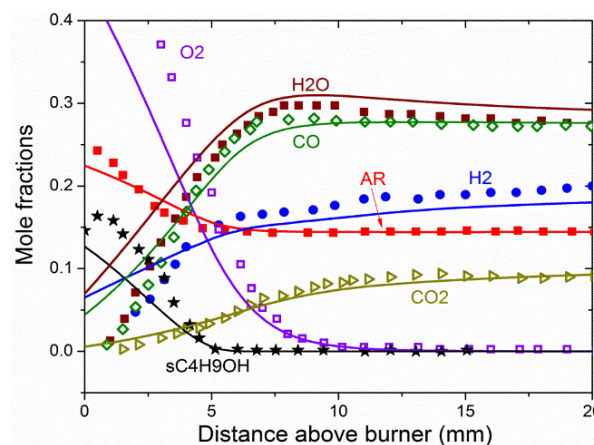
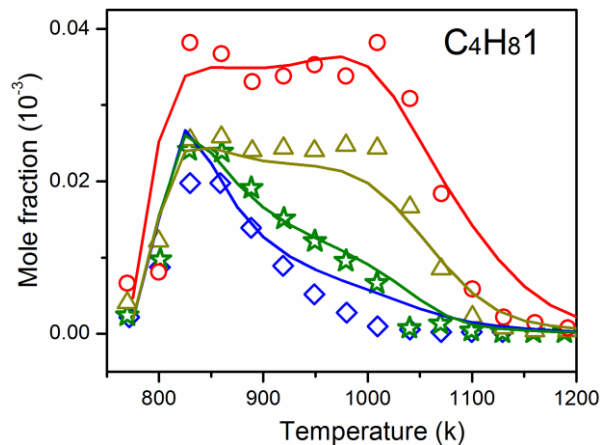
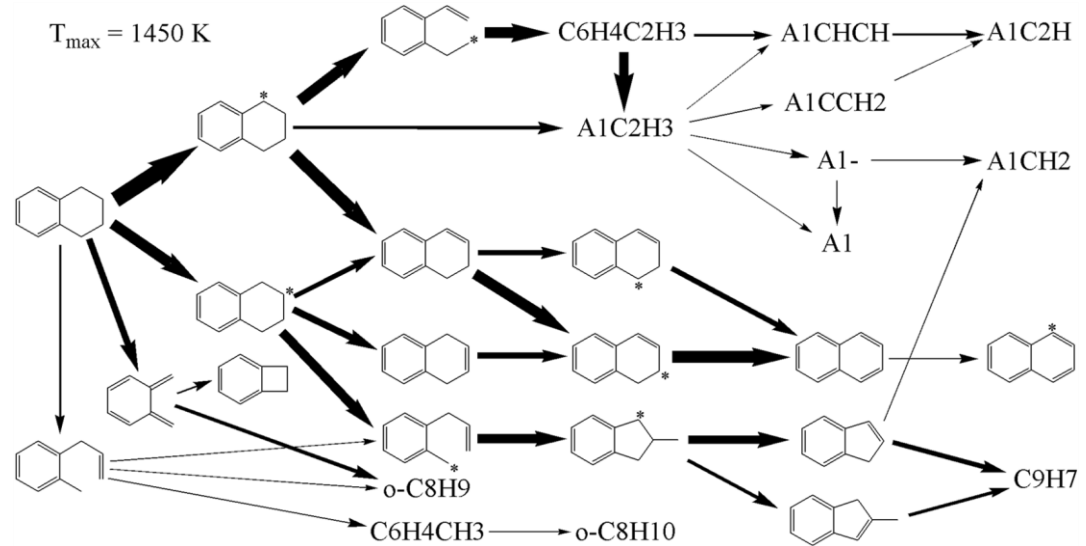


- ◆ 2D diffusion flame
- ◆ This type of flame is usually used for soot and PAH study

Experimental Methods—Reactors

Speciation and Global Parameters

- Speciation provides key kinetic information for the combustion reaction network
- The measurement of species concentration is crucial for kinetic model development

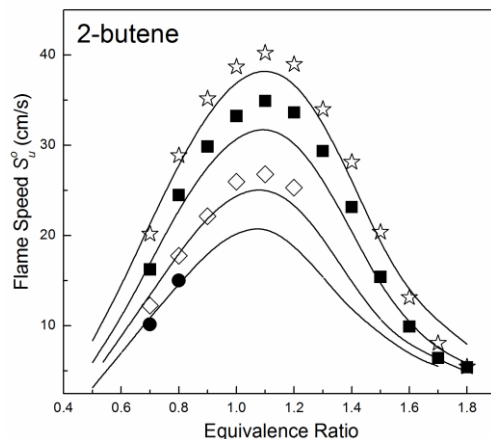


Experimental Methods—Reactors

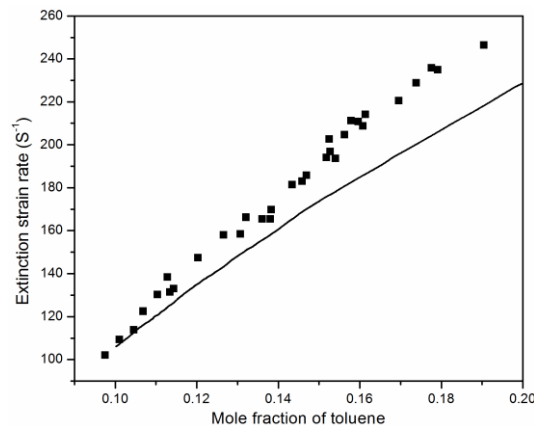
Speciation and Global Parameters

- Global parameters have specific engineering meanings.

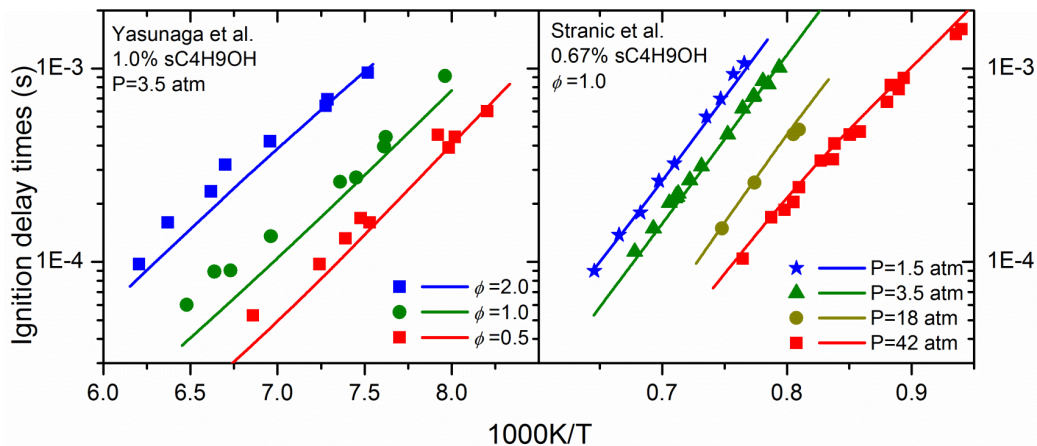
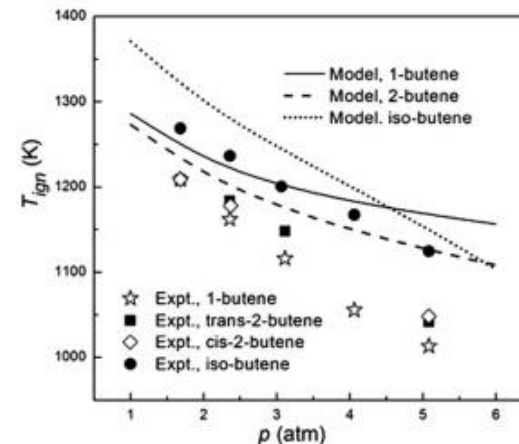
Laminar flame speeds



Extinction strain rate



Ignition temperature

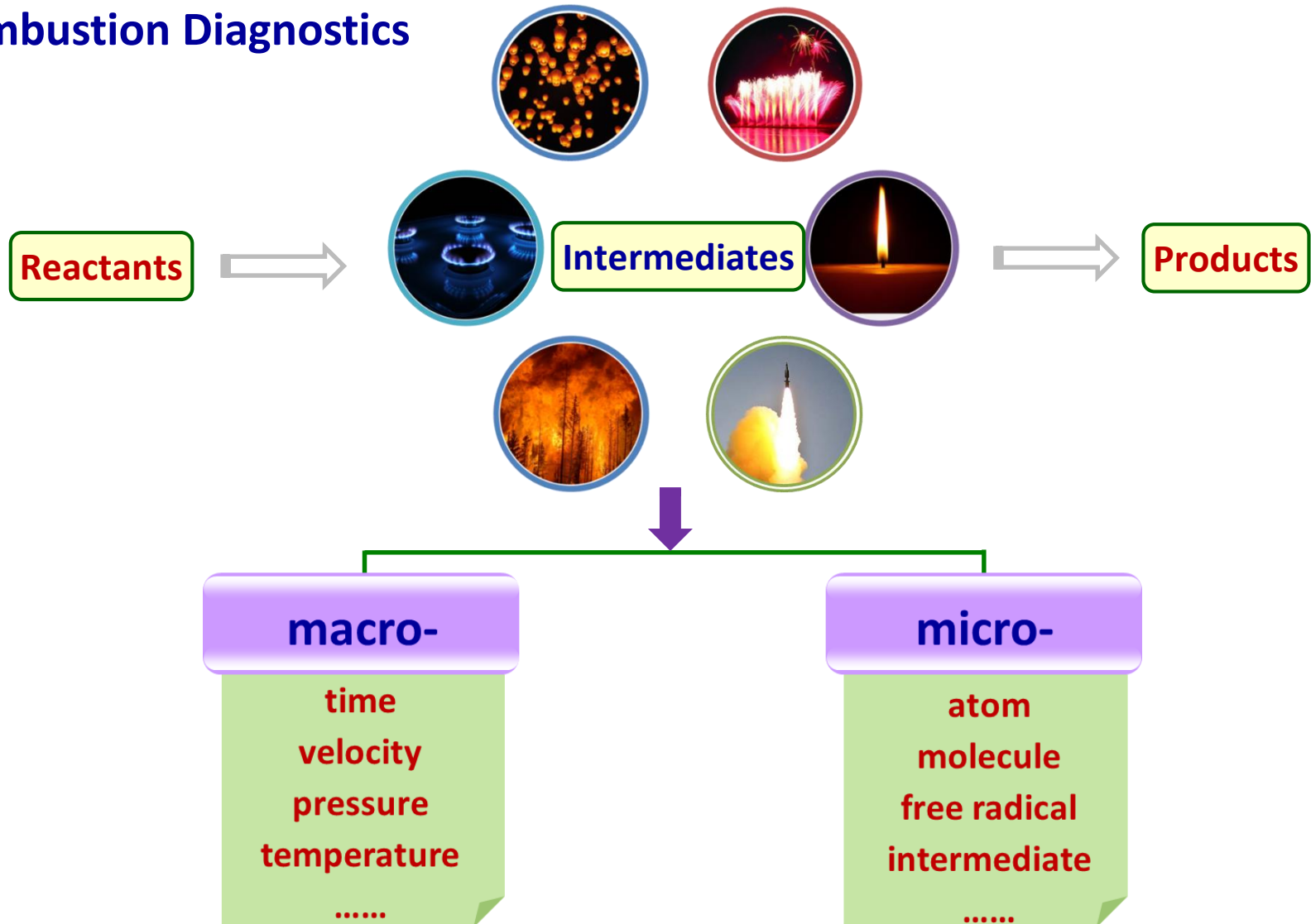


Ignition delay times

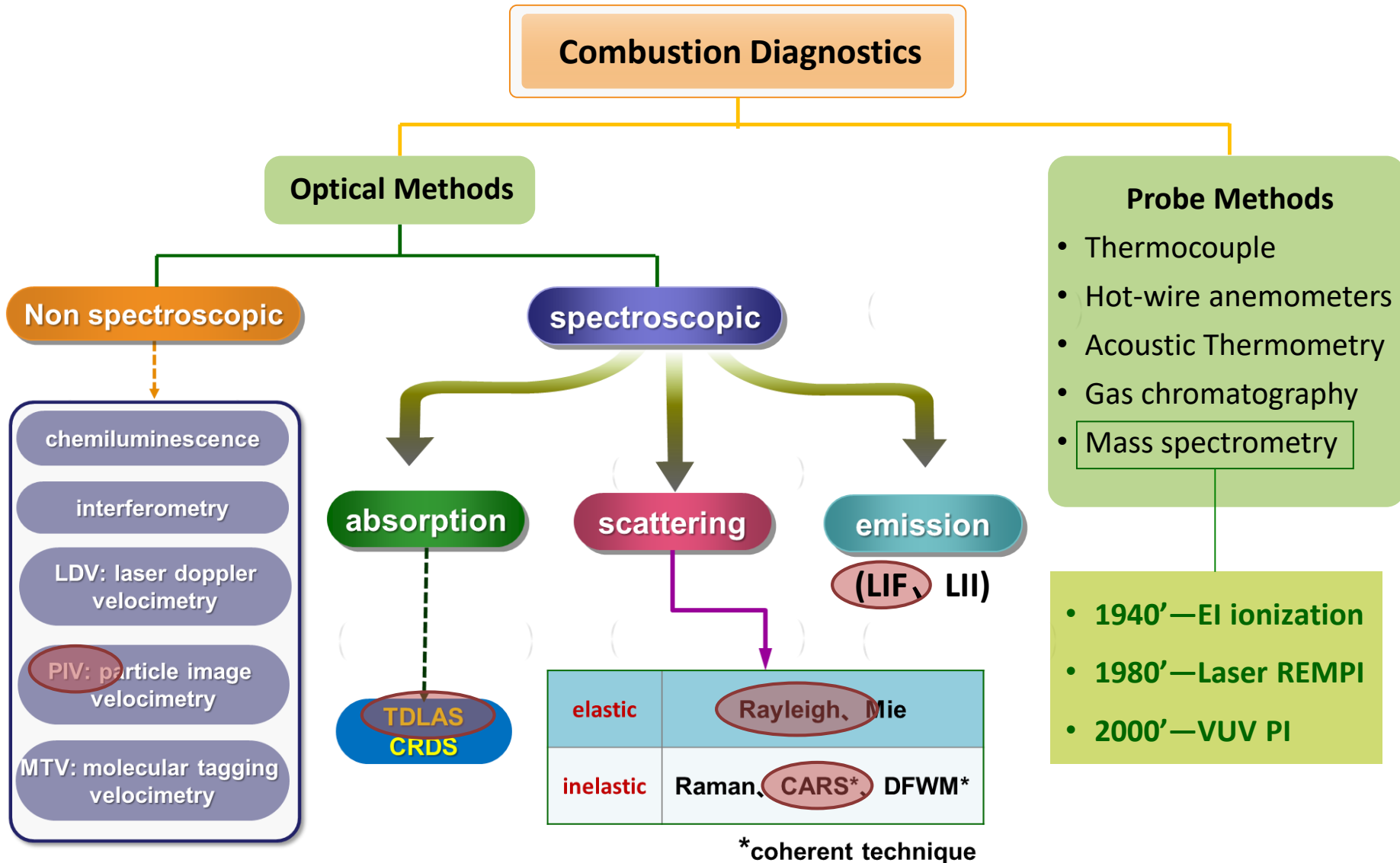
Experimental Methods—Diagnostics



Combustion Diagnostics



Experimental Methods—Diagnostics



Optical Techniques



Advantages

- Non-intrusive with high temporal/spatial resolution (laser techniques)
- Can measure simultaneously in many points
- Simple (not laser techniques!)

Disadvantages

- Mostly expensive and complicated equipment, theory (laser techniques)
- May have difficulties to give quantitative data

Laser in Combustion Diagnostics

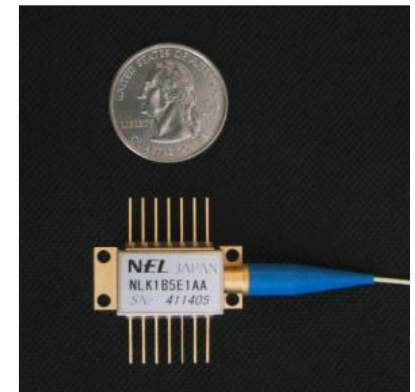
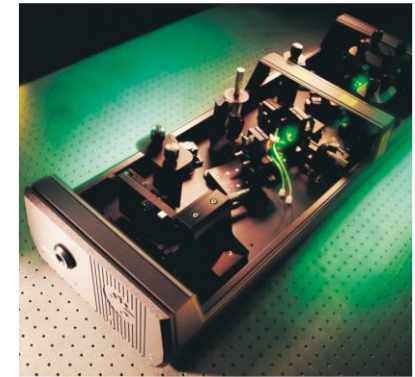


Advantages

- Non-intrusive
- High spatial resolution ($<0.001 \text{ mm}^3$)
- High temporal resolution ($<10 \text{ ns}$)
- High spectral resolution ($\sim \text{MHz}$)
- Multiplex (multi-species, multi-point)
- Can measure non-thermal equilibrium

Disadvantages

- Complicated
- Expensive
- Eye safety
- Optical access required
- Intrusive?
 - Laser-induced breakdown (LIBS)
 - Creation of molecular fragments, atoms
 - Optical pumping



Probe Techniques



Possible disturbances during probe (extraction) measurements

- Aerodynamic effects
- Effects of concentration gradients
- Thermal effects
- Catalytic effects
- Quenching effects



Probe Techniques



Advantages

- Relatively cheap (some)
- Robust and easy to use (some)
- Do mostly give quantitative results (correct?)
- Can measure atoms and radicals (some)
- Can measure larger molecules

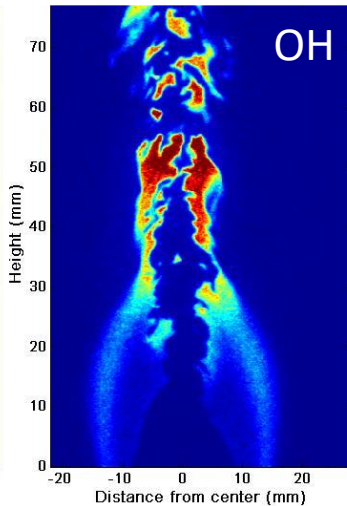
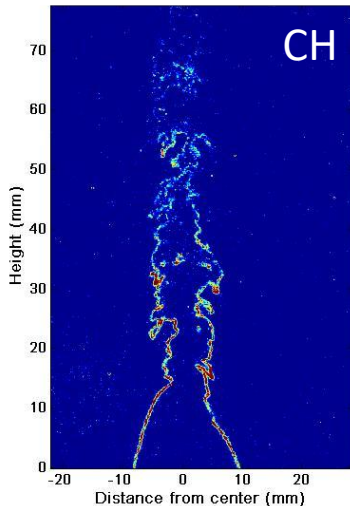
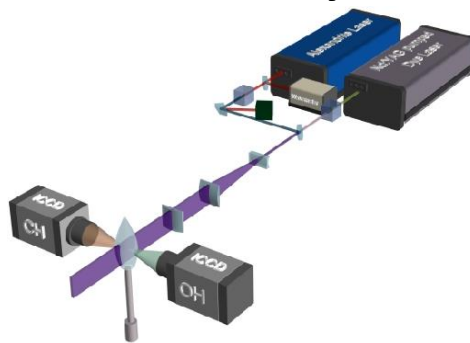
Disadvantages

- Intrusive
- May be difficult to estimate measuring errors
- Often low temporal and spatial resolution

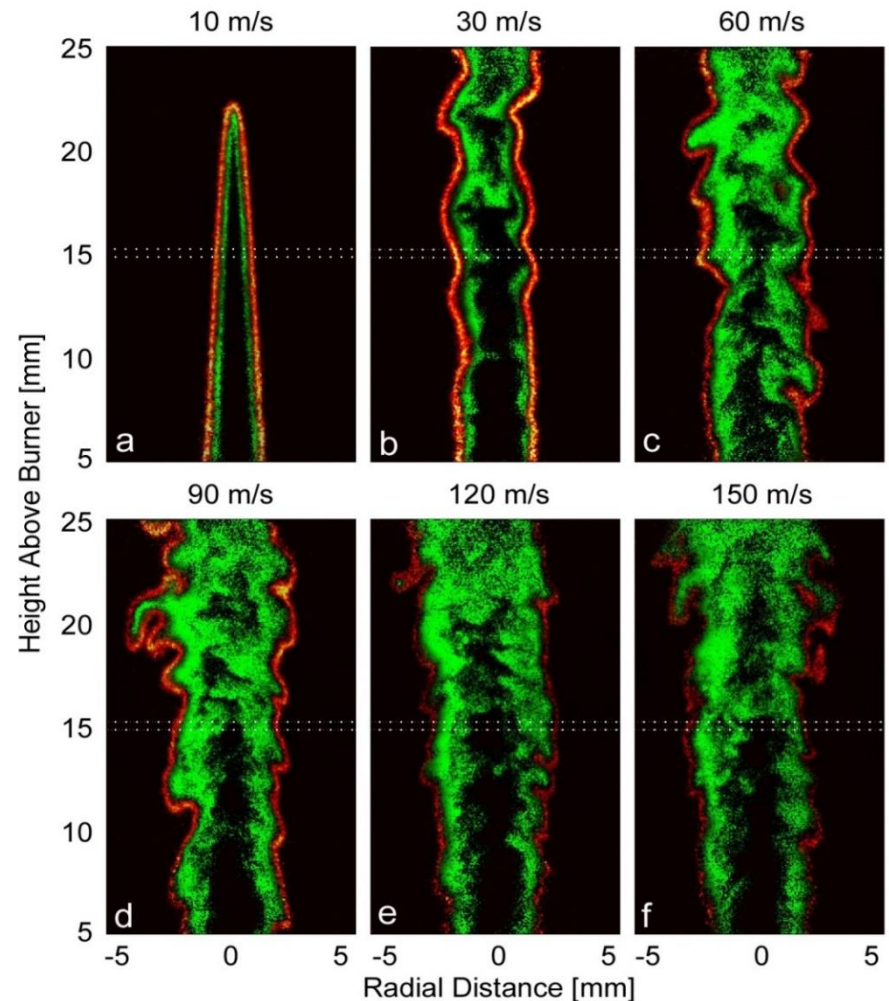
Planar Laser Induced Fluorescence (PLF)—2D

Local flame structures and statistics turbulence premixed jet flame

Lund University



CH/CH₂O

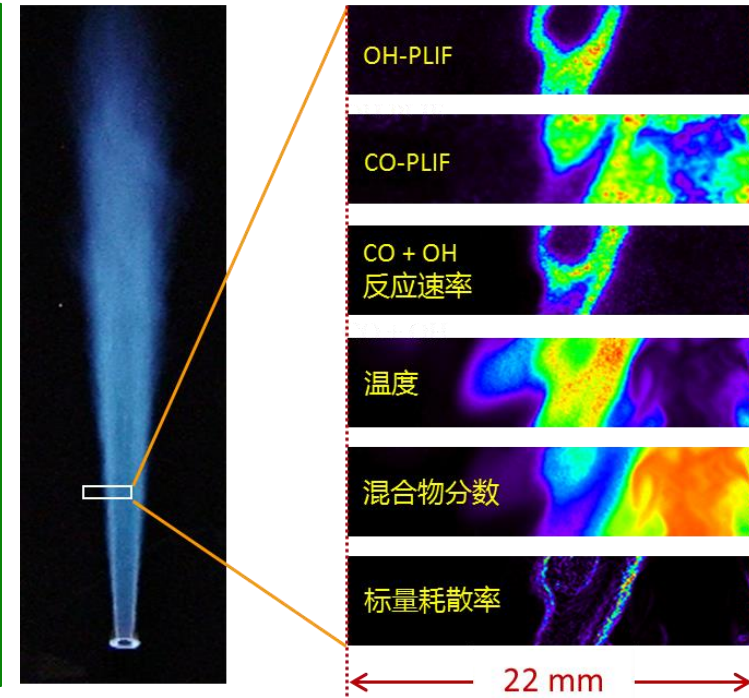
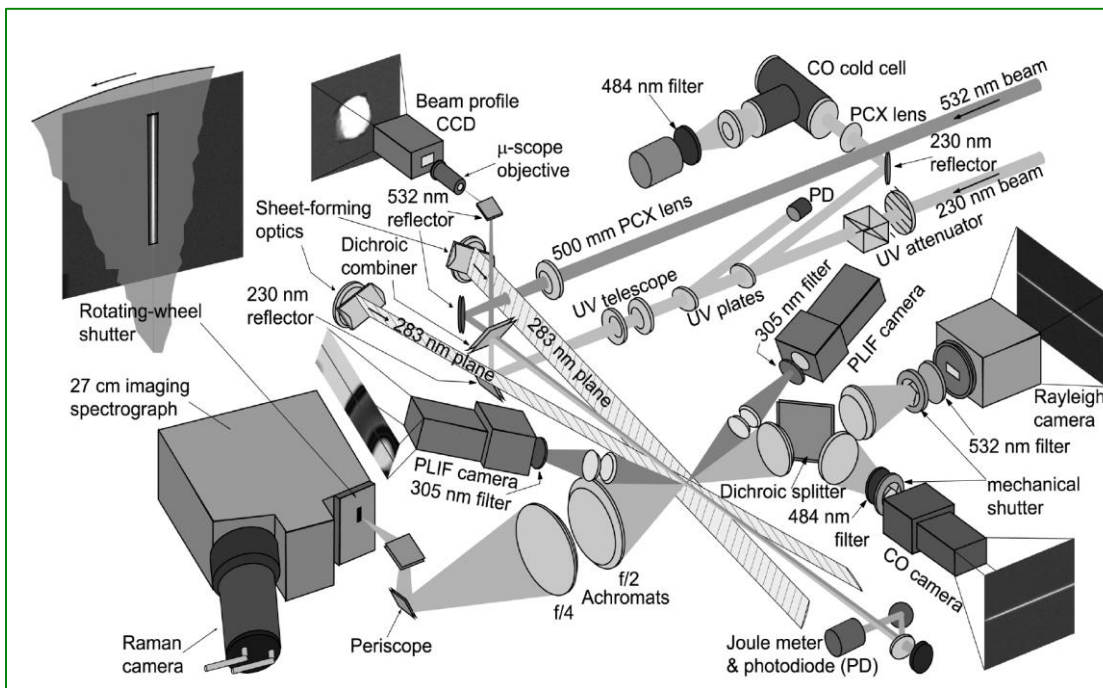


Kiefer et al., *Combust. Flame* 154 (2010) 802; Li et al., *Combust. Flame* 157 (2010) 1087.

Reaction-rate, Mixture-fraction, T & Species—2D

Reaction-rate, mixture-fraction, temperature turbulent CH₄/air flame

Sandia National Lab.

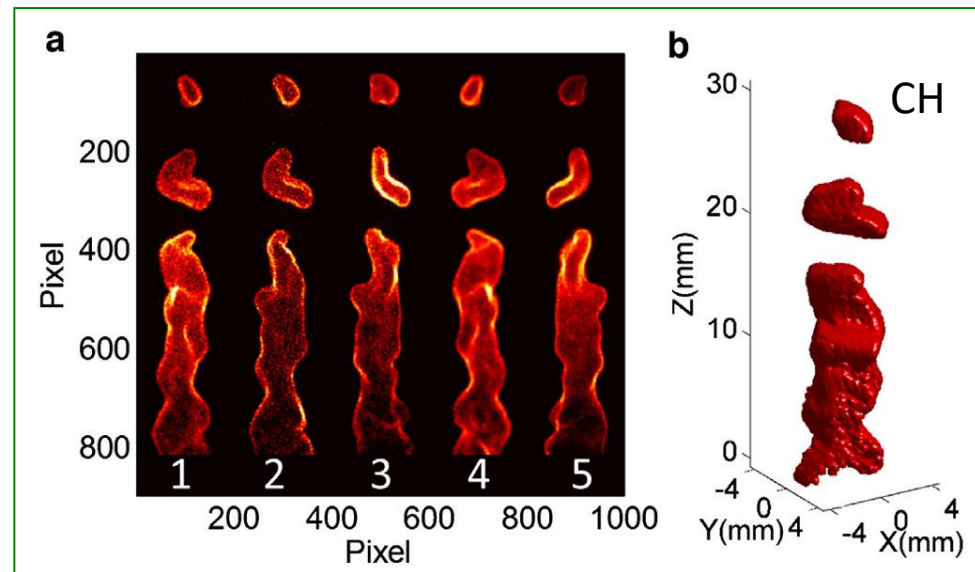
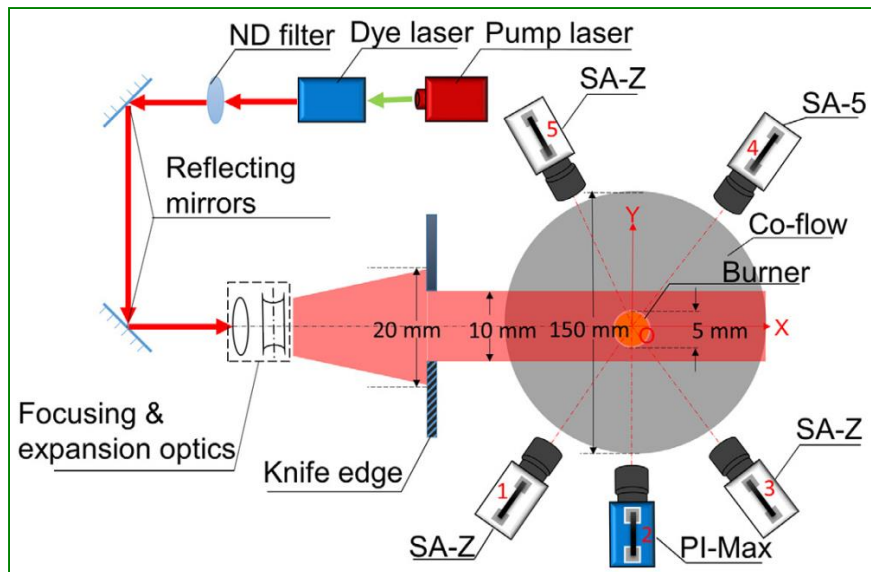


- Six lasers plus 5 cameras
- T, V, Species (OH, CO, N₂, O₂, CO₂, H₂O, H₂)

Volumetric Laser Induced Fluorescence (VLIF)—3D

Single-shot 3D flame diagnostic based on volumetric laser induced fluorescence (VLIF)

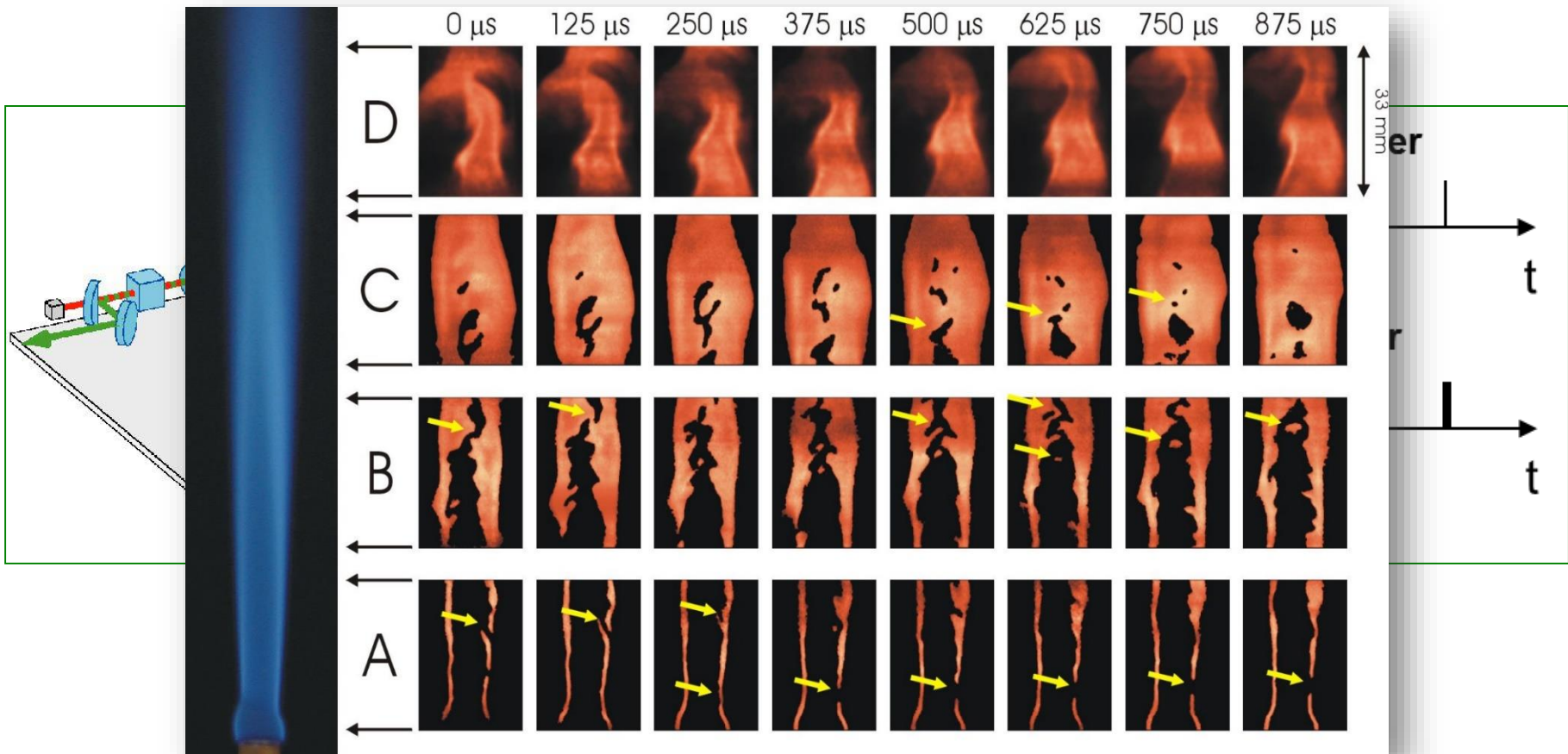
Virginia Tech, Air Force Research Laboratory



Temporally Resolved 2D Species Measurement

High repetition rate PLIF of OH in turbulent jet flame

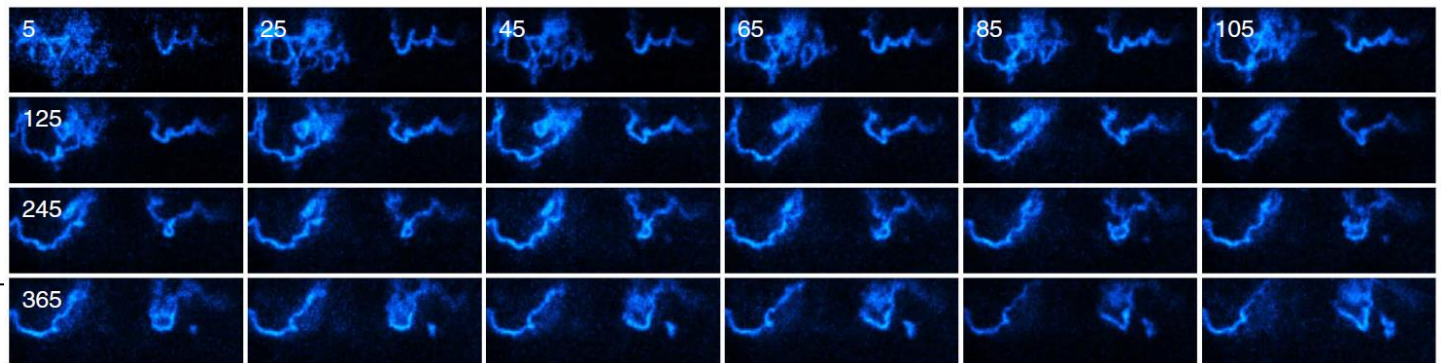
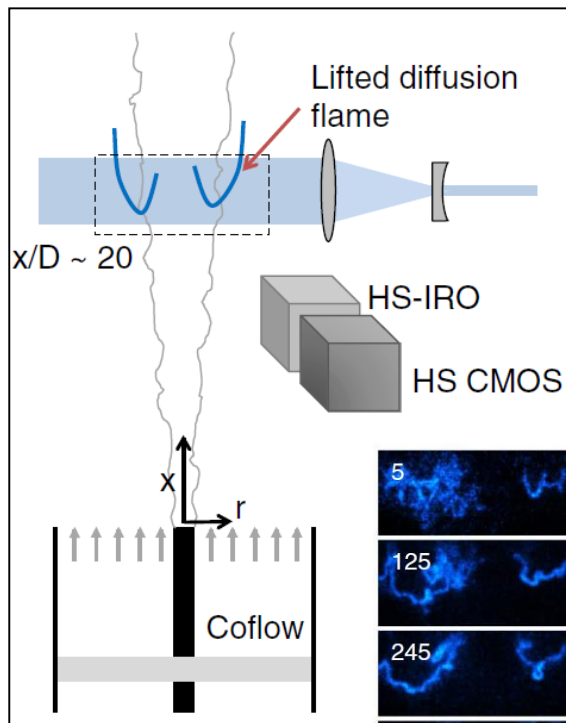
Lund University



Temporally Resolved 2D Species Measurement

100 kHz burst-mode planar imaging in turbulent flames

Air Force Research Laboratory, Purdue University, Spectral Energies

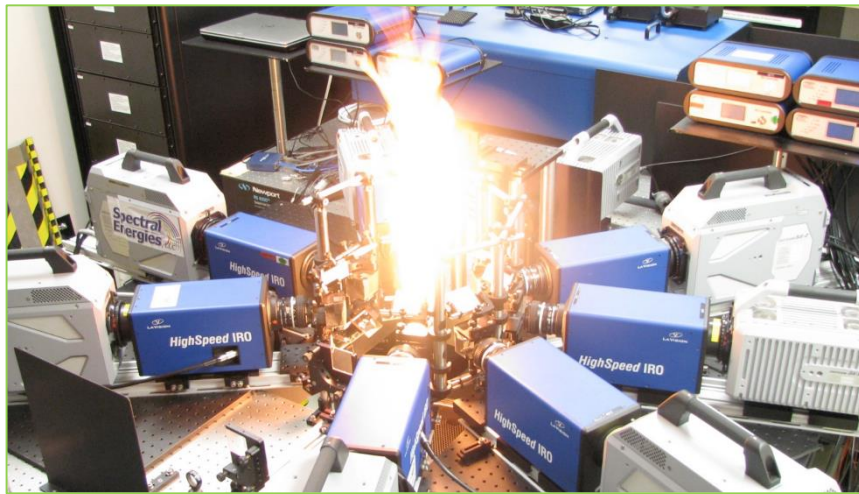


Sample sequence of CH₂O PLIF at 100 kHz showing every 20th image

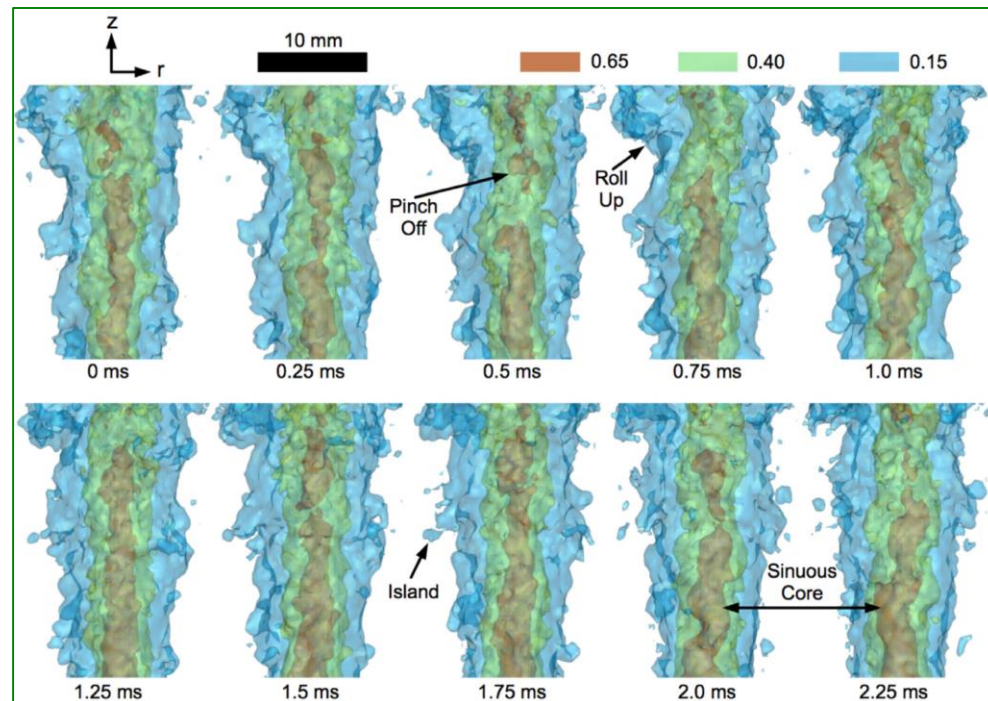
Temporally Resolved 3D Species Measurement

20-kHz-rate three-dimensional tomographic imaging of the concentration field in a turbulent jet

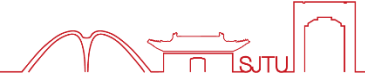
Air Force Research Laboratory, Purdue University, Spectral Energies



Acetone Fluorescence Imaging

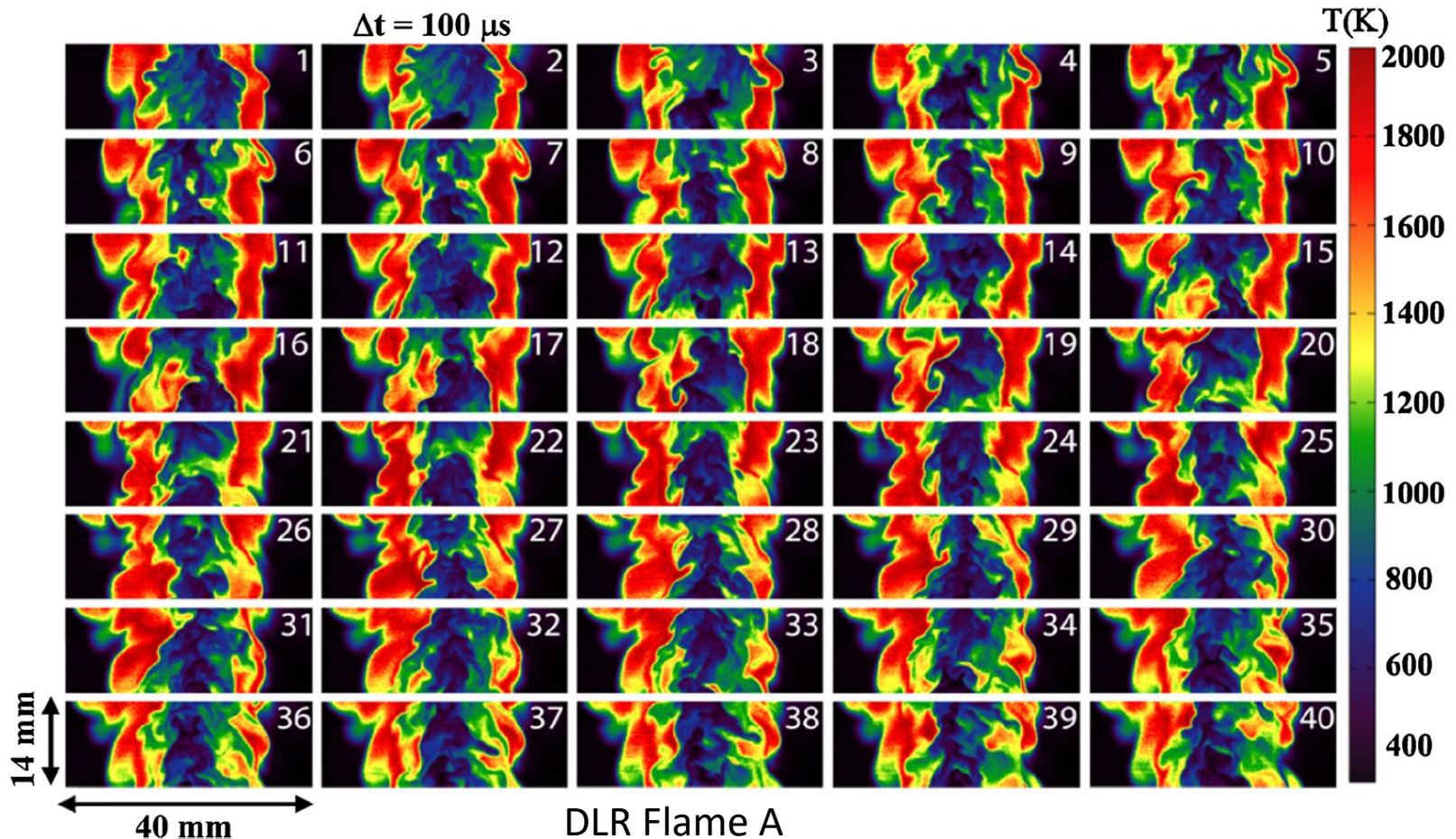


Temporally Resolved 2D T Measurement



High-speed planar Rayleigh scattering in turbulent jets and flames

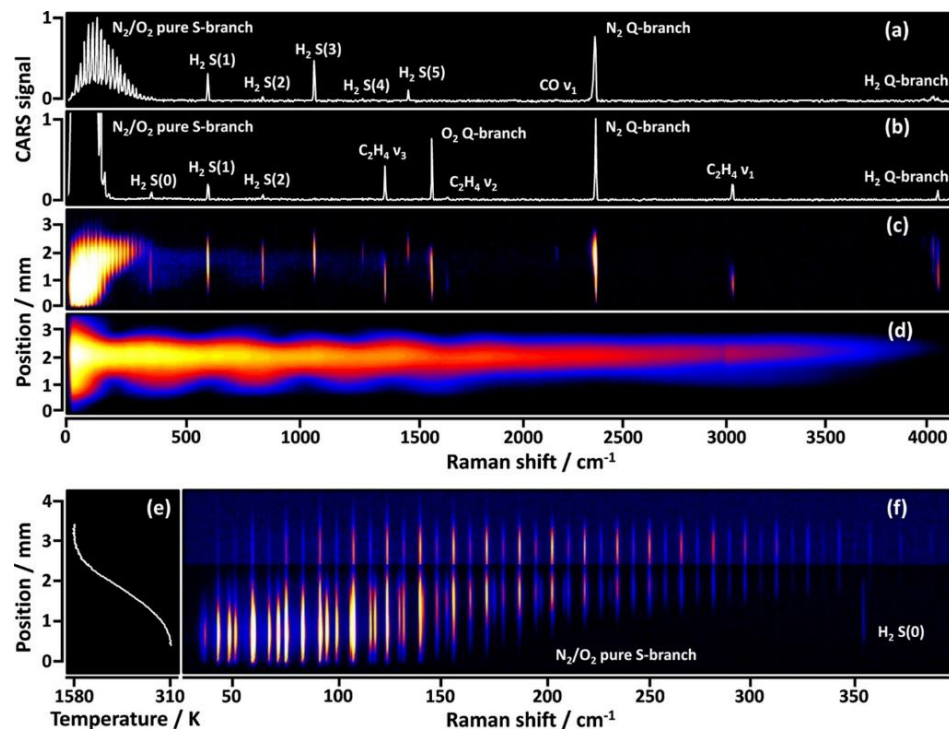
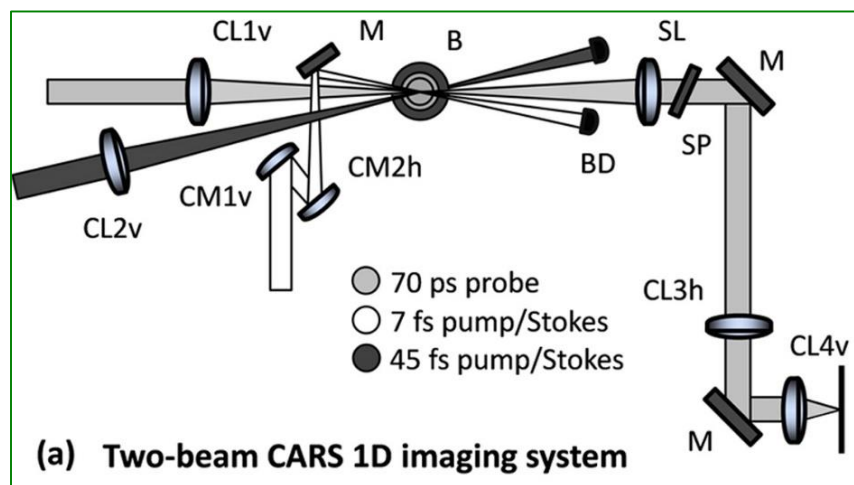
Ohio State University



Temporally Resolved 1D T and Species

Direct coherent Raman temperature imaging and wideband chemical detection in a hydrocarbon flat flame

Sandia National Lab.



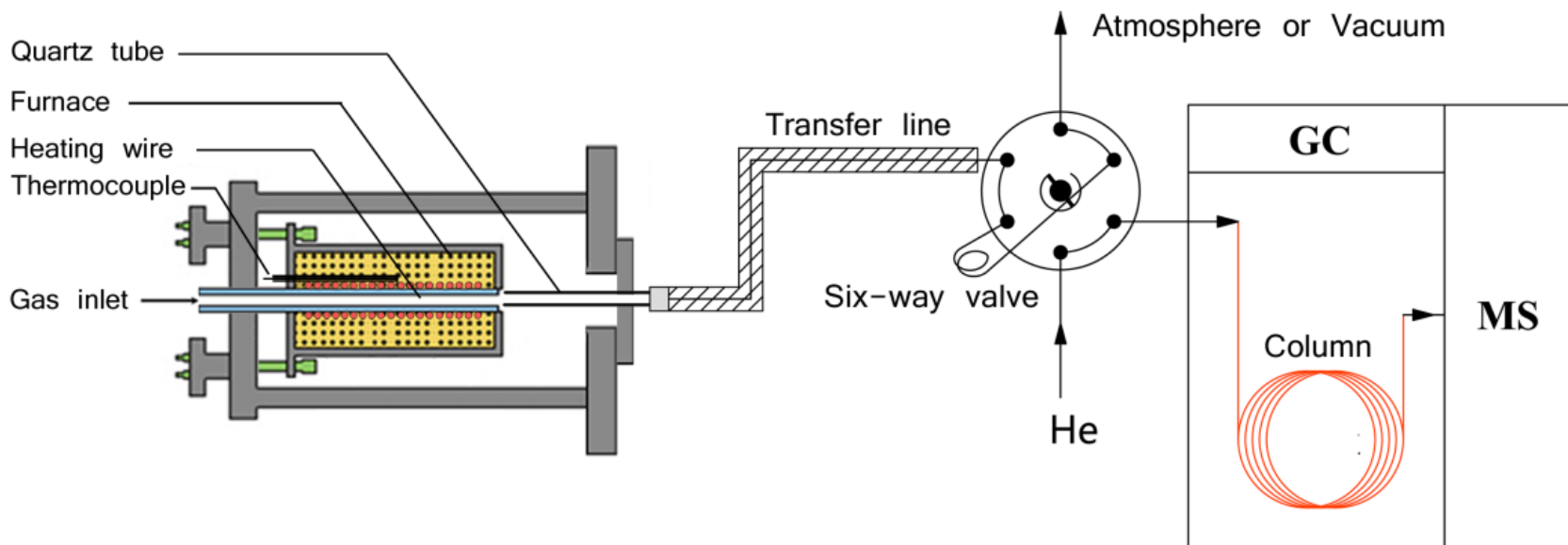
Single-shot two-beam fs/ps CARS 1D imaging system with quantified image retrieval

Experimental Methods—Diagnostics



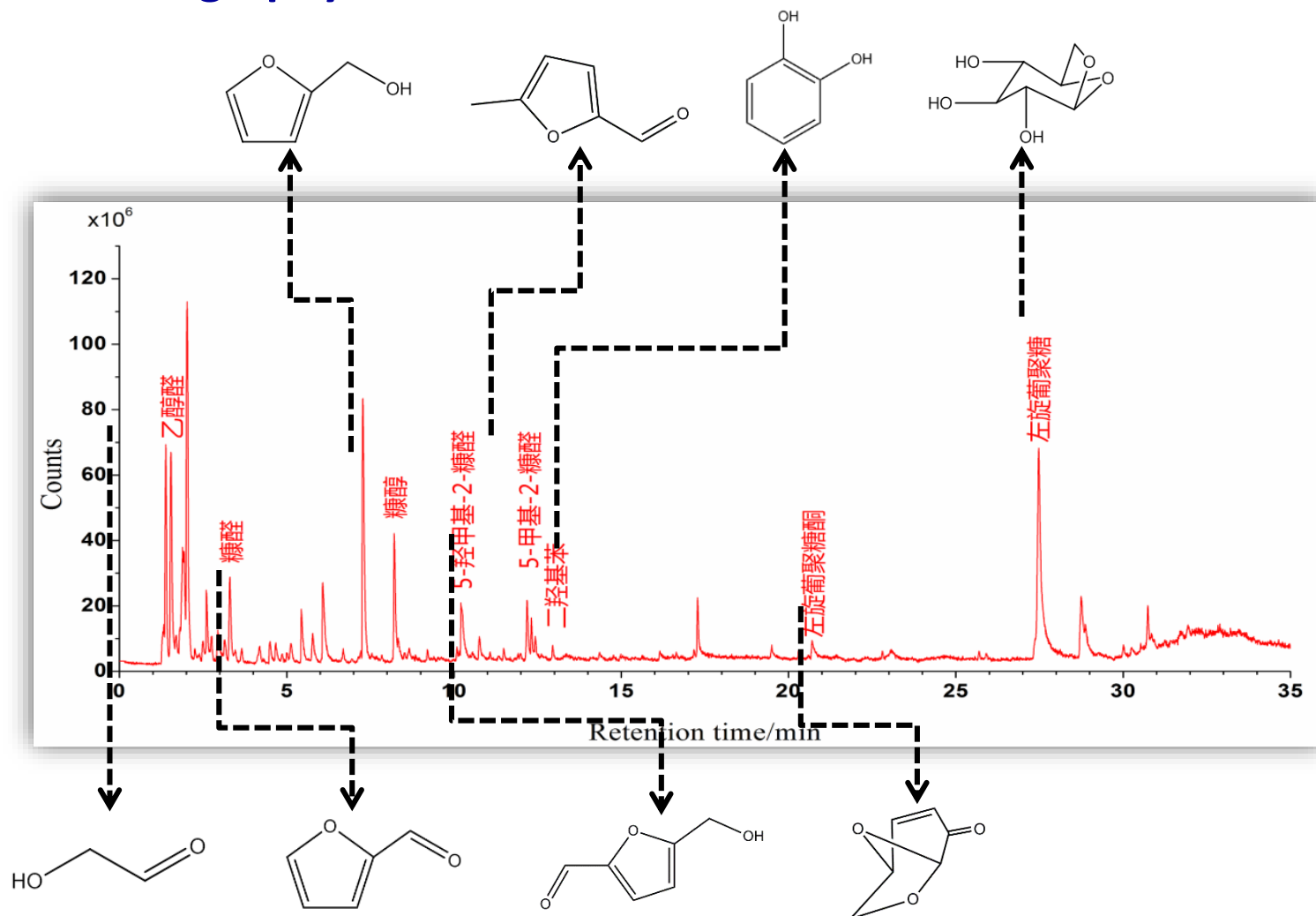
Gas Chromatography

- Gas chromatography is usually coupled with mass spectrometry for species measurement
- Cannot used to detect reactive species such as radicals and peroxides



Experimental Methods—Diagnostics

Gas Chromatography



Experimental Methods—Diagnostics



Mass Spectrometry

El ionization *Universal, not selective*

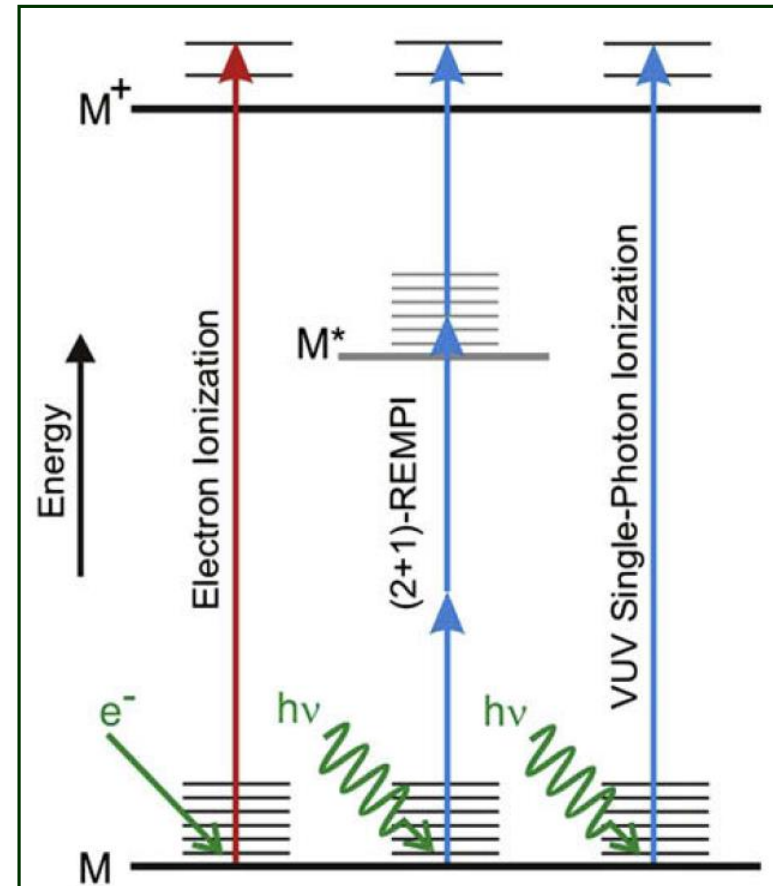
An ionization method in which energetic electrons interact with gas phase atoms or molecules to produce ions.

REMPI *Selective, not universal*

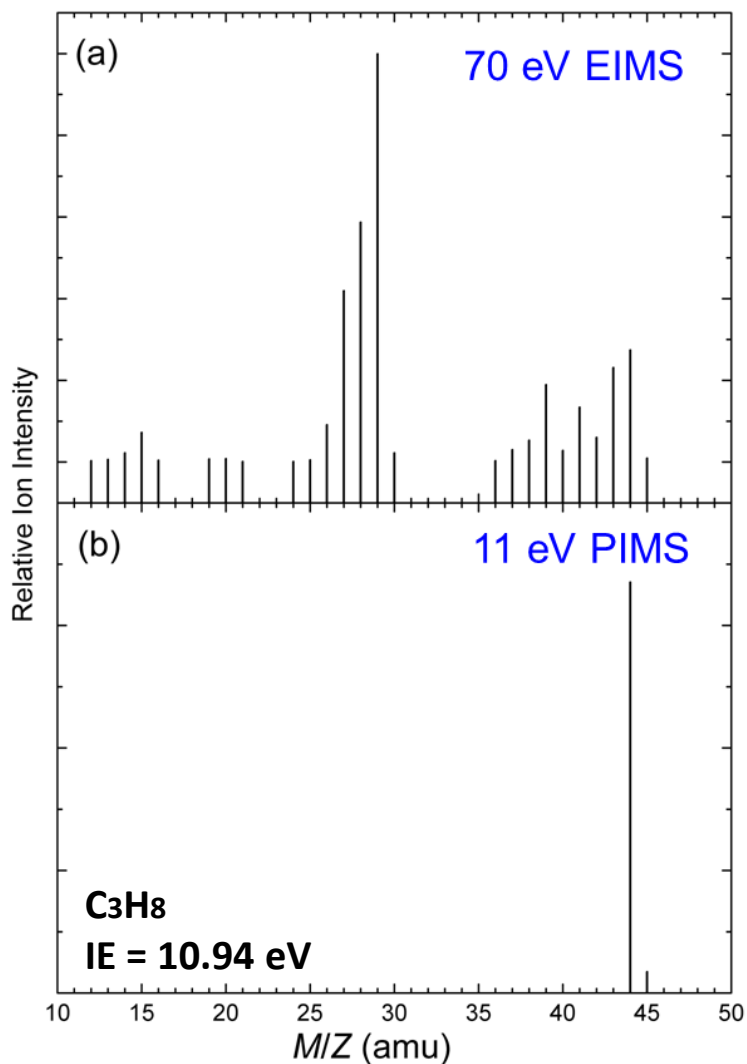
REMPI involves a resonant single or multiple photon absorption to an electronically excited intermediate state followed by another photon which ionizes the atom or molecule.

VUV PI *Universal, also selective*

Photoionization is the physical process in which an incident photon ejects one or more electrons from an atom or molecule.



Experimental Methods—Diagnostics

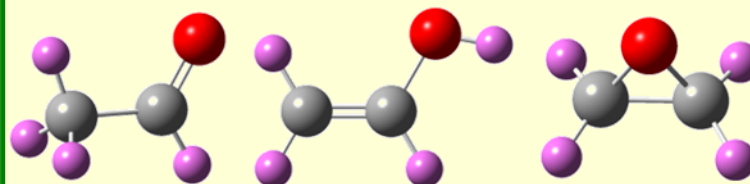


Photoionization Mass Spectrometry

PIMS can resolve different mass, and gives a single identifiable peak for each compound.

Question: how to distinguish isomers?

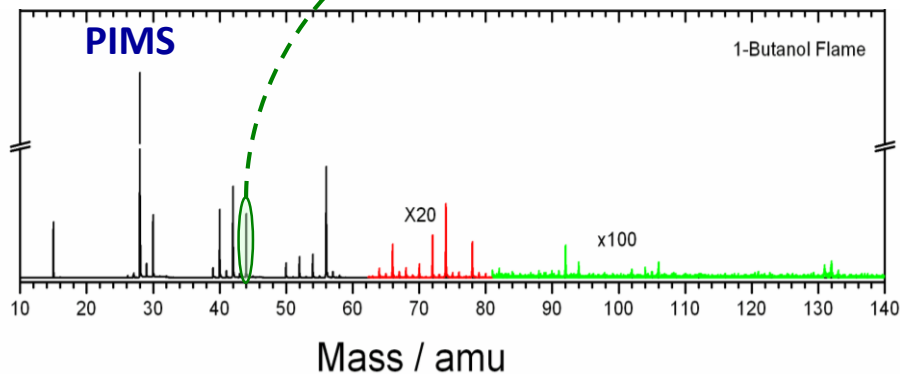
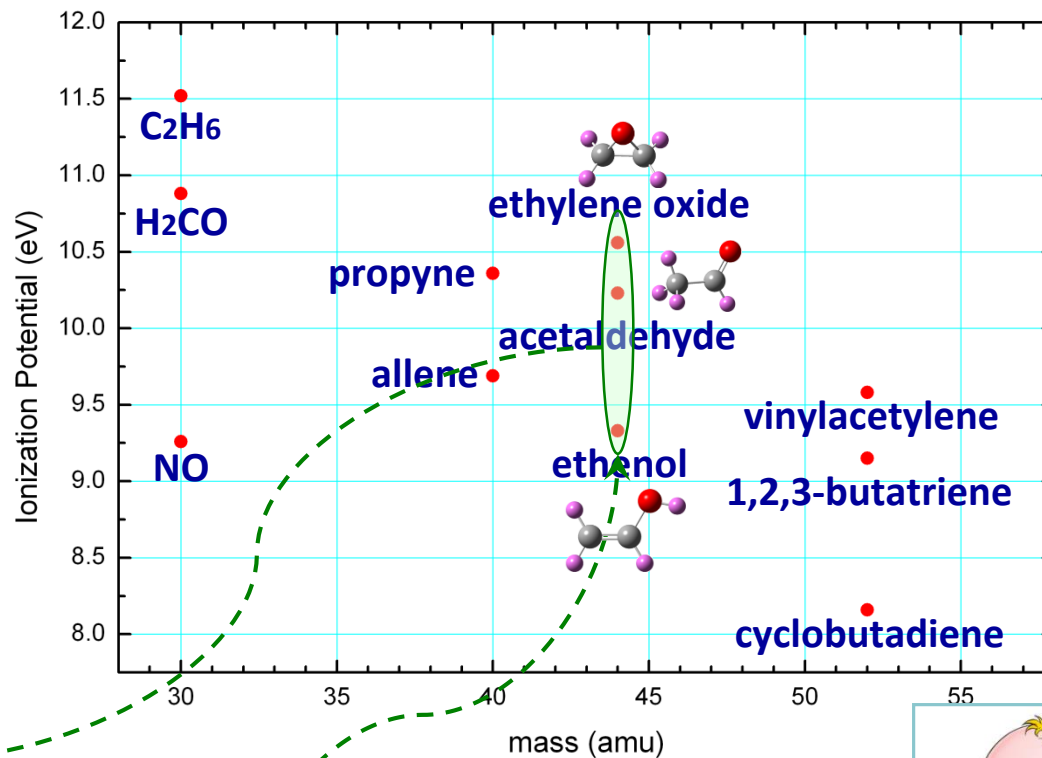
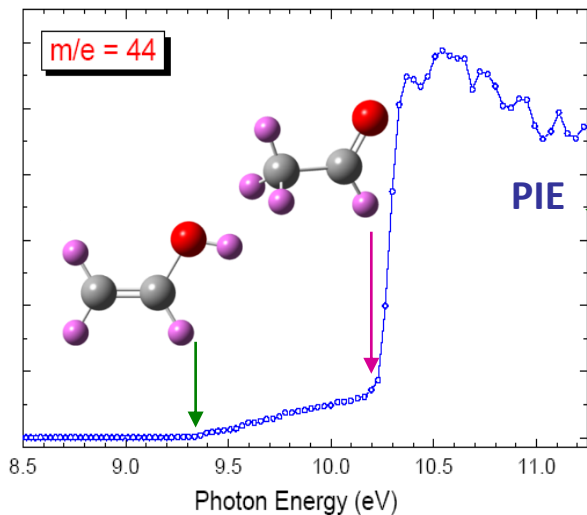
C_2H_4O ($m/z = 44$)



Experimental Methods—Diagnostics

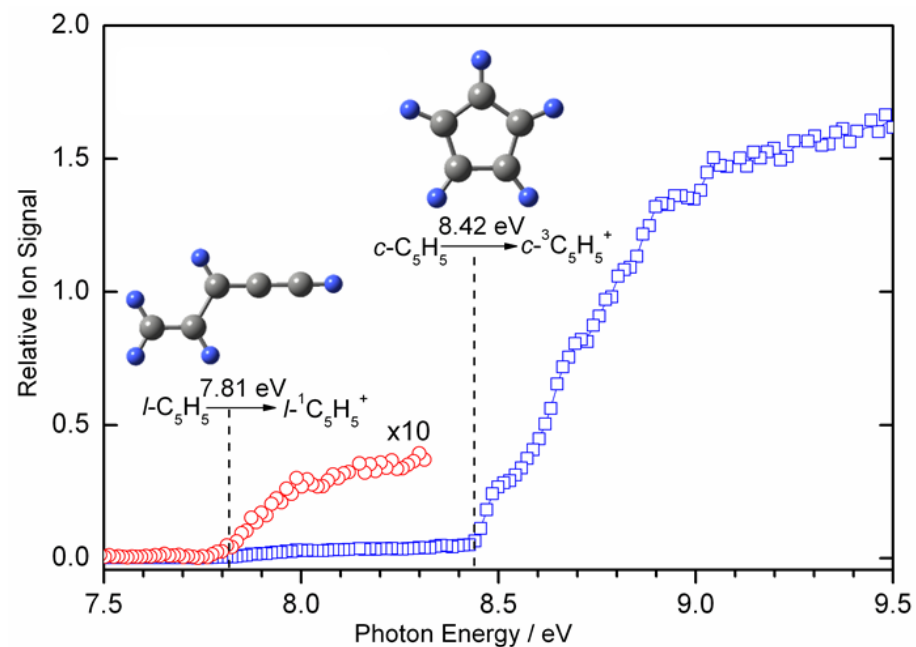
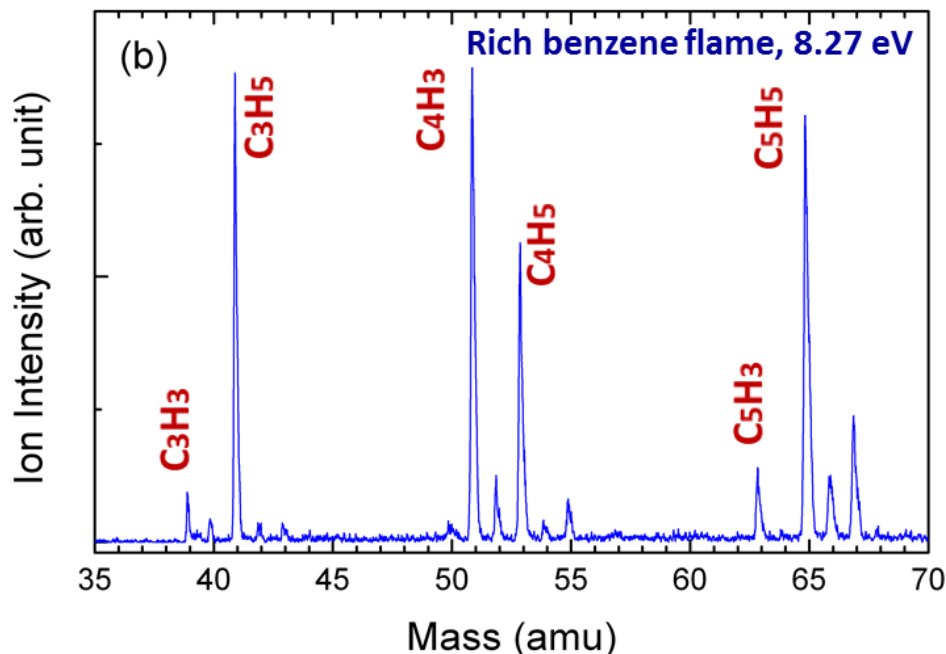
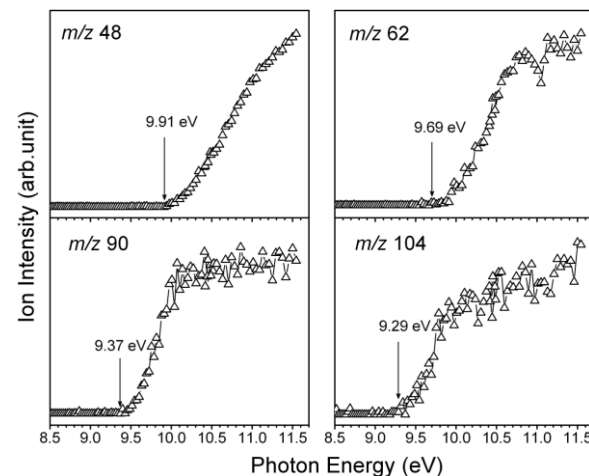
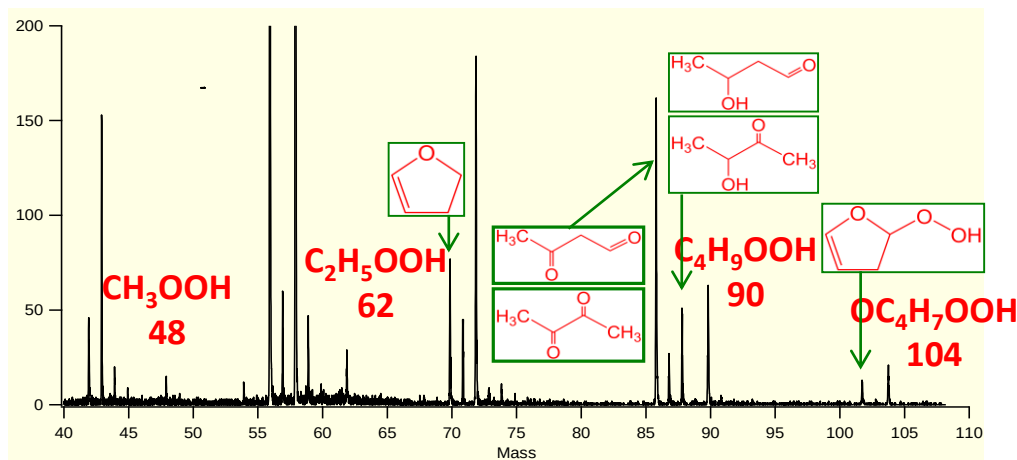


Height Scale
II



Weight Scale

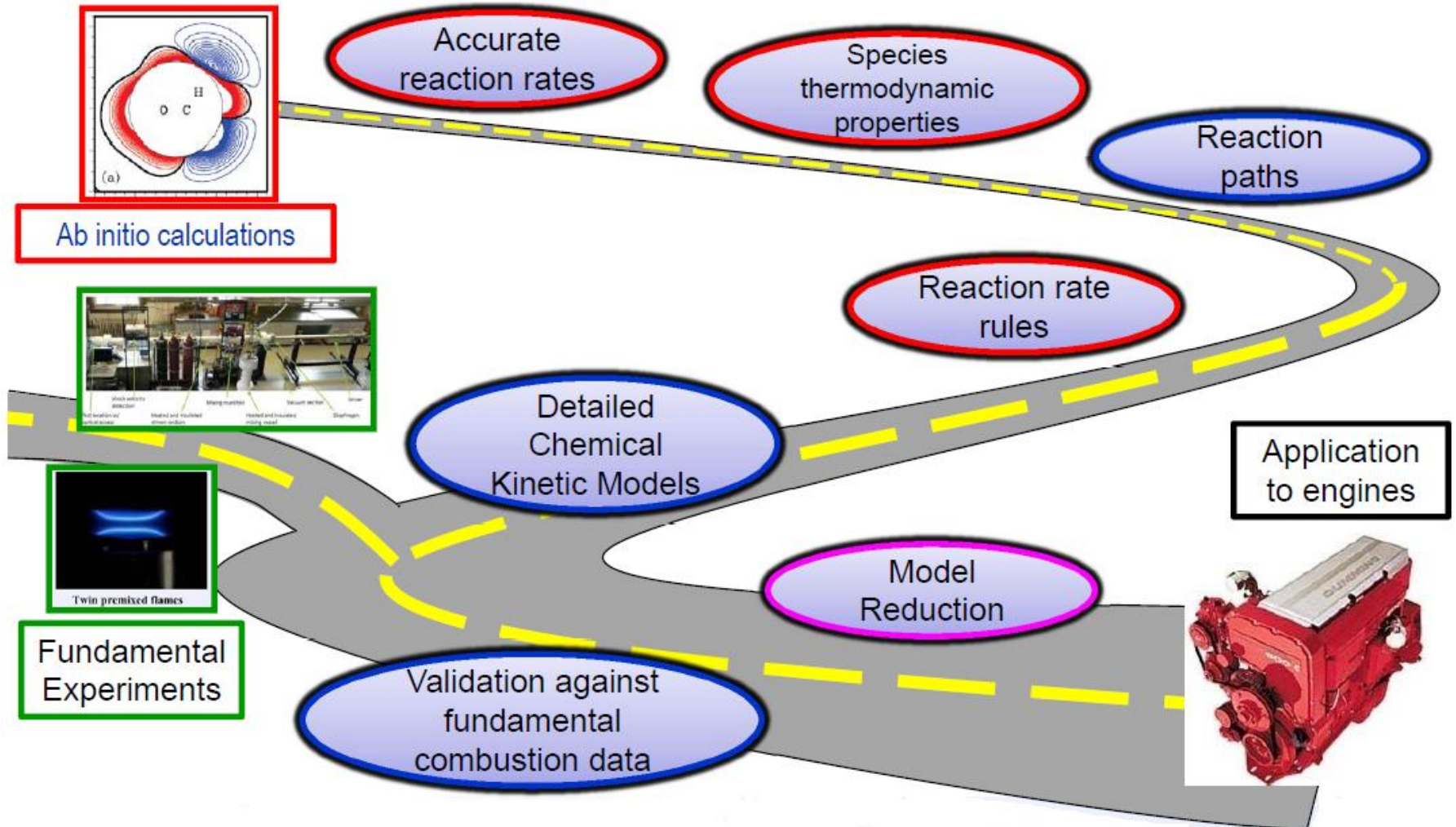
Experimental Methods—Diagnostics



Kinetic Modeling Methods

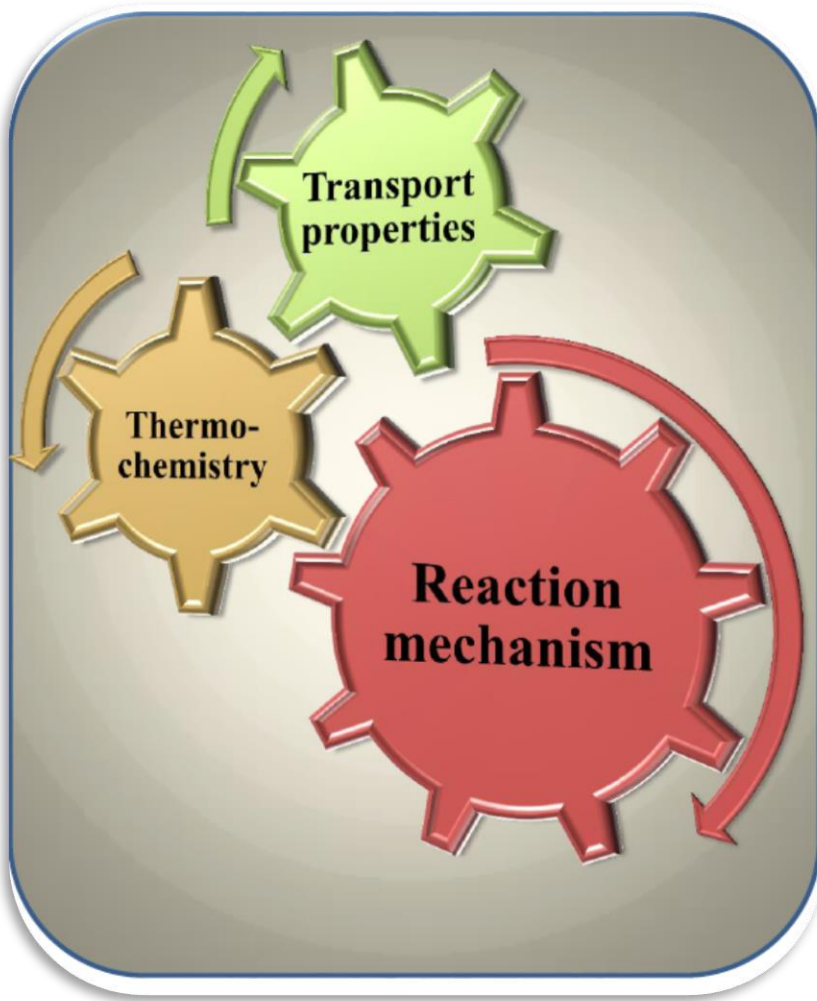


Roadmap for Model Development



Kinetic Modeling Methods

Key Components of Kinetic Model



Literature research

Google scholar

NIST



ACS
Chemistry for Life®

GRI-MECH 3.0

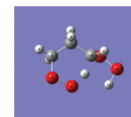
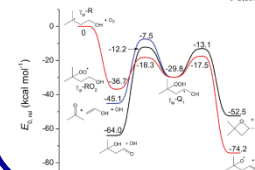
Combustion Kinetics Laboratory
 Aerospace and Mechanical Engineering
 Professor Hai Wang

QC calculation/estimation

$$k^{TST}(T) = \kappa \frac{k_B T}{h} \frac{Q^\ddagger(T)}{Q_{\text{react}}(T)} \exp\left(-\frac{E^\ddagger}{k_B T}\right)$$

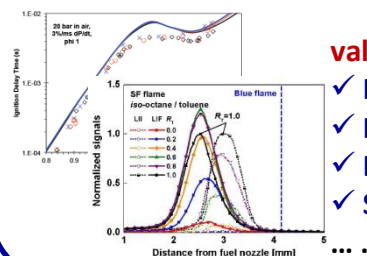
Estimation based on:

- ✓ Rate rules
- ✓ TST theory
- ✓ Structure and BDE
- ✓ Evans-Polanyi relationship



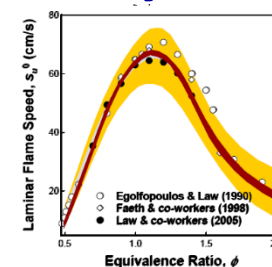
... ..

Validation/uncertainty



validation:

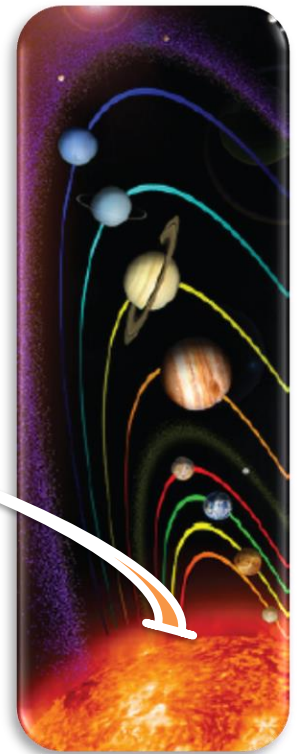
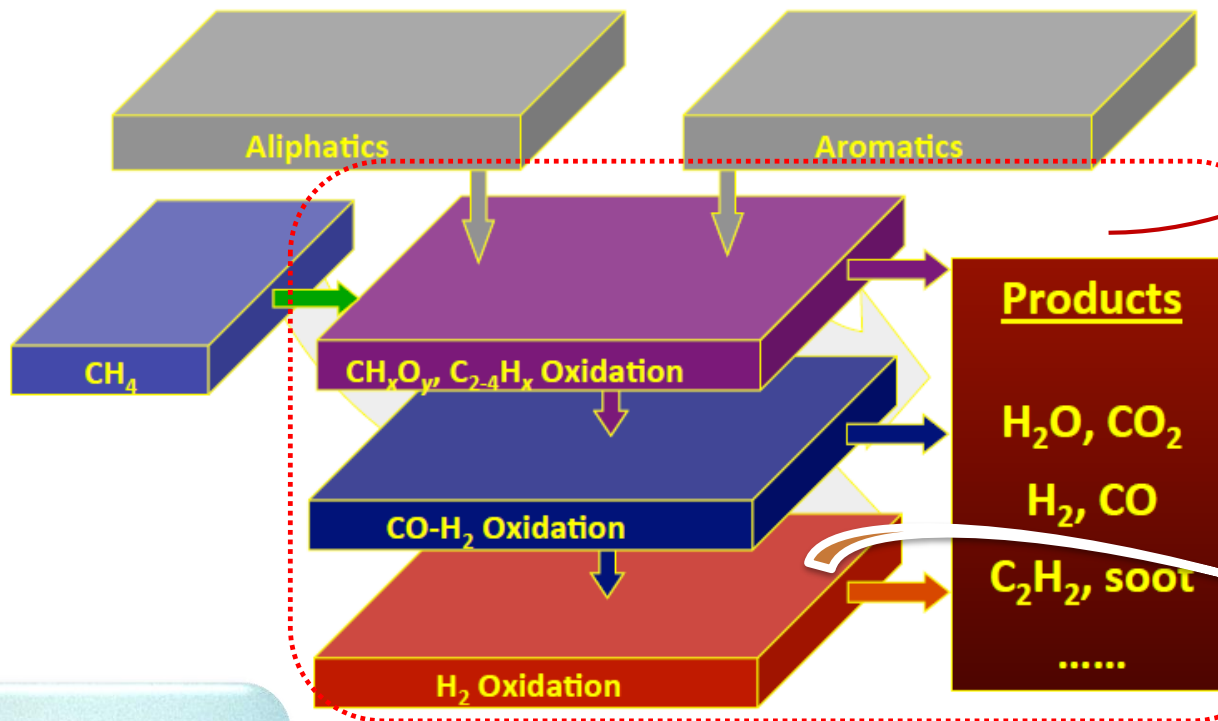
- ✓ Homogeneous
- ✓ Premixed flame
- ✓ Diffusion flame
- ✓ Soot formation



Kinetic Modeling Methods

Hierarchy Structure of Kinetic Model

Core mechanism



Kinetic Modeling Methods



Reaction Classes—High temperature mechanism

- Reaction class 1: Unimolecular fuel decomposition
- Reaction class 2: H atom abstractions from fuel
- Reaction class 3: Alkyl radical decomposition
- Reaction class 4: Alkyl radical + $O_2 = \text{olefin} + HO_2$
- Reaction class 5: Alkyl radical isomerization
- Reaction class 6: H atom abstraction from olefins
- Reaction class 7: Addition of radical species to olefins
- Reaction class 8: Alkenyl radical decomposition
- Reaction class 9: Olefin decomposition

Kinetic Modeling Methods



Reaction Classes—Low temperature mechanism

Reaction class 10: Alkyl radical addition to O_2 ($R + O_2$)

Reaction class 11: $R + R'O_2 = R\dot{O} + R'\dot{O}$

Reaction class 12: Alkylperoxy radical isomerization

Reaction class 13: $R\dot{O}_2 + H\dot{O}_2 = ROOH + O_2$

Reaction class 14: $R\dot{O}_2 + H_2O_2 = ROOH + H\dot{O}_2$

Reaction class 15: $R\dot{O}_2 + CH_3\dot{O}_2 = R\dot{O} + CH_3\dot{O} + O_2$

Reaction class 16: $R\dot{O}_2 + R'\dot{O}_2 = R\dot{O} + R'\dot{O} + O_2$

Reaction class 17: $ROOH = R\dot{O} + \dot{O}H$

Reaction class 18: $R\dot{O}$ Decomposition

Reaction class 19: $QOOH = \text{Cyclic Ether} + \dot{O}H$

Reaction class 20: $QOOH = \text{Olefin} + H\dot{O}_2$

Reaction class 21: $QOOH = \text{Olefin} + \text{Aldehyde or Carbonyl} + \dot{O}H$

Reaction class 22: Addition of $QOOH$ to molecular oxygen O_2

Reaction class 23: $\dot{O}_2QOOH$ isomerization to carbonylhydroperoxide + $\dot{O}H$

Reaction class 24: Carbonylhydroperoxide decomposition

Reaction class 25: Reactions of cyclic ethers with $\dot{O}H$ and $H\dot{O}_2$

Kinetic Modeling Methods



Physical Aspects

Flow pattern
Transport phenomena
Resident time distribution

Chemical Aspects

Reaction kinetics
Thermodynamics
Transport properties



Mathematical Model

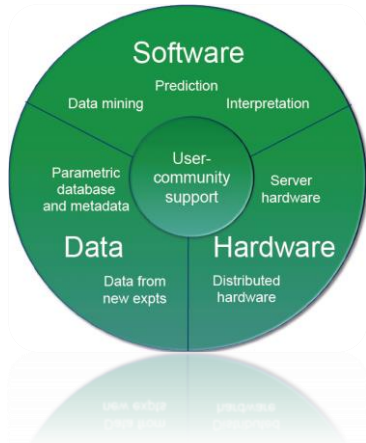
Reactor type
Balance equations
Numerical algorithms

Experimental Data

Measurements in flames:
temperature, velocity, species
mole fraction, ...

Comparison
?

- ☐ Kinetics?
- ☐ Thermodynamics?
- ☐ Exp. Uncertainties?
- ☐ Systematic error?
- ☐ Mathematical Model?
- ☐ Boundary conditions?
- ☐ Catalytic effects?
- ☐



Kinetic Modeling Methods



Model Validations

Homogeneous Reactors

Homogeneous

Shock Tube

Homogeneous

Turbulent (Plug) Flow Reactor

Homogeneous

Rapid Compression

Non Homogeneous

Laminar Flow Reactor

Non Homogeneous

Laminar Premixed Flame

Non Homogeneous

Laminar Diffusion Flame

Homogeneous

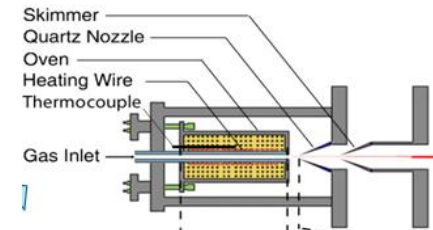
Perfectly Stirred Reactor

Degree of Mixing



Flow Reactors

Jet Stirred Reactor



Shock Tube

RCM

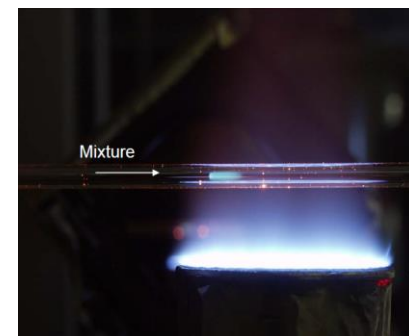
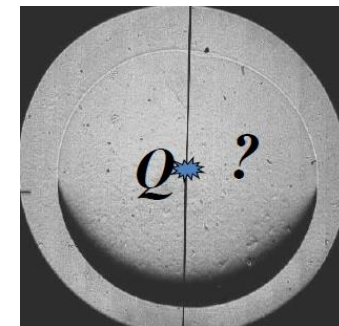
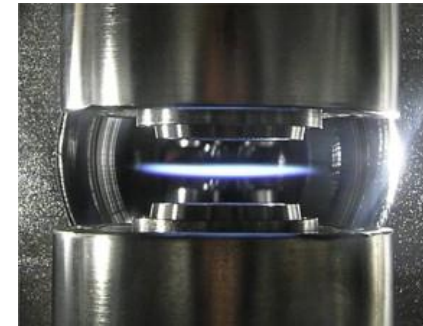
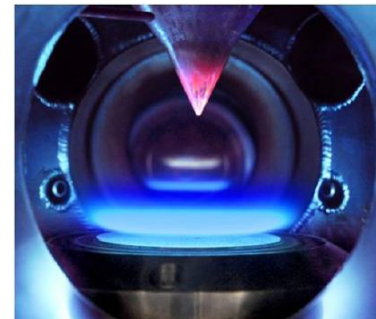
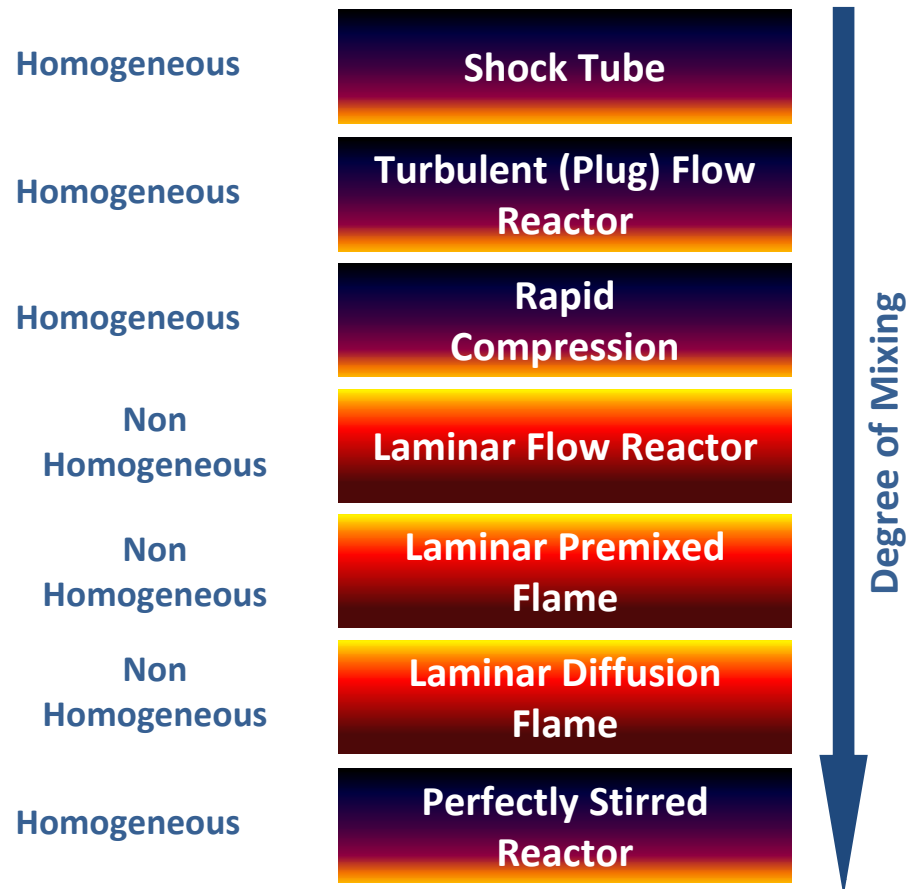


Kinetic Modeling Methods



Model Validations

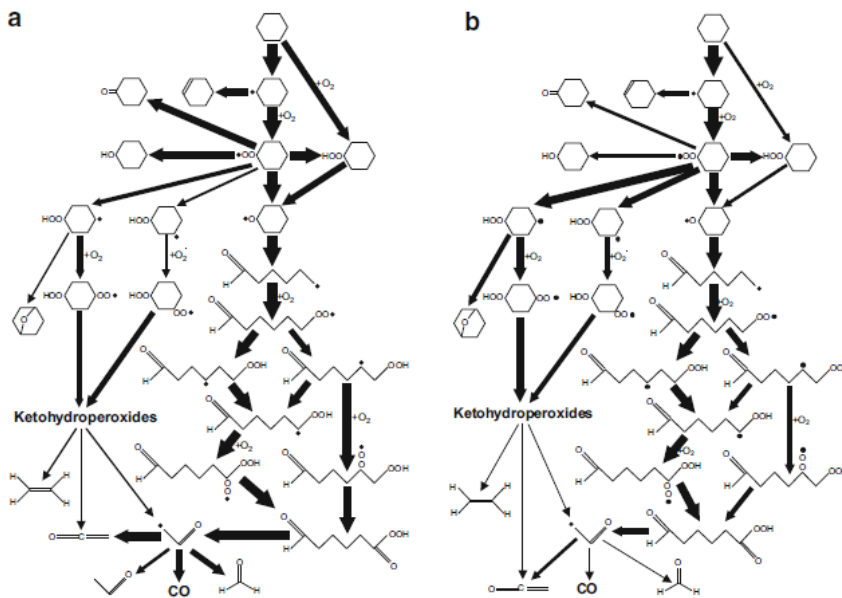
Flames



Kinetic Modeling Methods

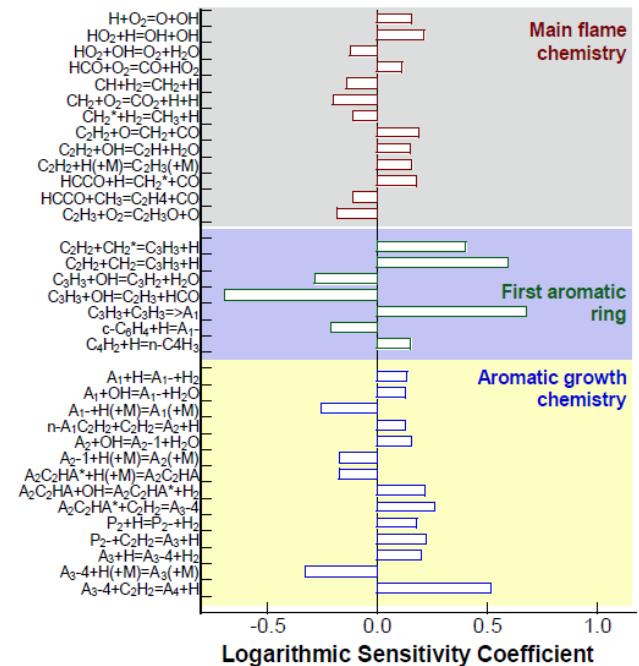
Model Analysis

Rate of production (ROP) analysis



- Species conversion pathways
- Pathways leading to the consumption or production of a species
- Flux balance analysis

Sensitivity analysis

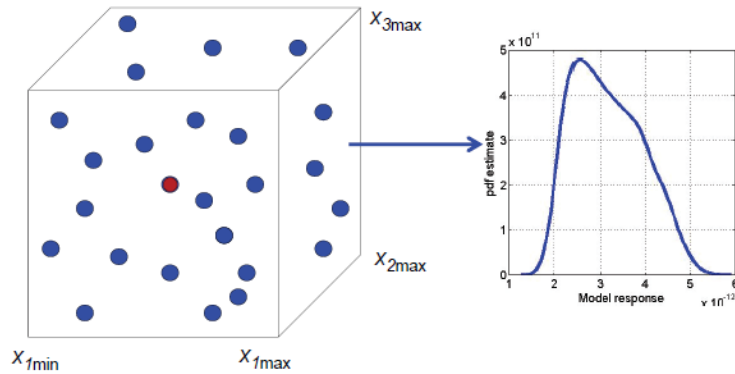


- Determine how much each input parameter contributes to the output uncertainty

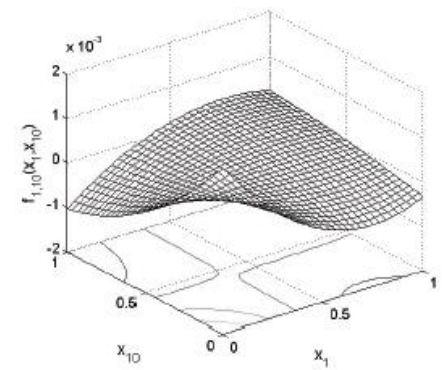
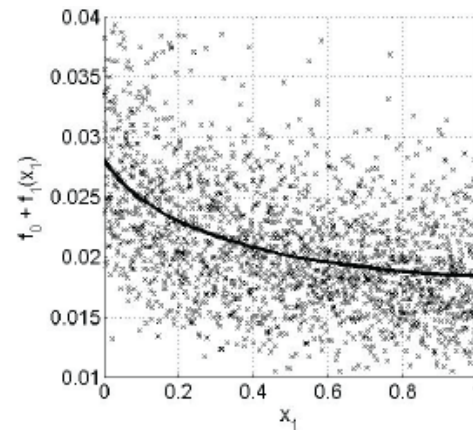
Kinetic Modeling Methods

Uncertainty analysis

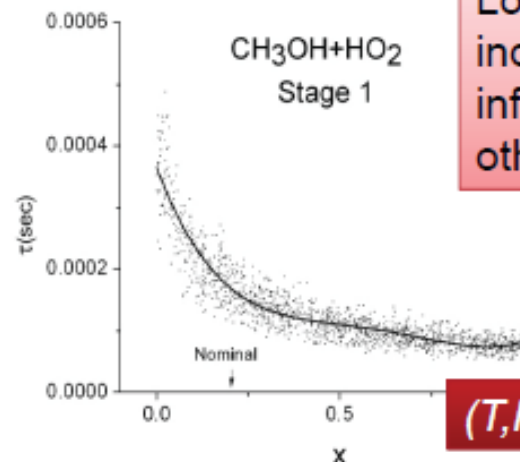
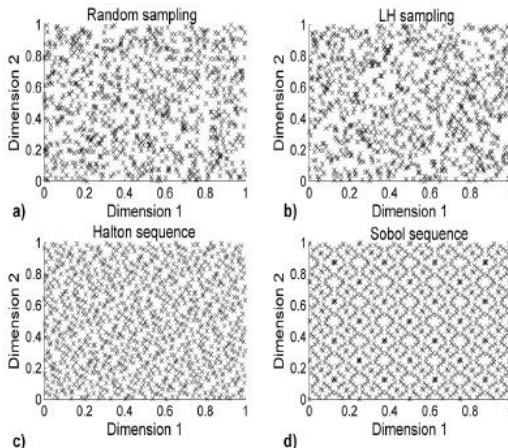
Attempts to cover whole input space using a sampling method



Perform many simulations (N) until output mean/variances converge



Sample from input distributions



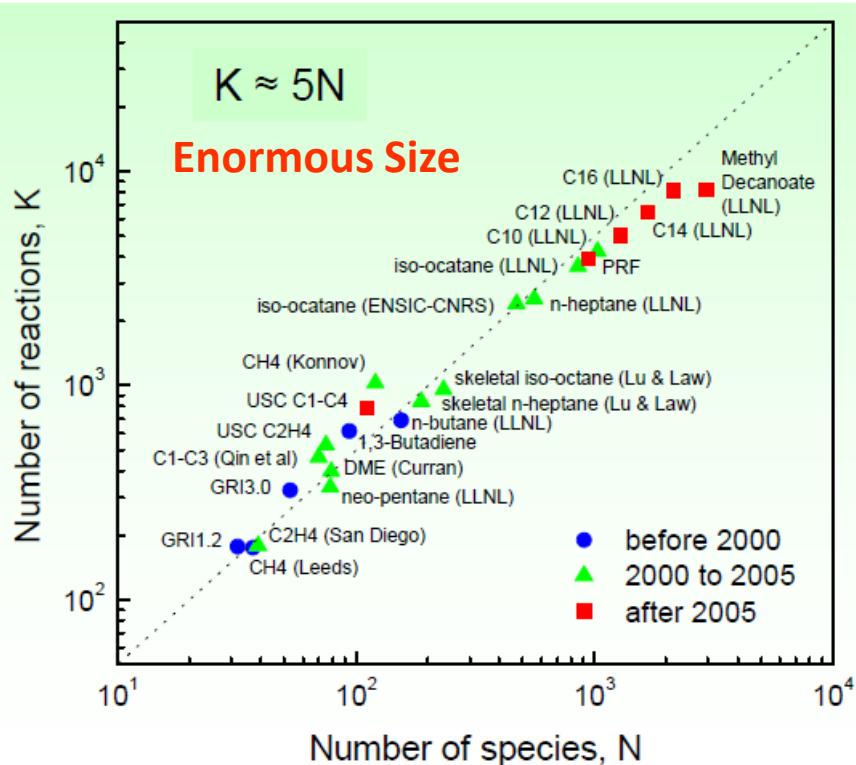
Low scatter indicates low influence of all other parameters

$(T, P, \phi) = (1150K, 5bar, 1)$

Kinetic Modeling Methods

Model reduction – Necessity

- Problem: Detailed mechanism cannot always be used in CFD simulations; computation time $\sim N^2$



Chemical time scales

Physical time scales

Kinetic post-processor

Slow Processes:

NO_x Formation

Intermediate Processes

PAH Formation

Soot Formation

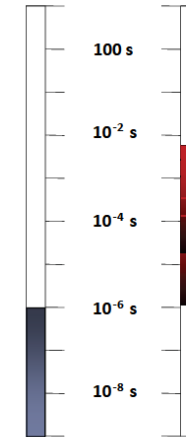
Fast Processes:

Partial Equilibrium

Steady State Cond.

Kinetics

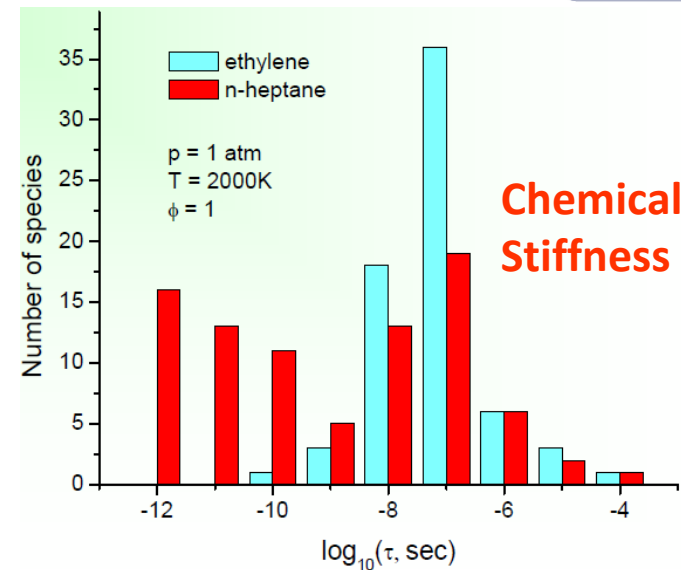
Physics



flux,
transport,
turbulence

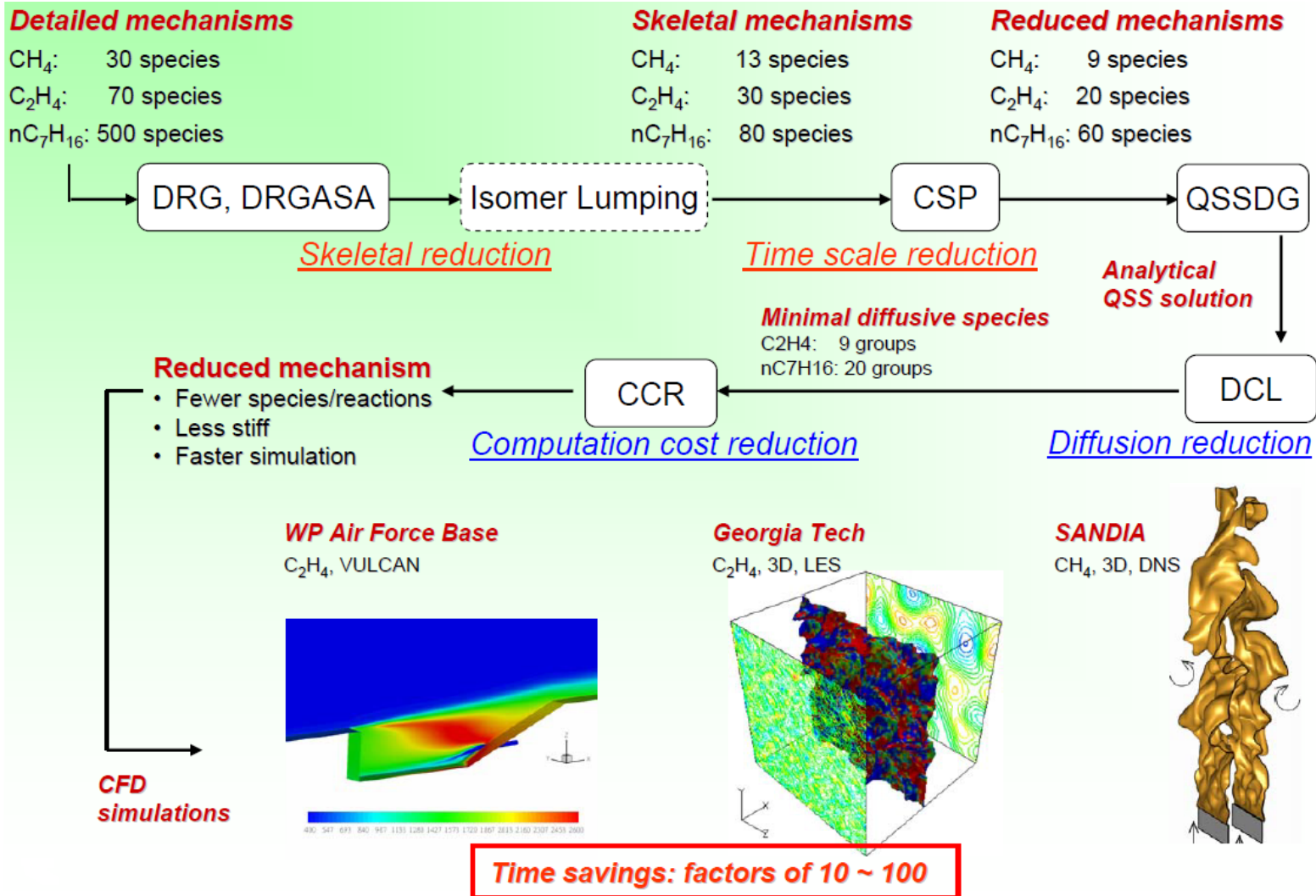
Mixed = Burned

→ CFD



Kinetic Modeling Methods

Model reduction – Strategy



Kinetic Modeling Methods

Model reduction – Methods

Skeletal reduction(Removal of redundant species)

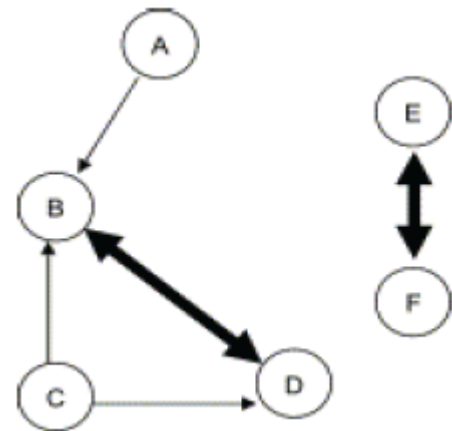
- Trial & error with validation
- Sensitivity analysis
- Principal component analysis (PCA)
- Graph based methods: DRG, DRGEP, PFA, DRGEPSA...

Timescale based reduction

- Quasi steady state approximations (QSSA)
- Partial equilibrium approximations (PEA)
- Rate controlled constrained equilibrium (RCCE)
- Computational singular perturbation (CSP)
- Intrinsic low dimensional manifold (ILDM)

Other methods

- In-situ adaptive tabulation (ISAT)
- Mathematically based lumping
- Dynamic adaptive chemistry (DAC)
-



Kinetic post-processor

Slow Processes:

NO_x Formation

Intermediate Processes

PAH Formation

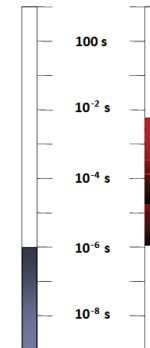
Soot Formation

Fast Processes:

Partial Equilibrium

Steady State Cond.

Kinetics Physics



flux,
transport,
turbulence

Mixed = Burned

→ CFD

Outline of Chapter 3



Part 1

- Fundamentals of Chemical Kinetics
- Reaction Kinetics Theory — Collision Theory
- Reaction Kinetics Theory — Transition State Theory
- Reaction Kinetics Theory — Unimolecular Reaction Theory
- Combustion Reaction Kinetics

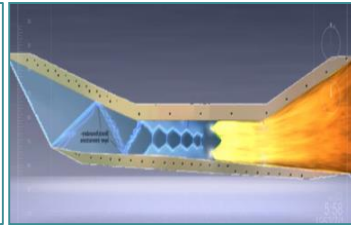
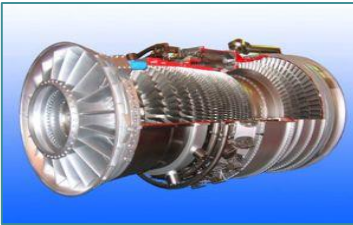
Part 2

- Research Methods of Combustion Reaction Kinetics
- Applications of Combustion Chemical Kinetics

Applications



Engine combustion

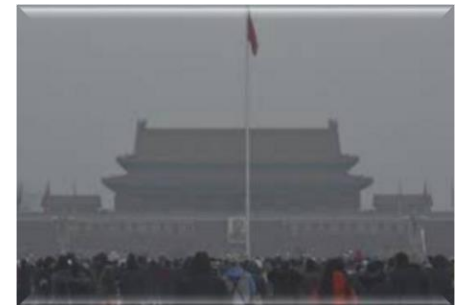
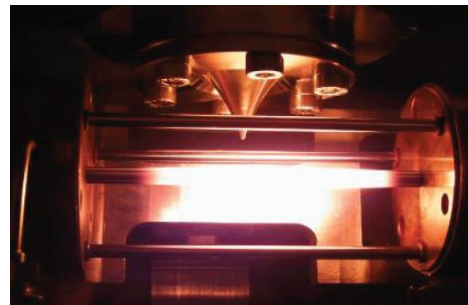


Clean combustion

New fuels

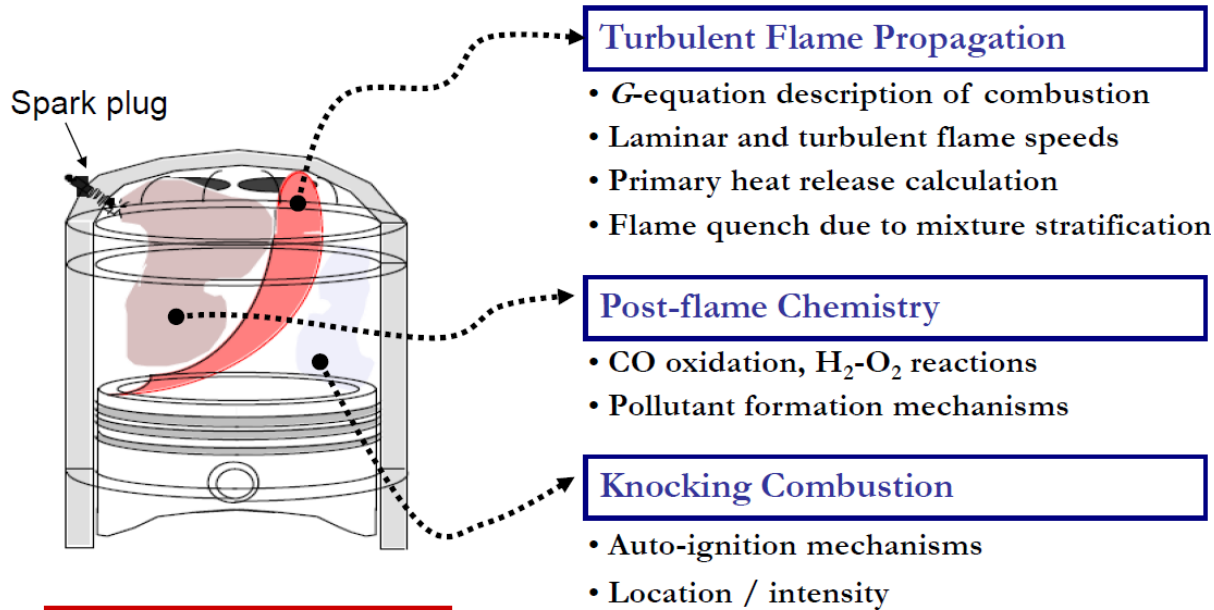
New combustion technology

Polluants control



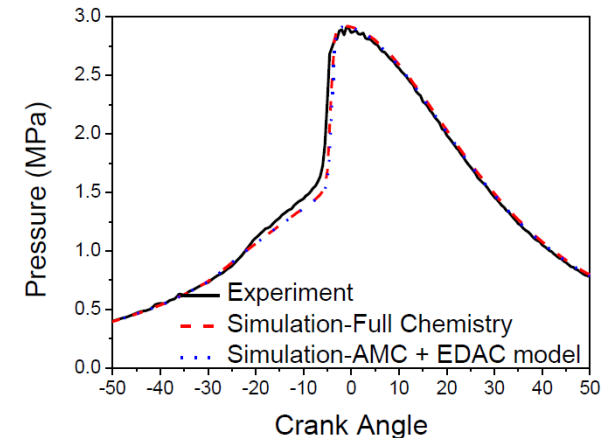
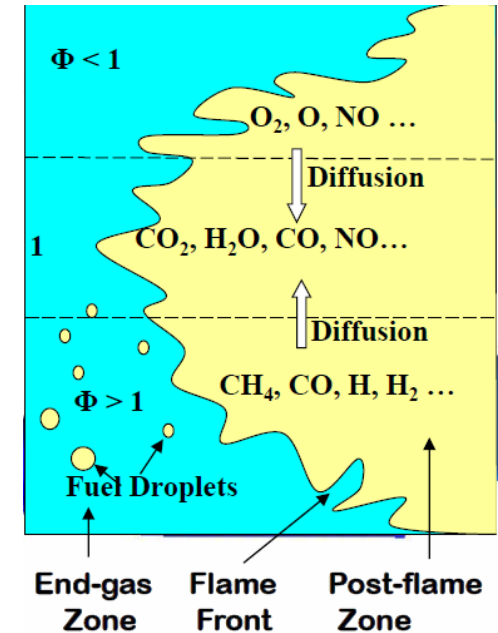
Gasoline Engine

Role of Chemistry in SI Engine



Spark Ignition Engine

- Improve efficiency
- Predict ignition and knock
- Control pollutant emission
- ...



Gasoline Engine

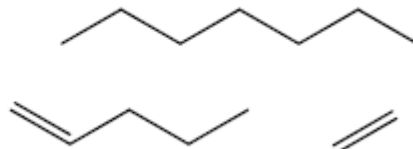
History of gasoline

1885-1910



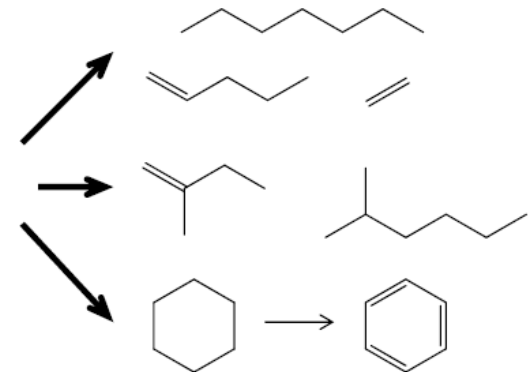
Petroleum, C_0 - C_5 alkanes

1920
Thermal cracking



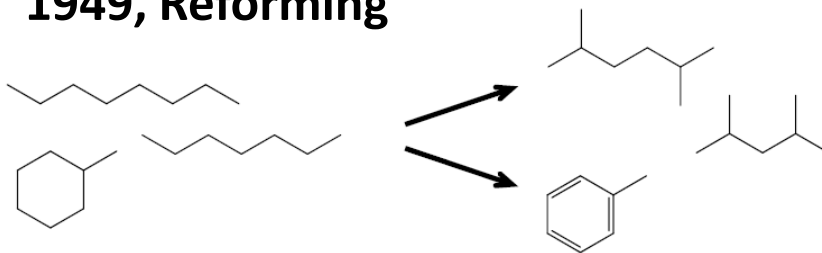
Heavier petroleum fractions to shorter (more volatile) alkanes and olefins

1937
Catalytic cracking



Better fuels (higher aromatic and lightly branched paraffins)

1949, Reforming



Upgrade the fuel quality by increasing the fraction of aromatics and branched paraffins

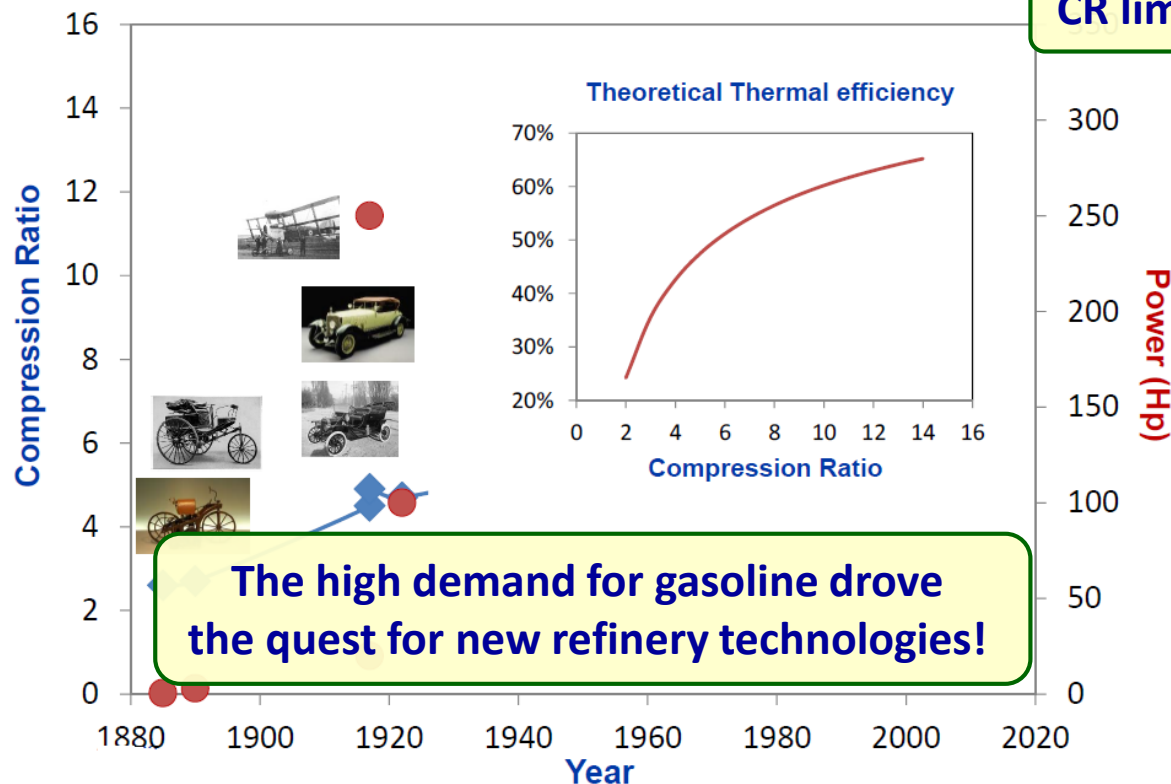


Gasoline Engine

Fuel and knock

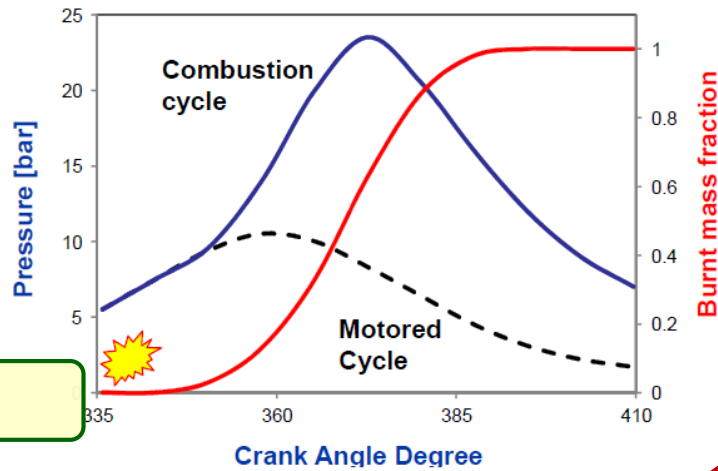
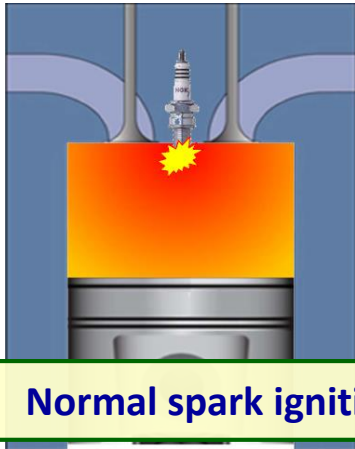
Increasing power output required bigger engine and higher compression ratio

When the compression ratio was increased, anomalous combustion behavior lead to engine knock

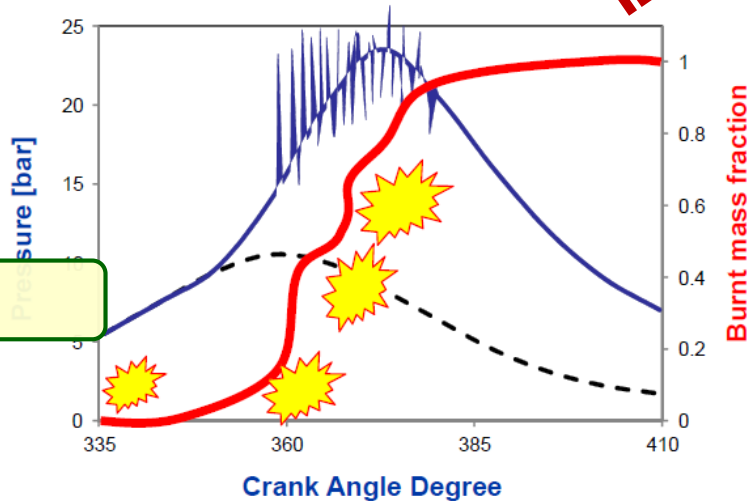
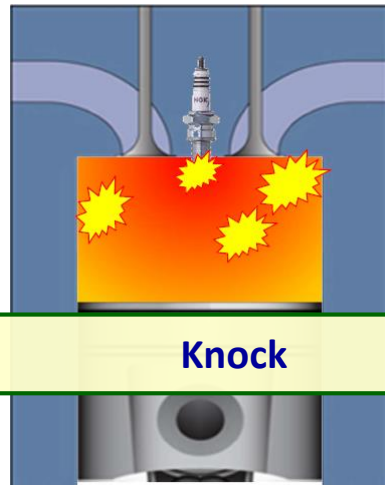


Gasoline Engine

Knock in SI Engine

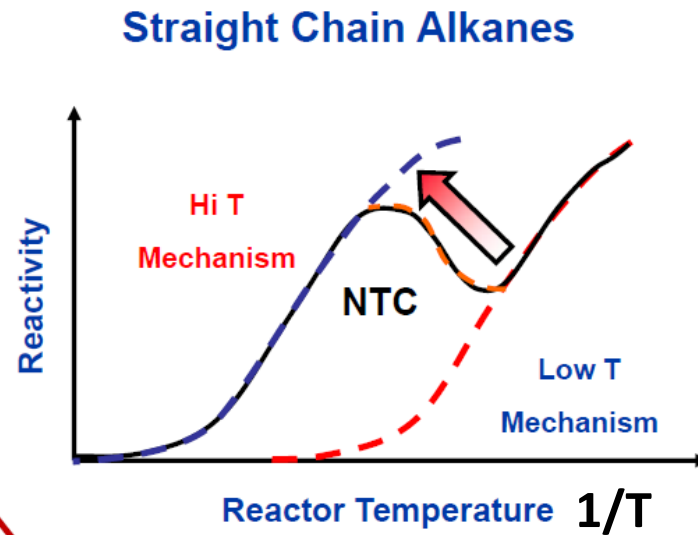


Knocking Cycle



The diagram illustrates the reaction scheme for the oxidation of 2-methylpentane, showing three competing pathways:

- Fast High Temperature Combustion:** This pathway is highlighted in a red oval. It starts with the abstraction of a hydrogen atom (RH) from 2-methylpentane to form a secondary radical. This radical then undergoes fast combustion to produce 2-methylpent-2-ene and a hydrogen atom ($\cdot\text{H}$).
- Branching Path:** This pathway is highlighted in an orange oval. It involves the addition of O_2 to the radical to form a peroxy radical. This peroxy radical can then undergo various reactions, including isomerization to a hydroperoxide radical, which can further decompose to form a carbonyl compound and a hydroxyl radical ($\cdot\text{OH}$).
- Degenerate Branching Path:** This pathway is highlighted in a blue oval. It shows the radical reacting with O_2 to form a peroxy radical, which then undergoes a degenerate branching reaction to produce a carbonyl compound and a hydroxyl radical ($\cdot\text{OH}$).



HO₂ Radicals lead to the formation of H₂O₂, when the temperature increases H₂O₂ → 2OH triggering the transition to high temperature ignition

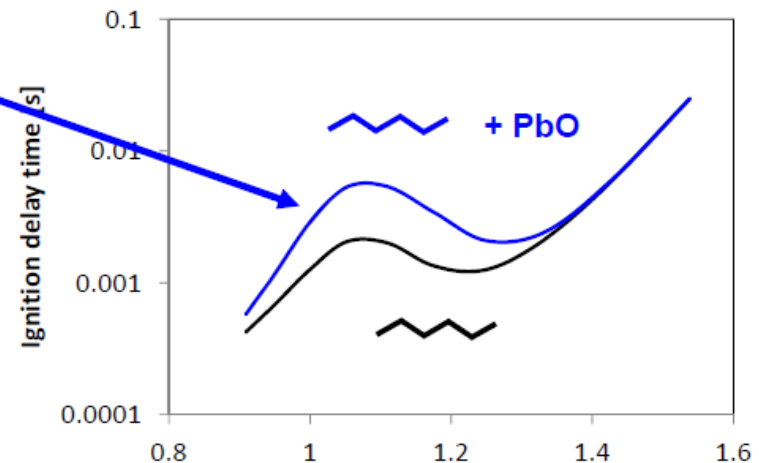
Gasoline Engine

Early way to reduce knock

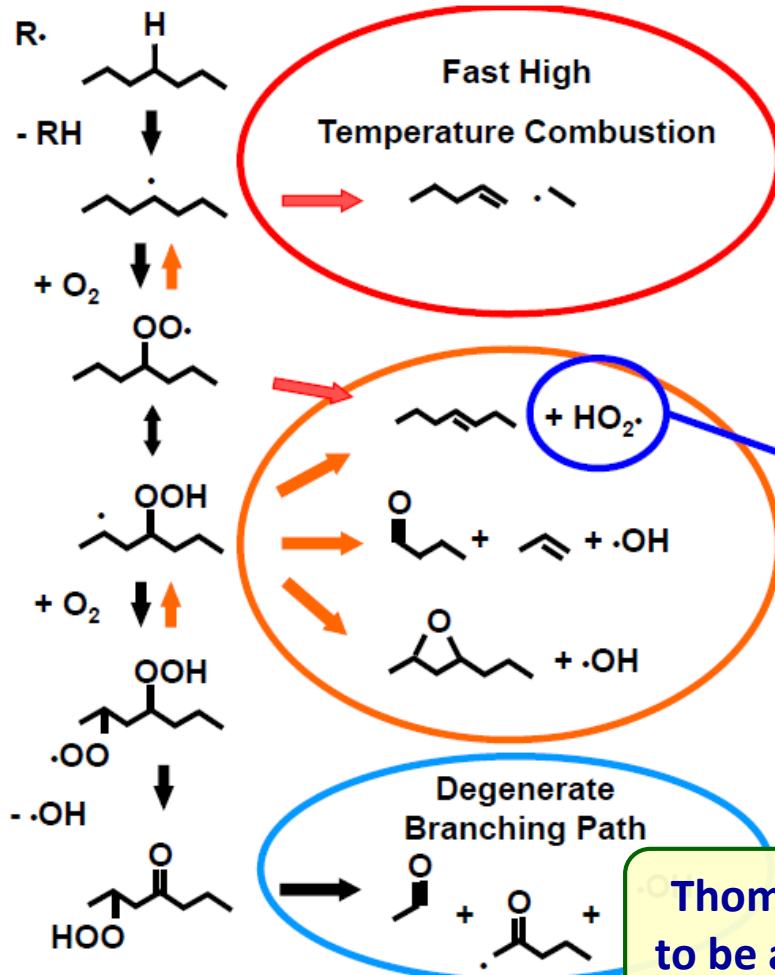
TEL (tetraethyl lead) additive

It is believed that TEL forms a “fog” of PbO which scavenges HO_2 radicals and prevents the build up of H_2O_2 inhibiting the transition to high temperature ignition

Pitz & Westbrook, Combustion and Flame 63: 113-133 (1986)



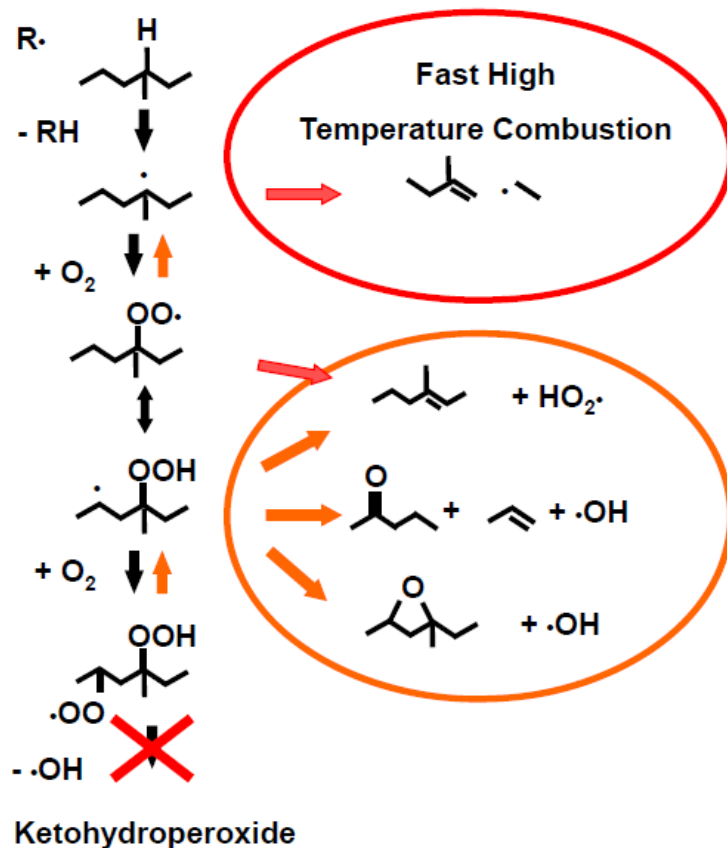
Thomas Midgley (1924) demonstrated tetraethyl lead to be an effective anti-knock agent (despite its toxicity)



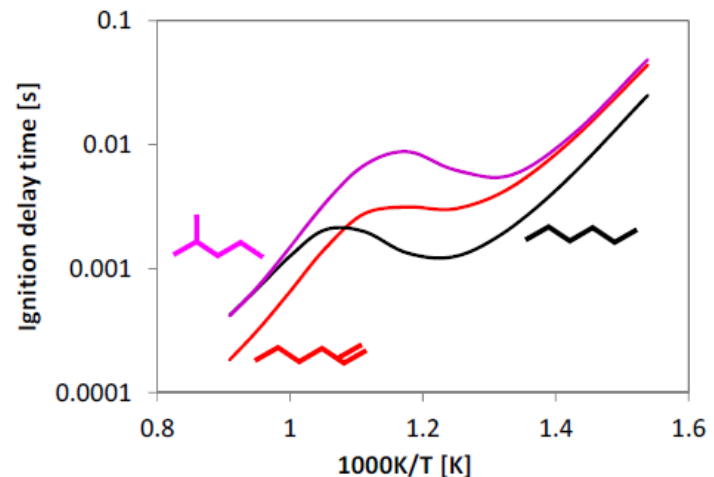
Gasoline Engine

Knock & Fuel – Linear Alkane Chemistry

- If the abstraction reaction happens at a tertiary site of an iso-alkane, the formation of ketohydroperoxides is not possible



The internal isomerization steps required by the low temperature degenerate branching mechanism are slower because of the higher activation energy required for primary hydrogens



Gasoline Engine

Octane number

n-Heptane (PRF0)

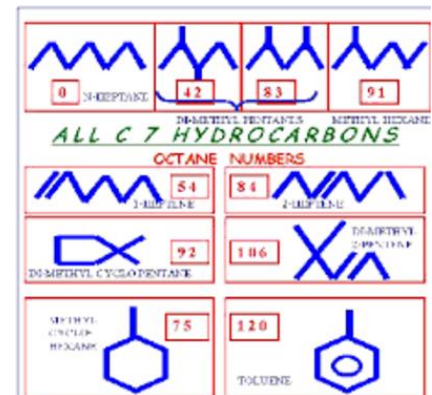
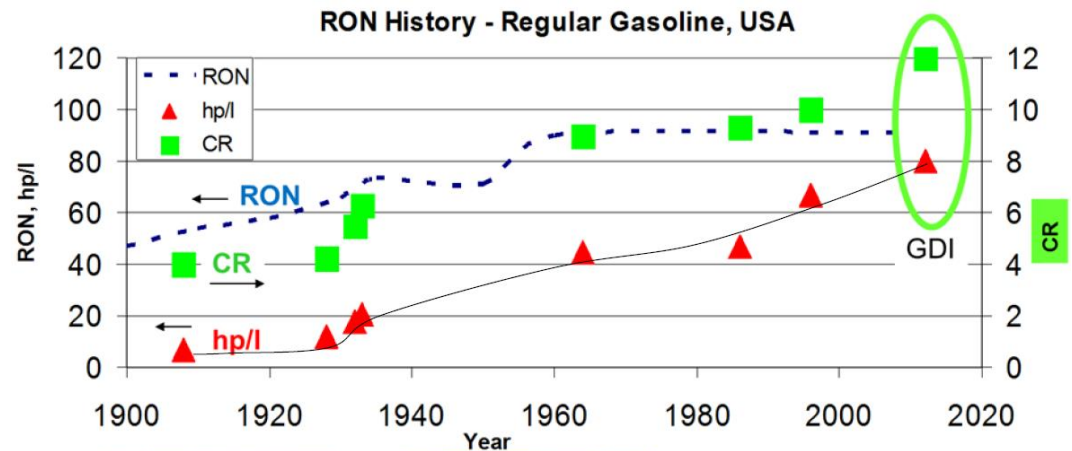
iso-Octane (PRF100)

Research Octane Number (RON)

- Speed 600 rpm
- Inlet Temperature 325K
- Spark Advance 13°
- Representative of idling/low load conditions

Motor Octane Number (MON)

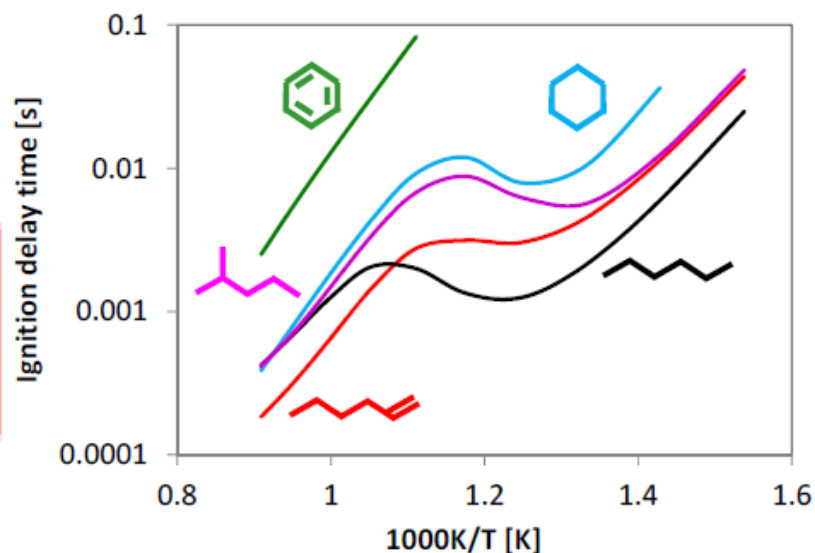
- Speed 900 rpm
- Inlet Temperature 422K
- Spark Advance 19-26°
- Representative of road/high load conditions



Gasoline Engine

Hydrocarbons with High Octane Number

Naphthenes

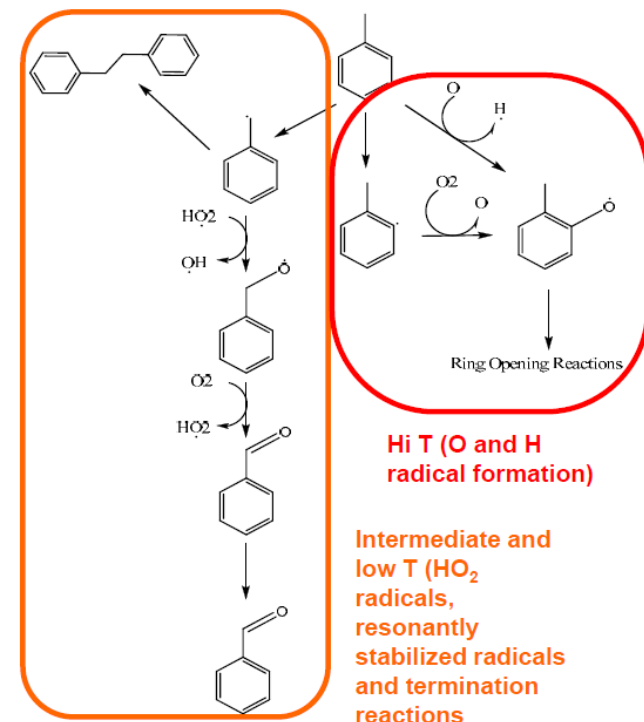


25 atm, $\Phi=1$, constant volume

The 3D structure of cycloalkane makes H atom unavailable for the internal abstraction reactions, reducing the reactivity of the fuel at low temperature

Aromatics

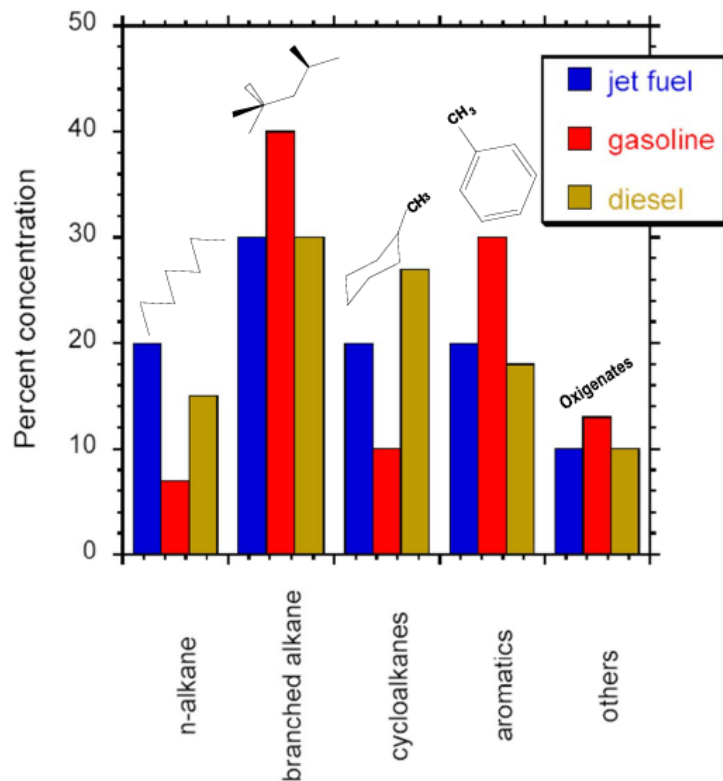
Aromatics do not have LT degenerate branching pathway and act as radical scavengers in mixtures (e.g. resonantly stabilized benzyl radical)



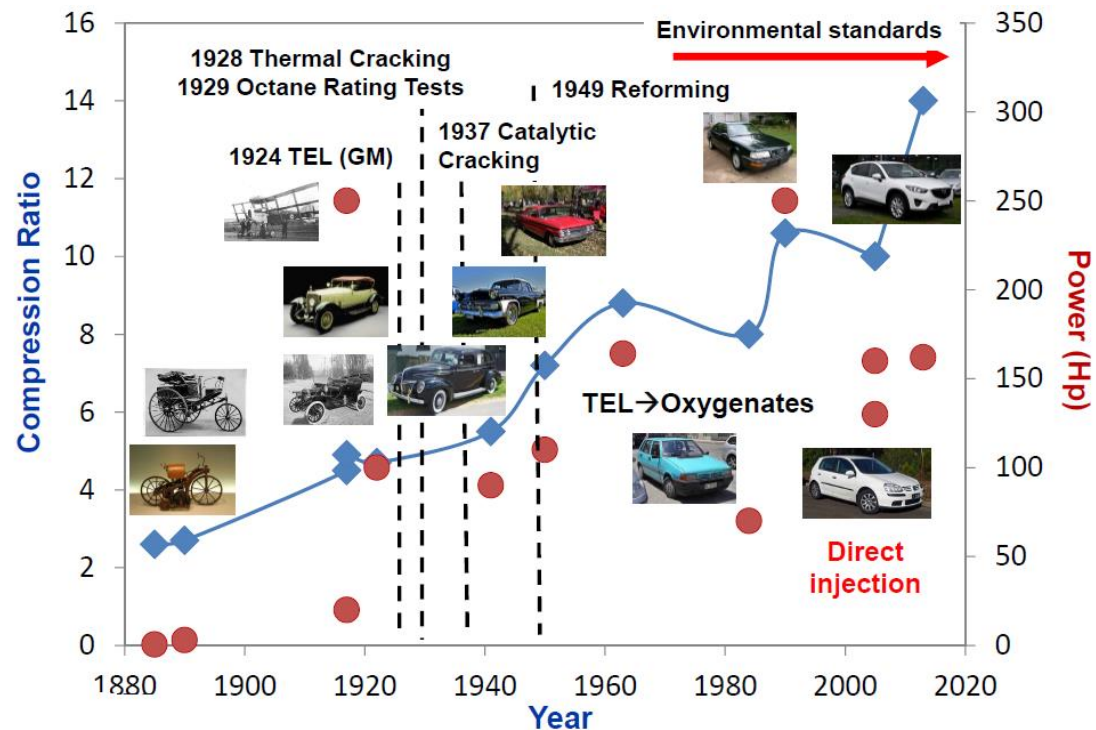
Gasoline Engine

Today's gasolines

Today gasolines are complex mixtures of hundreds of components



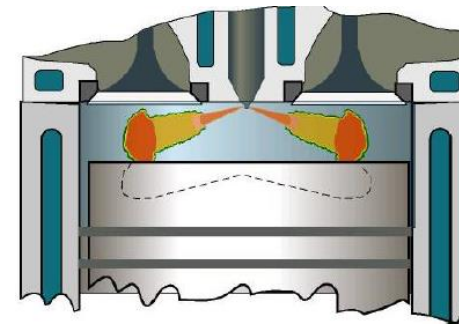
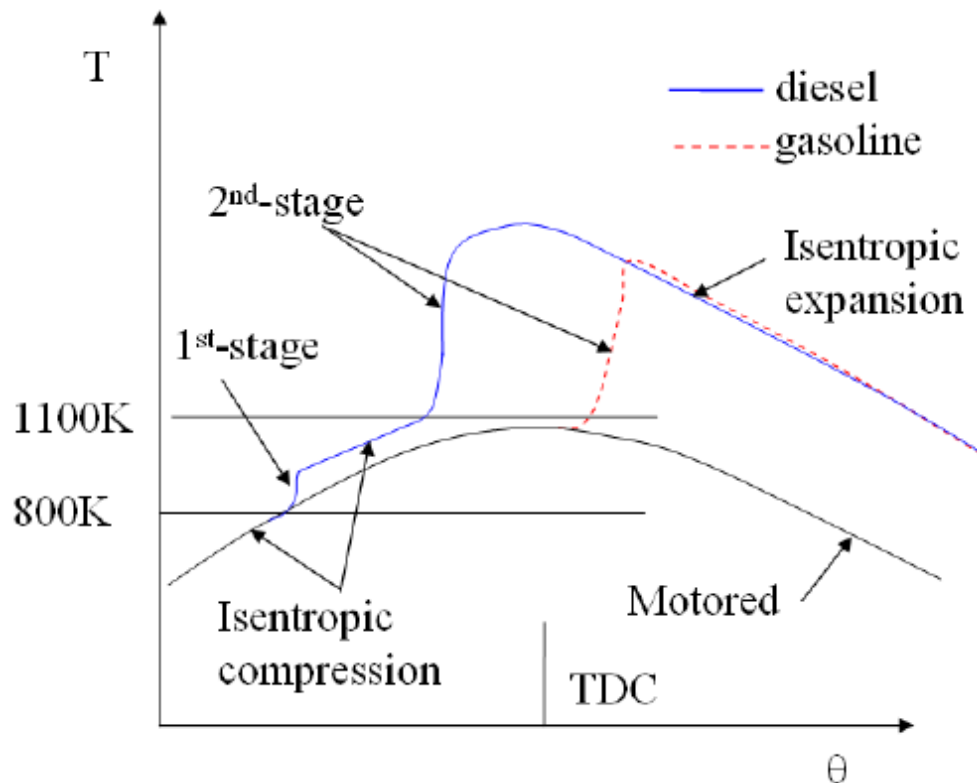
The introduction of additives, new refining technologies and fuel standards improved fuel quality, thus enabling the development of modern engines



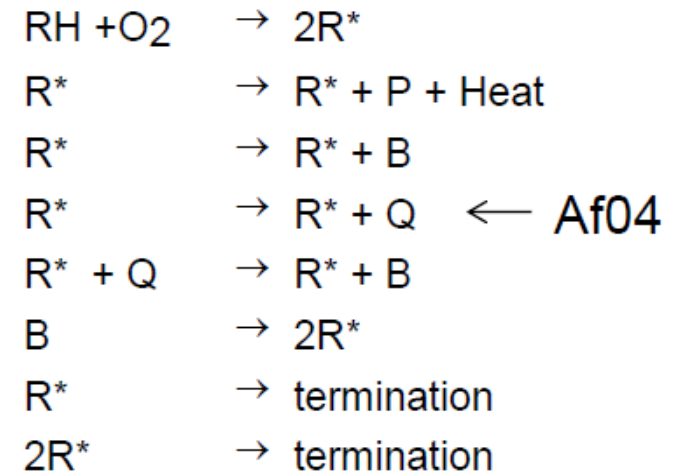
Diesel Engine

Diesel engine ignition

Ignition is controlled by chemistry



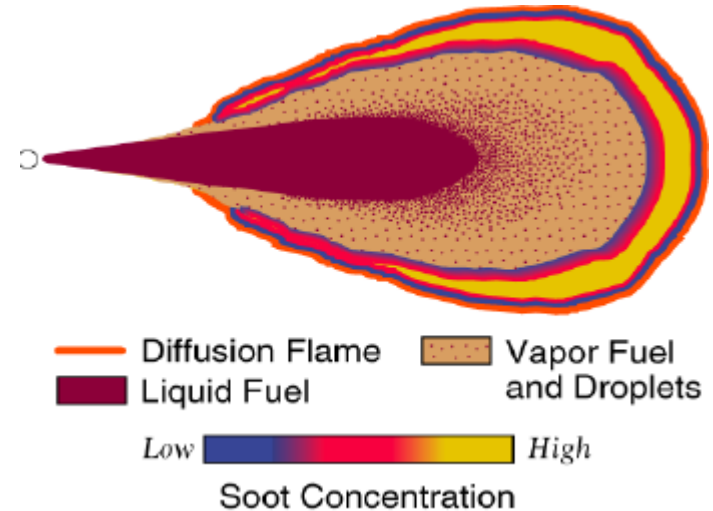
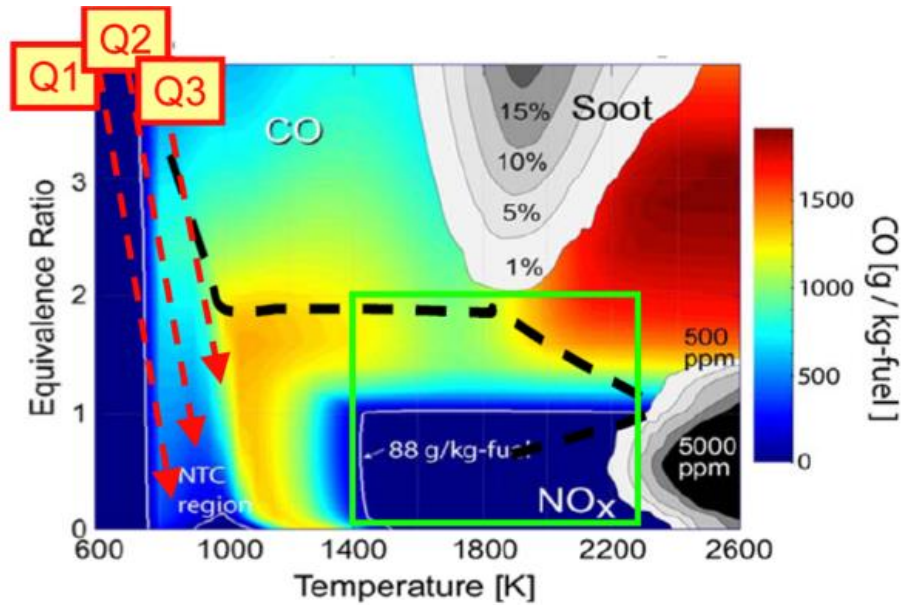
Shell Ignition Model



Prediction of low T auto-ignition needs detailed chemistry

Diesel Engine

Pollutant Formation in Diesel Engine



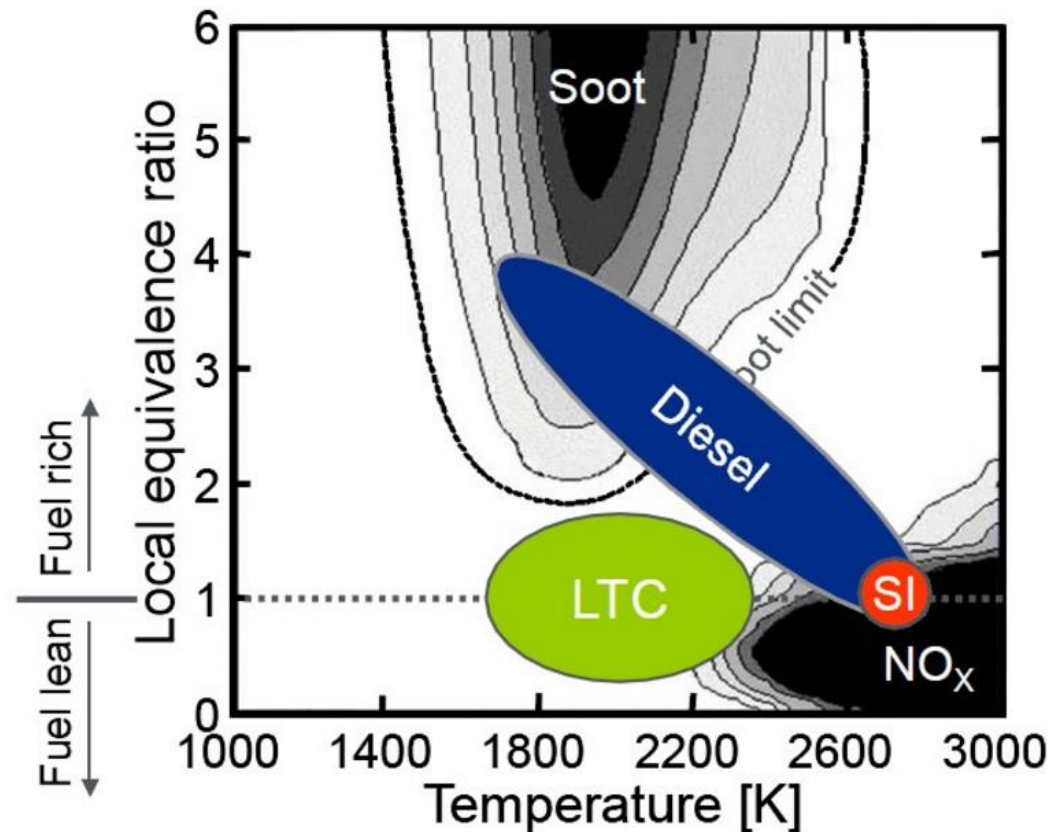
- High CR: high efficiency
- Rich combustion: soot production
- High temperature flame front: NO_x production

Diesel Engine

Control Strategy of NO_x Emission

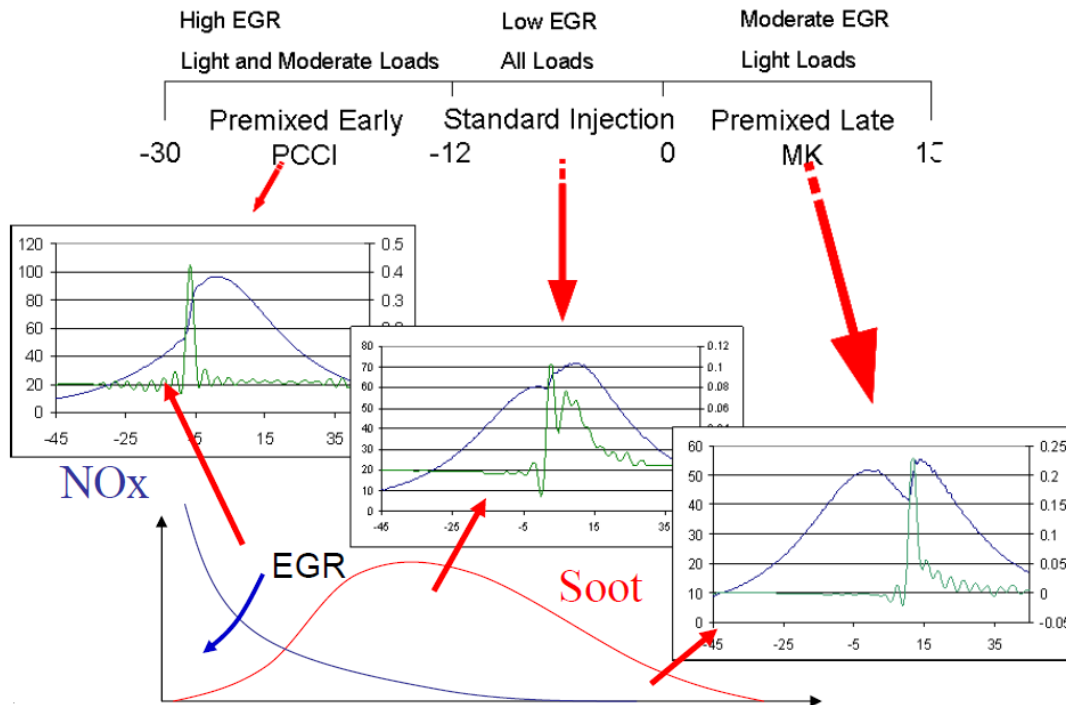
Low Temperature Combustion (LTC)

- The lower combustion temperature is useful for reducing engine-out NO_x emissions, which is highly sensitive to the flame temperature according to the Zel'dovich mechanism

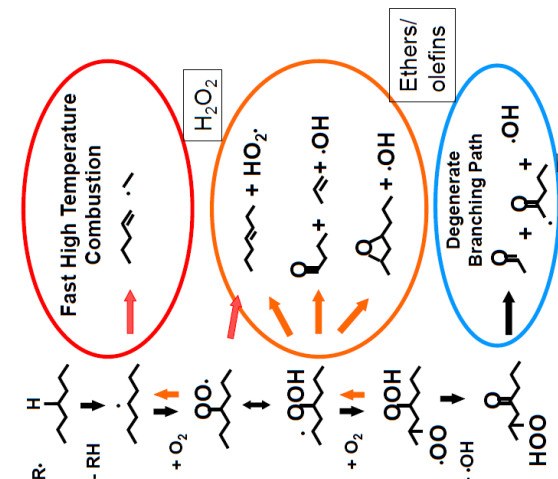
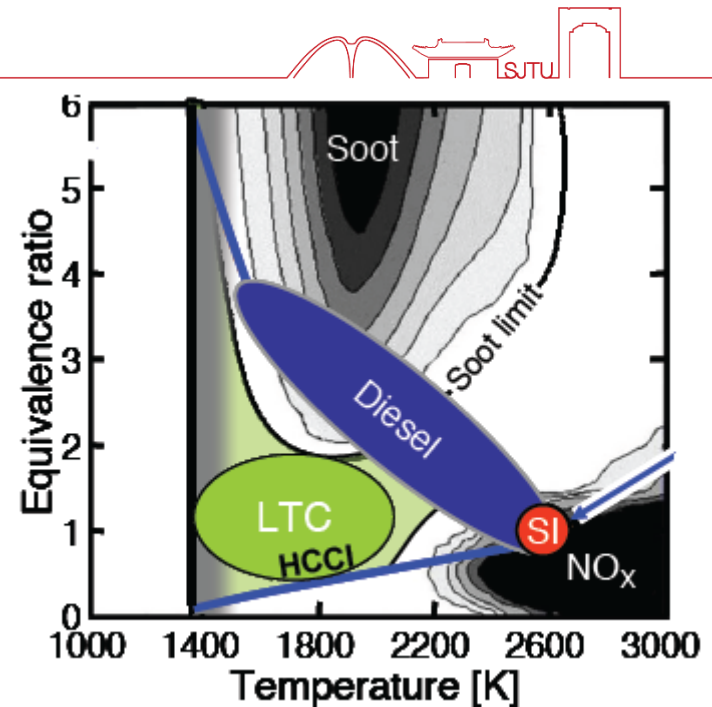


Diesel Engine

Low Temperature Combustion

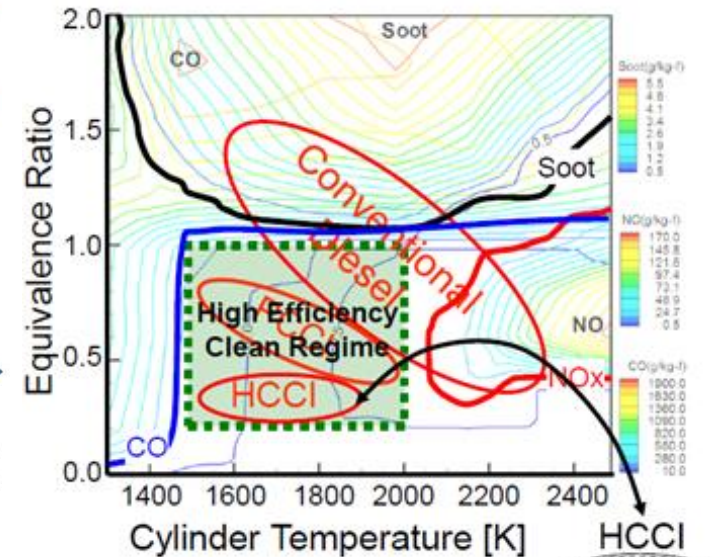
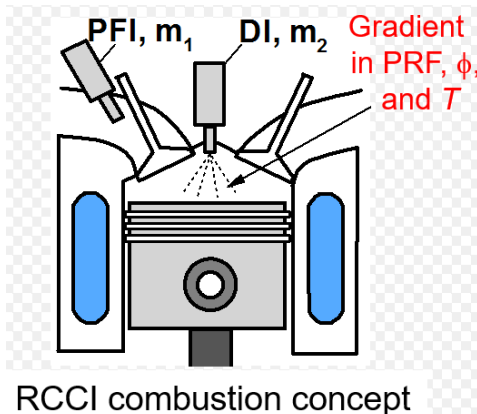
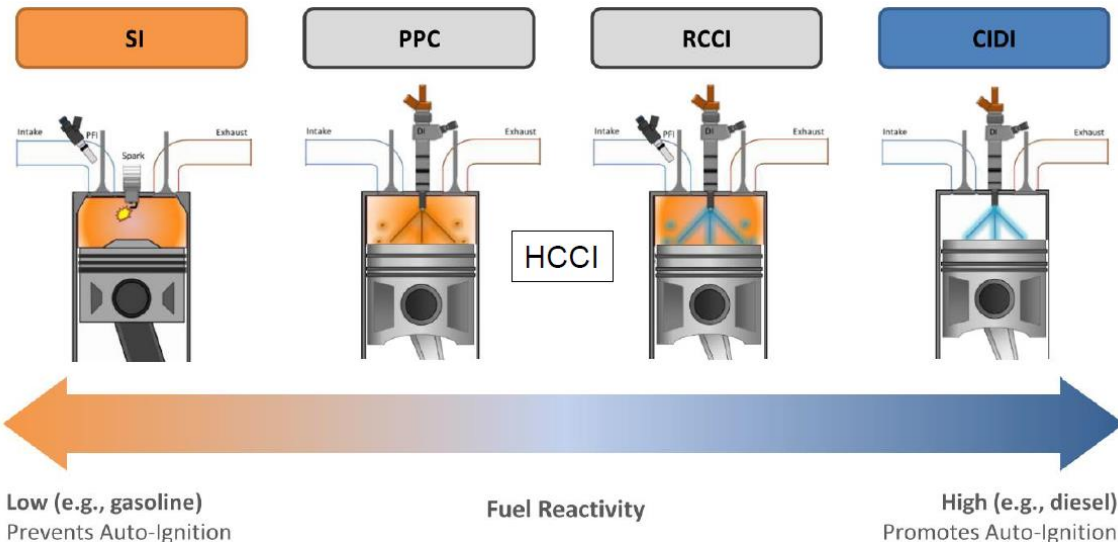


- Simultaneous reduction of NO_x and soot
- Increase the formation of CO and HC



Diesel Engine

Low Temperature Engine Combustion



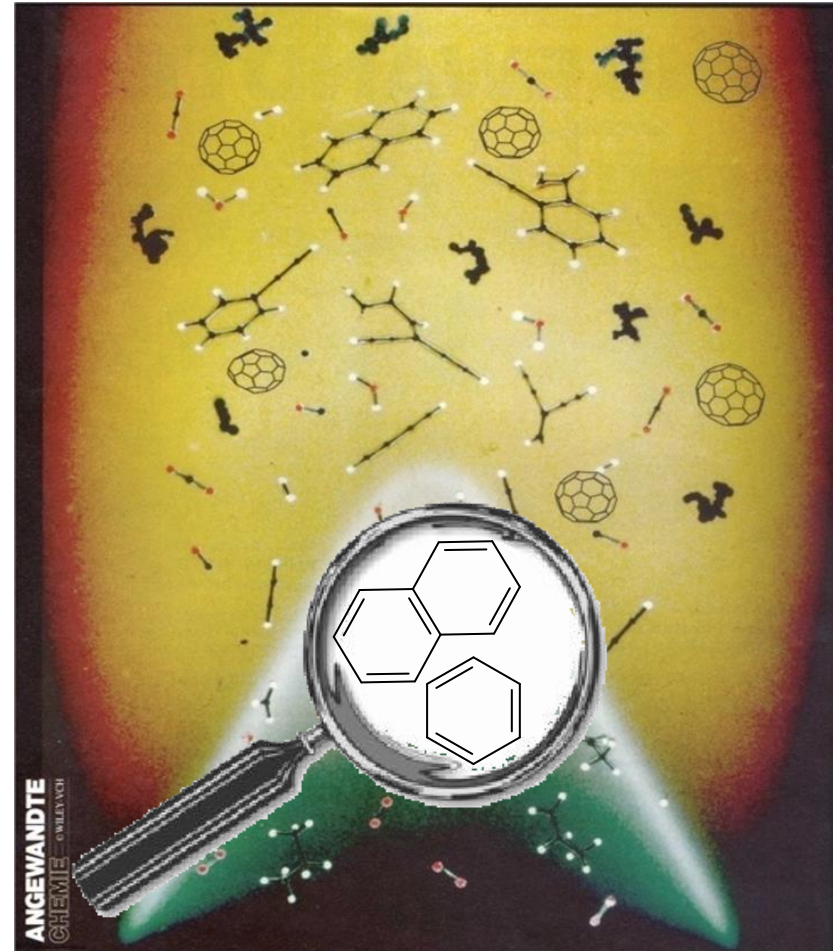
Requires precise charge preparation and combustion control mechanisms (for auto-ignition and combustion timing)



Diesel Engine

Pollutant Formation in Diesel Engine

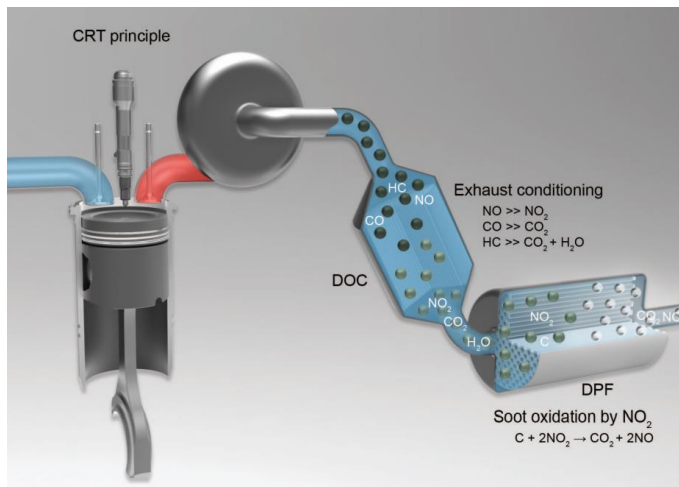
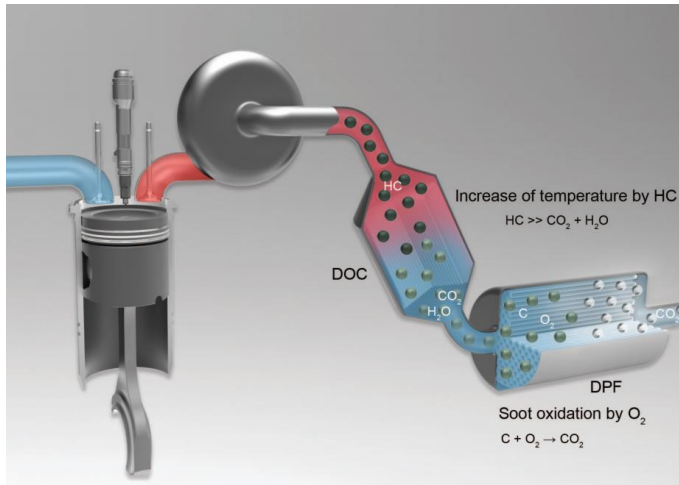
Soot formation in combustion



Diesel Engine

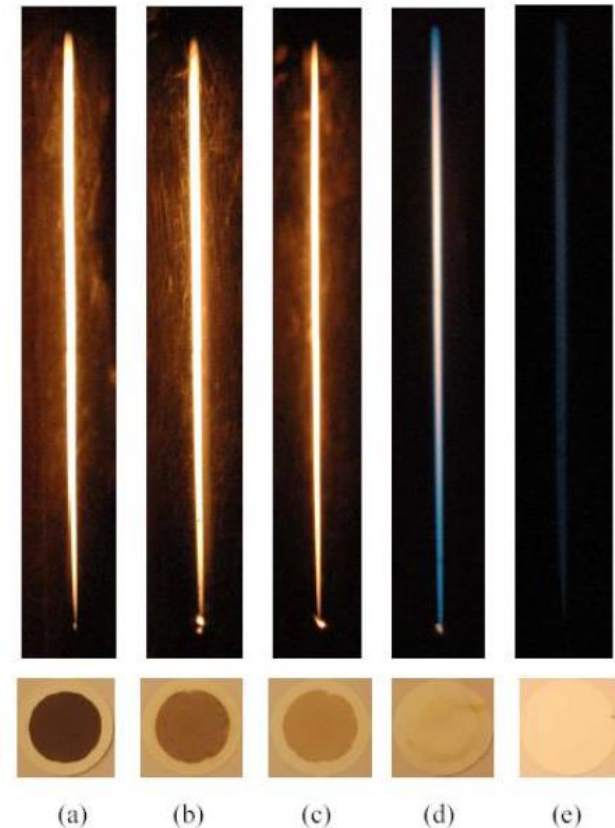
Control Strategy of Soot Emission

Diesel Particulate Filter

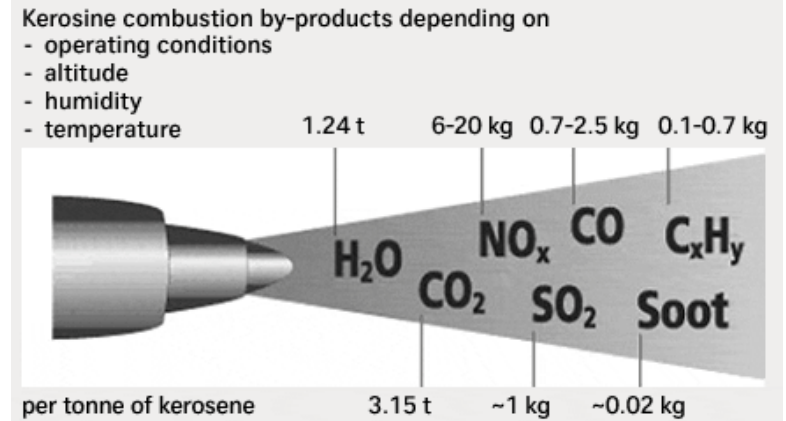


Oxygenated fuels blend

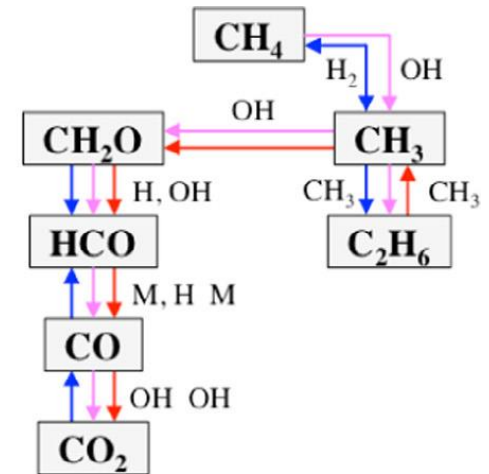
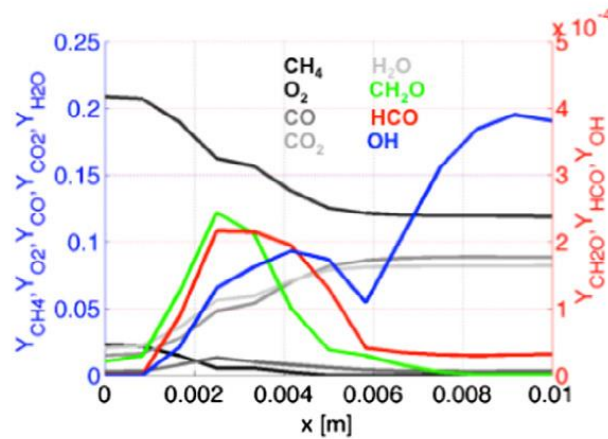
Biodiesel Blending: Soot Reduced



Soot catalytic oxidation technology



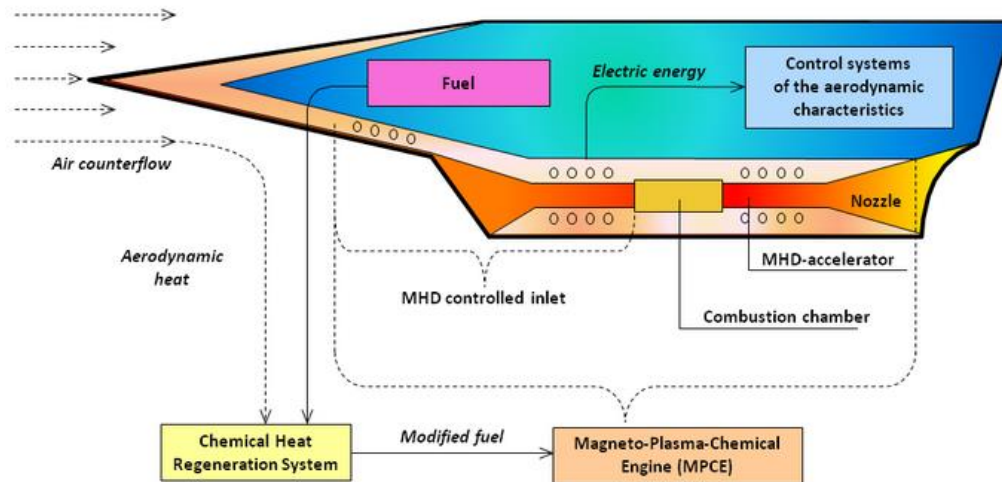
Pollutant formation



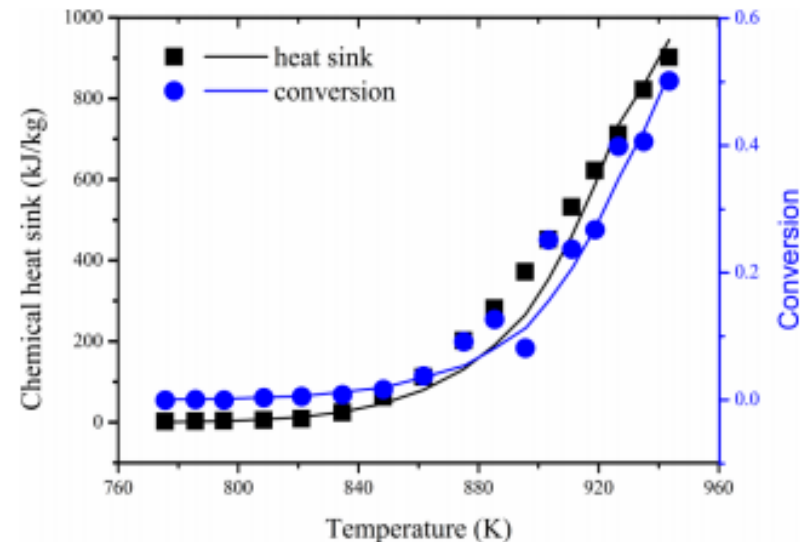
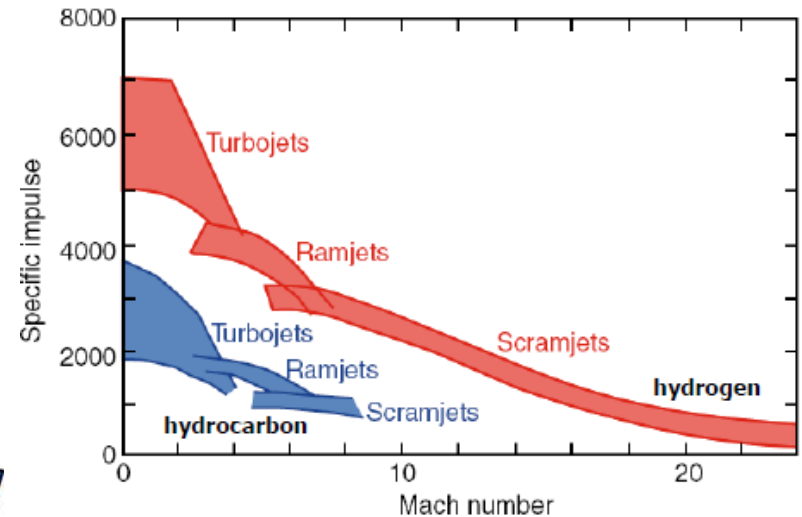
(c)

Jet Engine

Hypersonic engine and active cooling



Pyrolysis/catalytic pyrolysis of fuel

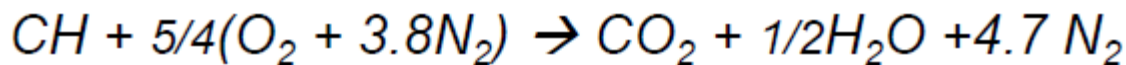


New Technology Applications



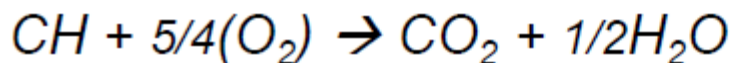
Oxy-fuel Combustion

Air-Combustion:



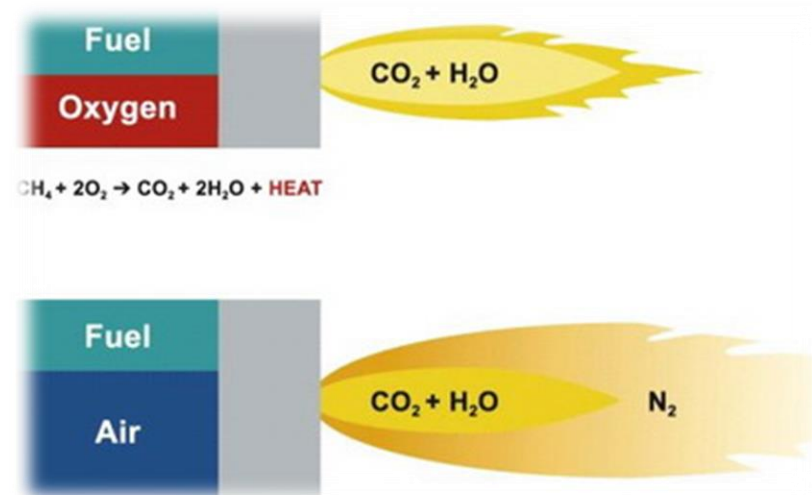
Costly to extract the CO_2 from the N_2 with amines

Oxy-Combustion:



Easy to extract the CO_2 from the H_2O via condensation

- ❖ Usual oxy-fuel approach: oxygen diluted with CO_2 or H_2O added to an existing boiler cycle. Dilution used to keep the temperatures same as existing cycle, efficiency of the plant is penalized by the energy needed to make oxygen.



New Technology Applications



Oxy-fuel Combustion

❖ Adiabatic Flame Temperature in Air versus Oxy-Combustion

TABLE 1.2 Approximate Flame Temperatures of Various Stoichiometric Mixtures, Initial Temperature 298 K

Fuel	Oxidizer	Pressure (atm)	Temperature (K)
Acetylene	Air	1	2600 ^a
Acetylene	Oxygen	1	3410 ^b
Carbon monoxide	Air	1	2400
Carbon monoxide	Oxygen	1	3220
Heptane	Air	1	2290
Heptane	Oxygen	1	3100
Hydrogen	Air	1	2400
Hydrogen	Oxygen	1	3080
Methane	Air	1	2210
Methane	Air	20	2270
Methane	Oxygen	1	3630
Methane	Oxygen	20	3460

^aThis maximum exists at $\phi = 1.3$.

^bThis maximum exists at $\phi = 1.7$.

- ❖ Flame temperature in oxy-fuel combustion is ~800 K higher than in air combustion (at 1 atm).
- ❖ At 20 atm flame temperature in oxy-fuel combustion is more than 1100 K higher.

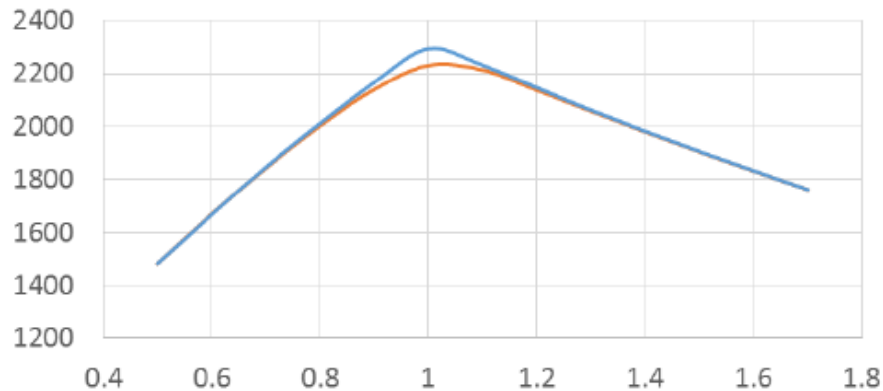
New Technology Applications



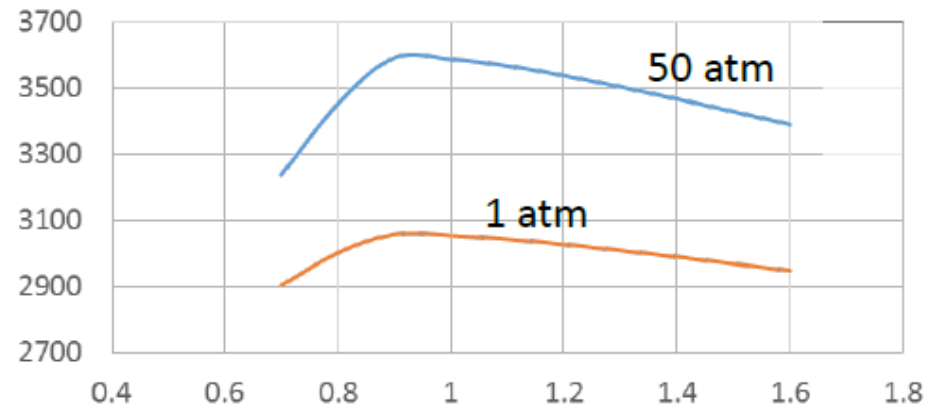
Oxy-fuel Combustion

❖ **Adiabatic Flame Temperature: Pressure effect → from 1 to 50 atm**

Flame Temperature [K] vs Φ
in Air at $p=1-50$ atm



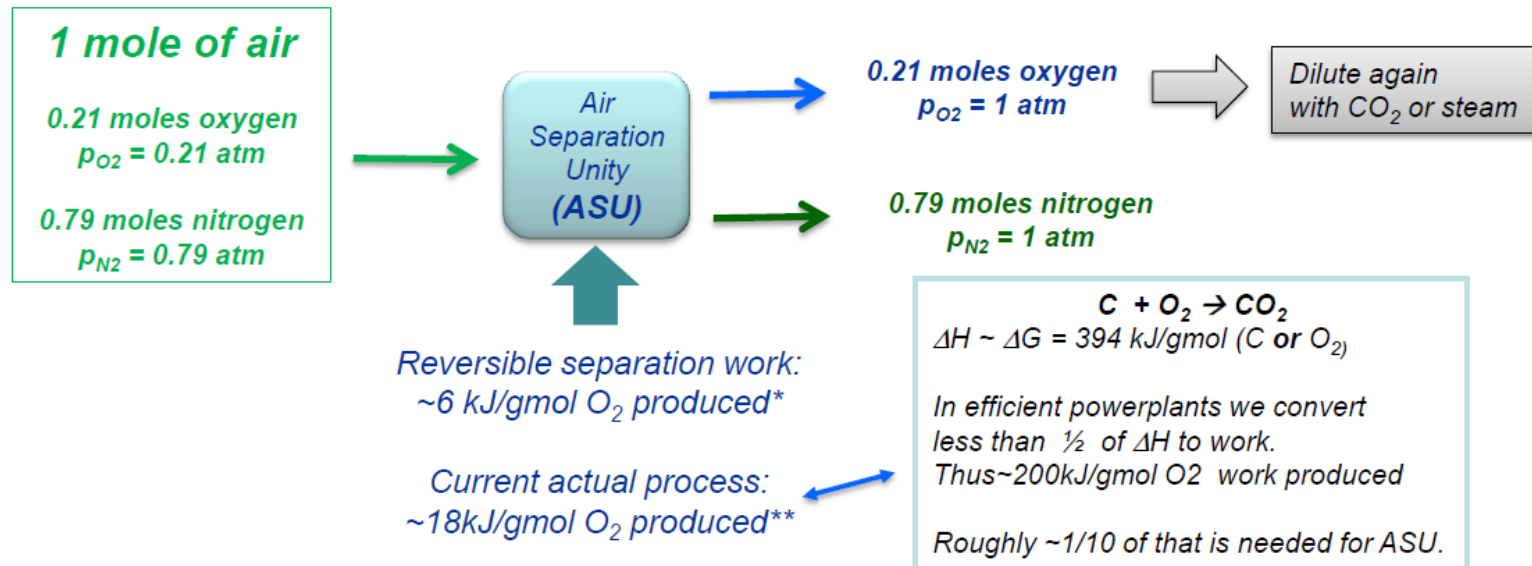
Flame Temperature [K] vs Φ
Oxycombustion at $p=1-50$ atm



- ❖ **CH_4/Air :** the max flame temperature increases ~ 60 K, because of a limited importance of dissociation reactions.
- ❖ **Oxy-fuel combustion:** maximum flame temperature increases even more than 500 K. At high pressures, dissociation reactions are less favored.

New Technology Applications

Oxy-fuel Combustion

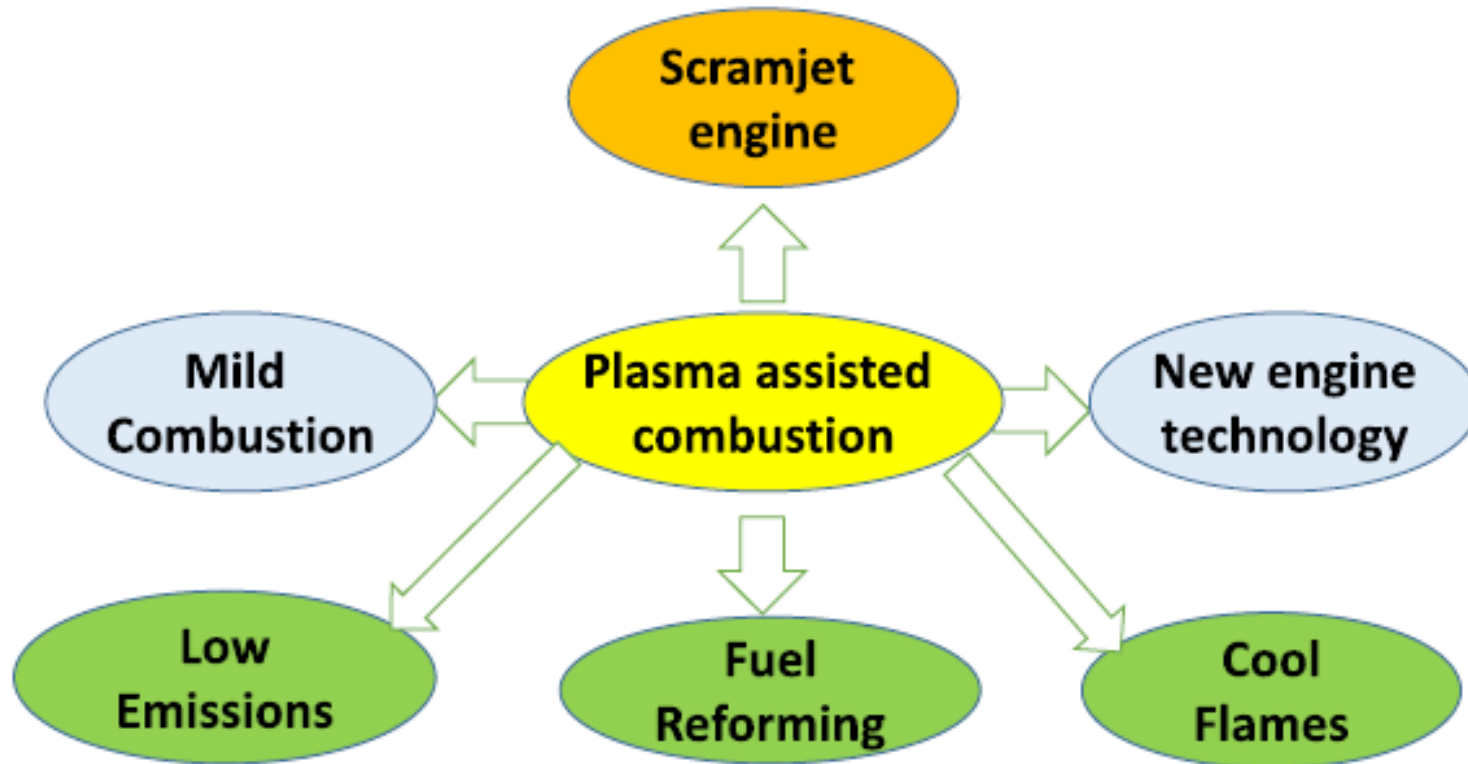


- ❖ Oxy-fuel already provides advantages in propulsion and process industries that benefit from high temperatures.
- ❖ Oxygen can be supplied today by commercial air separation units based on established cryogenic separation.
- ❖ The energy needed to separate oxygen from air is significant. Producing pure oxygen requires a lot of energy.

New Technology Applications



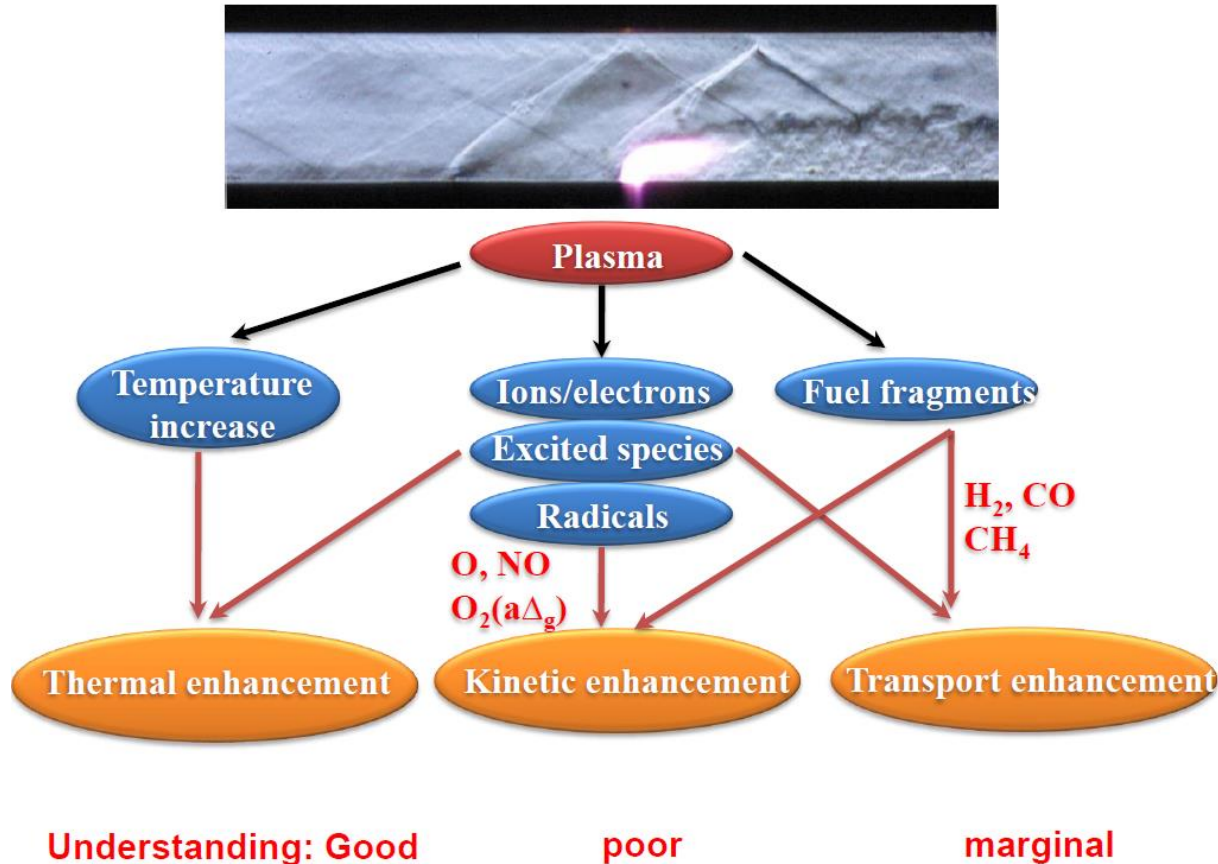
Plasma Assisted Combustion



- ❖ Plasma assisted combustion: a new way to control combustion, emissions, and fuel reforming.

New Technology Applications

Plasma Assisted Combustion

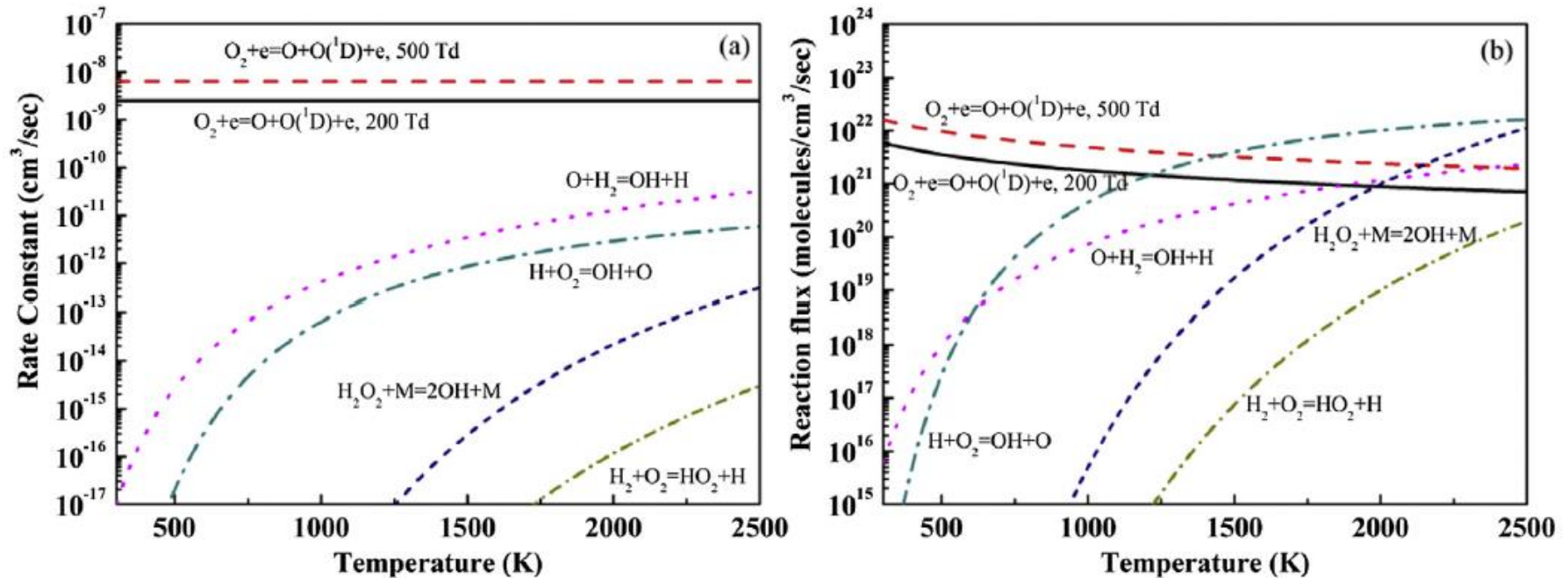


- ❖ Three major enhancement pathways of plasma interaction with combustion.

New Technology Applications



Plasma Assisted Combustion



- ❖ **Thermal enhancement:** plasma increases temperature and accelerates chemical reactions and fuel oxidation according to the Arrhenius law.

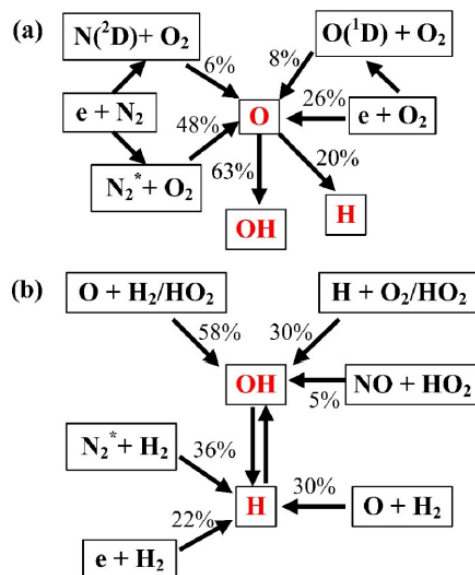
New Technology Applications



Plasma Assisted Combustion

- ❖ ***Kinetic enhancement***: produce high energy electrons and ions will further produce active radicals (O, OH, and H...) via direct electron impact dissociation, ion impact and recombination dissociation, and collisional dissociations of reactants with electronically excited and vibrationally excited molecules.

O, H and OH production and consumption pathways

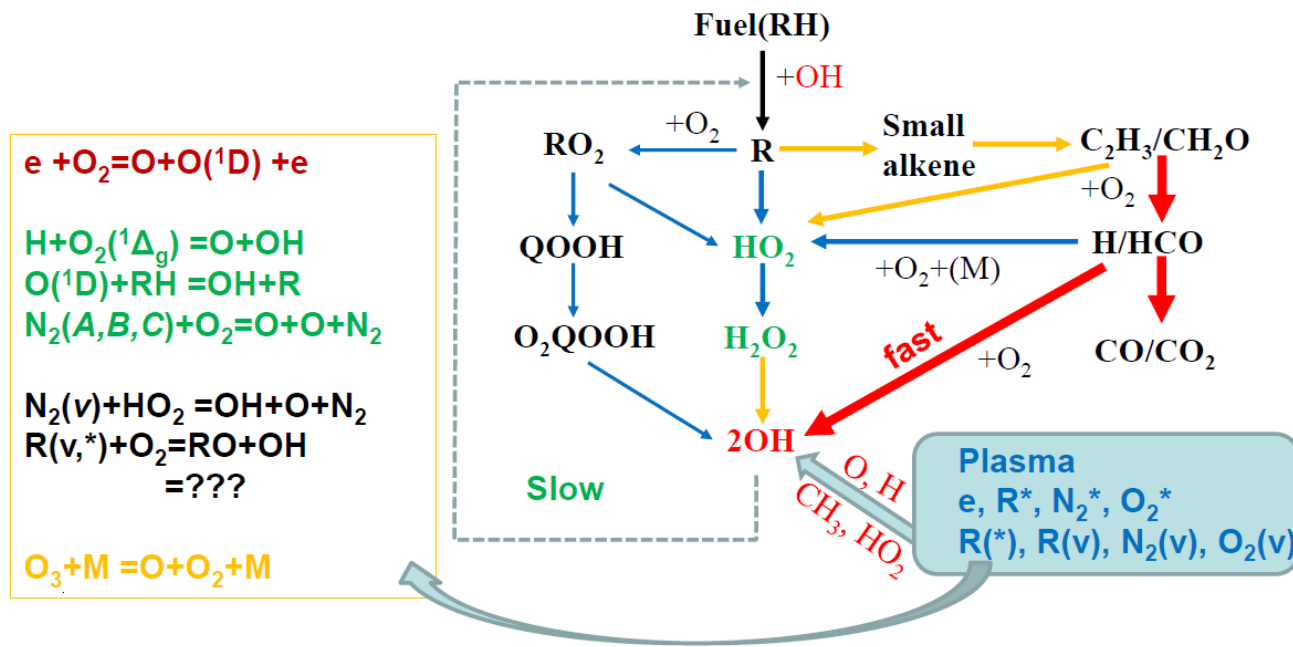


- ❖ O and H generated through direct electron impact/ excited species
- ❖ O and H are primary radicals produced via electron impact kinetics as well as thermal pathways
- ❖ OH is a secondary radical produced from O and H

New Technology Applications

Plasma Assisted Combustion

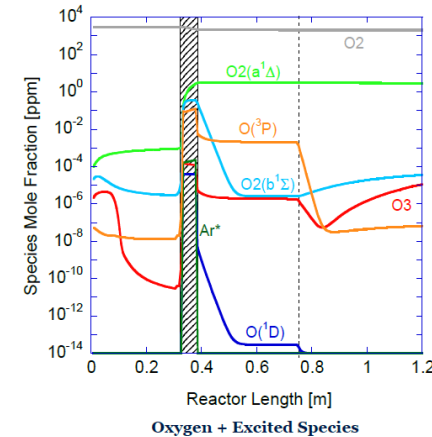
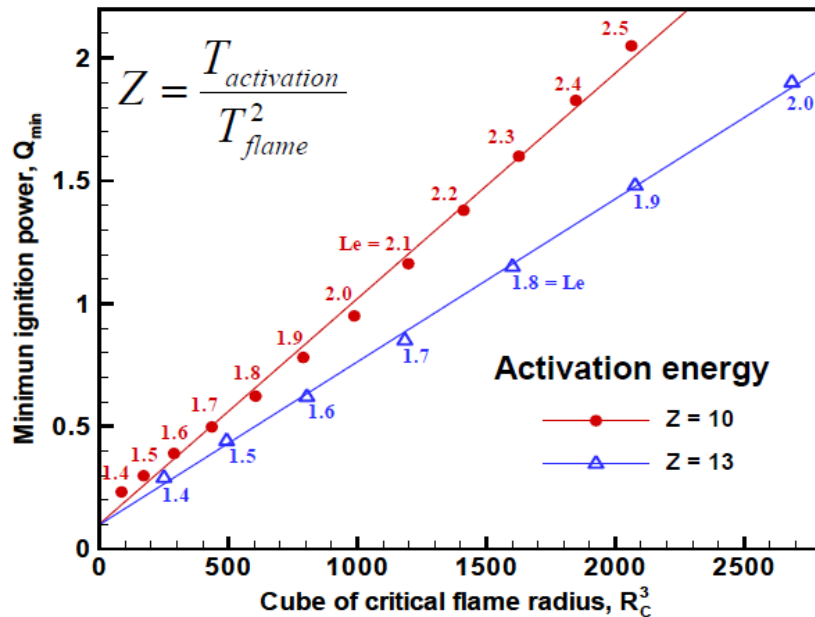
- ❖ **Kinetic enhancement:** plasma can produce long-lifetime reactive (e.g., ozone (O₃)) and catalytic intermediate species (e.g., NO) which can accelerate low temperature fuel oxidation.



- ❖ O₃ can kinetically activate cool flame chemistry by releasing atomic O at a low temperature.

New Technology Applications

Plasma Assisted Combustion

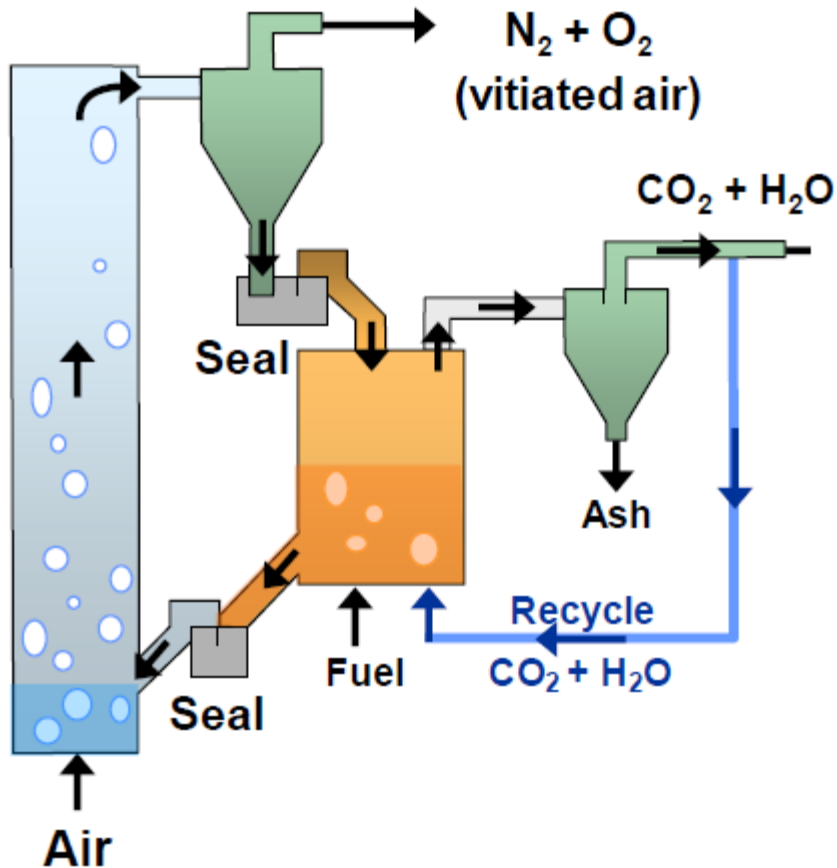


$$Le = \frac{\text{Thermal diffusivity}}{\text{Mass diffusivity}}$$

Molecule size of fuel $\uparrow$ = Mass Diffusivity $\downarrow$ = Le $\uparrow$

- ❖ In the *transport enhancement* pathway, plasma can breakdown the large fuel molecules to small fuel fragments changing the fuel diffusivity and thus modifying the combustion process.

Chemical Looping Combustion



Carbon + metal oxide = CO₂ + metal

Metal + air (oxygen) = metal oxide

- ❖ A metal oxide is used as an oxygen carrier to transfer oxygen from the combustion air to the fuel, which avoids direct contact between air and fuel.
- ❖ CO_2 is inherently separated during combustion from the rest of the flue gases with only small energy losses.
- ❖ Product is just CO_2 and H_2O .

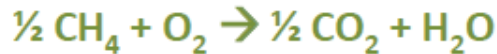


New Technology Applications



Chemical Looping Combustion

Methane overall



$\Delta H_{1000\text{K}} = -402\text{kJ}$ Exothermic overall reaction

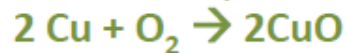
Copper carrier



$\Delta H_{1000\text{C}} = -283\text{kJ}$ Exothermic metal reduction

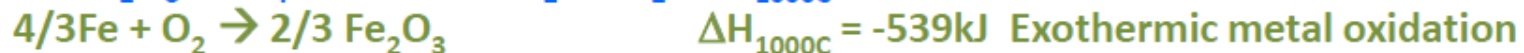
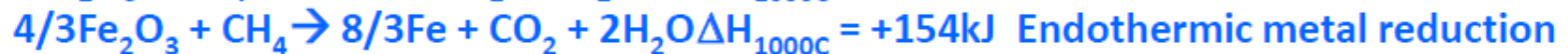
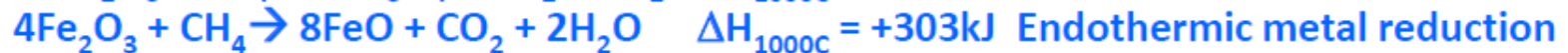


$\Delta H_{1000\text{C}} = -211\text{kJ}$ Exothermic metal reduction



$\Delta H_{1000\text{K}} = -274\text{kJ}$ Exothermic metal oxidation

Iron carrier



- ❖ In the first step, the fuel is oxidized to CO_2 and H_2O by a metal oxide (MexO_y) that is reduced to a metal (Me) or a reduced form MexO_{y-1} .
- ❖ The metal or reduced metal oxide is further oxidized with air in a second step, and the material regenerated is ready to start a new cycle.

Homework



n-heptane, iso-octane and toluene are commonly used as primary reference fuel (PRF) of gasoline:

- (1) How many fuel radicals (directly loss an H atom) can be produced from each one of them? Write the fuel radicals.*
- (2) If we directly use each one of them in spark ignition engine, what will happen? Give your explanations from combustion kinetics aspect.*

Homework



Open Question:

The 2013 Nobel Prize in Chemistry was awarded to Martin Karplus, Michael Levitt and Arieh Warshel, for "*the development of multiscale models for complex chemical systems*". Combustion reaction process is a typical complex reaction system which coupled flow and chemical reaction. Give your opinions about the importance of multiscale (theoretical, laboratory experiment, simulation, technology) research methods in combustion reaction kinetics.



上海交通大学
SHANGHAI JIAO TONG UNIVERSITY

Thanks for your attention



fqi@sjtu.edu.cn
lilei81@sjtu.edu.cn